LM01 Rates and Returns

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Introductory Note

Financial Calculator: CFA Institute allows only two calculator models during the exam:

- Texas Instruments BA II Plus (including BA II Plus Professional) and
- Hewlett Packard 12C (including the HP 12C Platinum, 12C Platinum 25th anniversary edition, 12C 30th anniversary edition, and HP 12C Prestige)

Unless you are already comfortable with the HP financial calculator, we recommend using the Texas Instruments financial calculator. Explanations and keystrokes in our study materials are based on the Texas Instruments BA II Plus calculator.
Before you start using the calculator to solve problems, we recommend that you set the number of decimal places to 'floating decimal'.

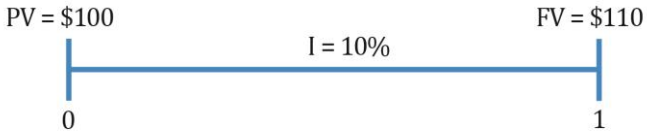
1. Introduction

If you have $100 today, versus an option to receive $100 after three years, what would you prefer?



Obviously, you would prefer $100 today. Even though you have the same amount ($100) in both cases, you prefer $100 today. This means that there has to be some value associated with time, because you are putting more value on the $100 that you are getting today, relative to the $100 at a later point in time. This is known as 'time value of money.'

Let us say that you are indifferent between $100 dollars today versus $ 110 after one year.



Present value (PV): The money today or the value today is called the present value (PV = 100). This could be an investment which you make at time 0.

Future value (FV): The value at a future point in time is called the future value (FV = 110).

Interest rate (I): The relationship or the link between present value and future value is established through an interest rate (I = 10%).

In this reading we will cover the meaning and interpretation of interest rates, and learn how to calculate, interpret, and compare different return measures.

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2. Interest Rates and Time Value of Money

Let's discuss the different interpretations of interest rates using an example. Say you lend $900 today and receive $990 after one year (–ve sign indicates outflow).



Interest rates can be interpreted as:

1. Required rate of return: The fact that you are willing to give $900 today on the condition that you get $990 after one year means that to engage in this transaction, you require a return of 10%. (Simple calculation will show you that the interest rate in this transaction is 10%).
2. Discount rate: You can discount the money that you will receive after one year i.e. $990 at 10% to get the present value of $900 (990/1.1 = 900). Therefore, the 10% can also be thought of as a discount rate.
3. Opportunity cost: Let's say instead of lending the $900, you spent it on something else. You have then forgone the opportunity to earn 10% interest. Therefore, 10% can also be thought of as an opportunity cost.

Determinants of Interest Rates

As an investor, we can think of the interest rate as a sum of the following components:

Interest rate = Real risk-free interest rate + Inflation premium + Default risk premium + Liquidity premium + Maturity premium

Let's look at the different components.

- Real risk-free interest rate: This is the rate that you get on a security that has no risk and is extremely liquid. We make an assumption here that there is no inflation.
- Inflation premium: We can then add on an inflation premium. Inflation premium is the expected annual inflation in the upcoming period.
- Default risk premium: We can also then add a default risk premium. This is the additional premium that investors require because of the risk of default.
  Example: Let's say that you lend $100 each to person A and person B. However, B has a high risk of default, so you are worried that he might not pay. Therefore you might demand a higher return from B as compared to A, because of the risk of default. This additional return that you demand is called the default risk premium.
- Liquidity premium: Liquidity premium compensates investors for the risk of receiving less than the fair value for an investment if it must be converted to cash quickly.

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Example: Think of two investments C and D which are similar in all regards. The only difference is that investment C is extremely liquid, whereas investment D is not that liquid. Clearly as investors, we will demand a higher return on D because it is not easy to sell. This additional return that we demand is called the liquidity premium.

- Maturity premium: Finally, we have the maturity premium. This is the premium that investors demand on a security with long maturity. The maturity premium compensates investors for the increased sensitivity of the market value of debt to a change in market interest rates as maturity is extended.
  Example: Let's say we have two securities, E and F. Security E has a maturity of 1 year and security F has a maturity of 4 years. Because of the longer maturity, F has more risk, in terms of its price being more sensitive to changes in interest rate.
  Instructor's Note: You will understand this concept better when you study fixed income securities. But for now, you can take it as a given that F has higher risk because of the longer maturity.
  Obviously, investors will demand some compensation for the higher level of risk. This additional return that investors demand is called the maturity premium.

Nominal risk-free rate:

Nominal risk-free rate = Real risk-free interest rate + Inflation premium.

So if the real risk-free rate is 3% and the inflation premium is 2%, then the nominal risk-free rate is 5%.

Instructor's Note: On the exam if you get a term 'risk-free rate' with no mention of whether the rate is real or nominal, then the assumption is that we are talking about the nominal risk-free rate.

Example

Investments	*Maturity (in years)*	*Liquidity*	*Default risk*	*Interest Rates(%)*
A	1	High	Low	2.0
B	1	Low	Low	2.5
C	2	Low	Low	r
D	3	High	Low	3.0
E	3	Low	High	4.0

1. Explain the difference between the interest rates on Investment A and Investment B.
2. Estimate the default risk premium.
3. Calculate upper and lower limits for the interest rate on Investment C, r.

Solution:

1. Investments A and B have the same maturity and the same default risk. However, B has a lower liquidity as compared to A. Hence, investors will demand a liquidity premium on B.

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The difference between their interest rates i.e. 2.5 – 2.0 = 0.5% is equal to the liquidity premium.

2. Consider investments D and E, they have the same maturity, but different liquidity and different default risk. Let's make liquidity the same and create a new low liquidity version of D. This version will have a higher interest rate, because now investors will demand a liquidity premium. We have already determined that the liquidity premium is 0.5%. Therefore, the low liquidity version of D will have an interest rate of 3.0 + 0.5 = 3.5%.
   Now compare this version of D with investment E. The only difference between the two is default risk. E has a higher default risk. Therefore, the difference between their interest rates i.e. 4.0 – 3.5 = 0.5% must be equal to the default risk premium.
3. Notice that between B and C, the only difference is that C has a longer maturity. Therefore, interest rate of C must be higher than B (2.5%).
   Also notice that between C and the low liquidity version of D, the only difference is that C has a shorter maturity. Therefore, interest rate on C has to be lower than the low liquidity version of D (3.5%).
   So, the range for C is 2.5 < r < 3.5.

3. Rates of Return

A financial asset's total return consists of two components: income and capital appreciation.

Holding Period Return

Holding period return (HPR) is the return that an investor earns over a specified holding period. The holding period can range from days to years. The formula for calculating HPR for an investment that makes one-time cash payment at the end of the holding period is given below:

$$\text{HPR} = \frac{P_1 - P_0 + D_1}{P_0} = \frac{\text{ending value} - \text{beginning value} + \text{cash flow}}{\text{beginning value}}$$

where:

P_0 = initial investment

P_1 = price received at the end of the holding period

D_1 = cash paid by the investment at the end of the holding period

Example

Assume we buy a stock for $50. Six months later, the stock price goes up to $53 and we receive a dividend of $2. Calculate the holding period return.

Solution:

The return for the six-month holding period is given below:

$$\text{HPR} = \frac{53 + 2 - 50}{50} = 0.10 = 10\%$$

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Sometimes a holding period return can be computed for a period longer than a year. For example, an analyst may need to calculate the holding period return from three annual returns. In this case, the holding period return is calculated as:

$R = [(1 + R_1) \times (1 + R_2) \times (1 + R_3)] - 1$

where:
R_1, R_2, and R_3 are the three annual returns

Example:

The annual returns of a mutual fund for the past three years are presented below.

2020 20%

2021 -8%

2022 -1%

Calculate the fund's holding period return over the three-year period.

Solution:

$R = [(1 + R_1) \times (1 + R_2) \times (1 + R_3)] - 1$

$R = [(1 + 0.20)\ (1 - 0.08)\ (1 - 0.01)] - 1 = 0.0929 = 9.296\%$

Arithmetic or Mean Return

The arithmetic mean is the sum of the observations divided by the number of observations. It is expressed as:

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

For example, if the annual returns of a mutual fund for the past three years are 20%, - 8%, and -1%, the mean return is $\frac{20-8-1}{3} = 3.66\%$

A drawback of the arithmetic mean is that it is sensitive to extreme values (outliers). It can be pulled sharply upward or downward by extremely large or small observations, respectively.

Outliers

When data contains outliers, there are three options to deal with the extreme values:

Option 1: Do nothing; use the data without any adjustment.

Option 2: Delete all the outliers.

Option 3: Replace the outliers with another value.

Option 1 is appropriate in cases when the extreme values are genuine.

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Option 2 excludes extreme observations. A *trimmed mean* excludes a stated percentage of the lowest and highest values and then calculates the arithmetic mean of the remaining values.

Option 3 replaces extreme observations with observations closest to them. A *winsorized mean* assigns a stated percentage of the lowest values equal to one specified low value and a stated percentage of the highest values equal to one specified high value, and then computes a mean from the restated data.

Geometric Mean Return

The geometric mean is calculated as the nth root of a product of n numbers. The most common application of the geometric mean is to calculate the average return of an investment. The formula is:

$$R_G = [(1 + R_1)(1 + R_2) \ldots (1 + R_n)]^{\frac{1}{n}} - 1$$

Example

The return over the last four periods for a given stock is: 10%, 8%, -5% and 2%. Calculate the geometric mean.

Solution:

$$[(1 + 0.10)(1 + 0.08)(1 - 0.05)(1 + 0.02)]^{\frac{1}{4}} - 1 = 0.0358 = 3.58\%$$

Given the returns shown above, $1 invested at the start of period 1 grew to:

$1.00 x 1.10 x 1.08 x 0.95 x 1.02 = $1.151. If the investment had grown at 3.58% every period, $1.00 invested at the start of period 1 would have increased to:

$1.00 x 1.0358 x 1.0358 x 1.0358 x 1.0358 = $1.151. As expected, both scenarios give the same answer. 3.58% is simply the average growth rate per period.

Other applications of the geometric mean involve the use of a second formula:

$$\ln\overline{X}_G = \frac{\sum_{i=1}^{n} \ln X_i}{n}$$

Instructor's Note: This formula is less testable.

Example:

The P/E ratio of a stock over the past four years has been: 10, 15, 14, 13. Calculate the geometric mean P/E.

Solution:

$$\ln\overline{X}_G = \frac{\sum_{i=1}^{n} \ln X_i}{n}$$

$$\ln\overline{X}_G = \frac{\ln 10 + \ln 15 + \ln 14 + \ln 13}{4} = 2.55$$

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$\overline{X}_G = e^{2.55} = 12.807$

Using Geometric and Arithmetic Means

The geometric mean is appropriate to measure past performance over multiple periods.

Example
The portfolio returns for the past two years were 100% in year 1 and -50% in year 2. What was the mean return?

Solution:
Past return = geometric mean = $((1+1.0) \times (1-0.5))^{0.5} - 1 = 0\%$

The arithmetic mean is appropriate for forecasting single period returns.

Example
Two possible returns for the next year are 100% and -50%. What is the expected return?

Solution:
Expected return = Arithmetic mean = $(100 - 50)/2 = 25\%$

The Harmonic Mean

The harmonic mean is a special type of weighted mean in which an observation's weight is inversely proportional to its magnitude. The formula for a harmonic mean is:

$$X_H = \frac{n}{\sum_{i=1}^{n} \frac{1}{X_i}}$$

where: $X_i > 0$ for i = 1, 2 … n, and n is the number of observations

The harmonic mean is used to find average purchase price for equal periodic investments.

Example
An investor purchases $1,000 of a security each month for three months. The share prices are $10, $15 and $20 at the three purchase dates. Calculate the average purchase price per share for the security purchased.

Solution:

The average purchase price is simply the harmonic mean of $10, $15 and $20.

The harmonic mean is:

$$\frac{3}{\frac{1}{\$10} + \frac{1}{\$15} + \frac{1}{\$20}} = \$13.85.$$

A more intuitive way of solving this is total money spent purchasing the shares divided by the total number of shares purchased.

Total money spent purchasing the shares = $1,000 x 3 = $3,000

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Total shares purchased = sum of shares bought each month

$$= \frac{\$1{,}000}{10} + \frac{\$1{,}000}{15} + \frac{\$1{,}000}{20}$$

= 100 + 66.67 + 50 = 216.67

$$\text{Average purchase price per share} = \frac{\$3{,}000}{216.67} = \$13.85$$

Comparison of AM, GM and HM

- Arithmetic mean × Harmonic mean = Geometric mean^2
- If the returns are constant over time: AM = GM = HM.
- If the returns are variable: AM > GM > HM.
- The greater the variability of returns over time, the more the arithmetic mean will exceed the geometric mean.

Which mean to use?

- Arithmetic mean: Should be used with single period or cross-sectional data.
- Geometric mean: Should be used with time-series data.
- Harmonic mean: Should be used to find average purchase price for equal periodic investments.
- Trimmed mean: Should be used when the data has extreme outliers.
- Winsorized mean: Should be used when the data has extreme outliers.

4. Money-Weighted and Time-Weighted Return

Internal Rate of Return

The internal rate of return (IRR) is the discount rate that makes the net present value of an investment equal to zero. It is 'internal' because it depends only on the cash flows of the investment; no external data is needed. The formula for IRR is as follows:

$$\text{NPV} = \text{CF}_0 + \left[\frac{\text{CF}_1}{(1 + \text{IRR})^1}\right] + \left[\frac{\text{CF}_2}{(1 + \text{IRR})^2}\right] + \ldots + \left[\frac{\text{CF}_\text{N}}{(1 + \text{IRR})^\text{N}}\right] = 0$$

where:
CF_0 = usually the initial investment which is a cash outflow
CF_t = the expected net cash flow at time t
NPV = net present value of the investment
IRR = internal rate of return

The IRR is a single number which represents the return generated by an investment. Consider a very simple example where the initial investment is $100. One year later the amount received from this investment is $110. There are no other cash flows. If we apply the formula for IRR we get:

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$$-100 + \frac{110}{1 + \text{IRR}} = 0$$

Solving this equation shows that IRR = 0.1 or 10%.

Example

Consider an initial investment of $150,000. Estimated cash flows for the following three years from this investment are $50,000, $100,000 and $40,000 respectively. What is the IRR?

Solution:

We can set up an equation with the initial outflow equal to the present value of future cash flows and solve for the IRR:

$$\text{CF0} = \left[\frac{\text{CF1}}{(1 + \text{IRR})^1}\right] + \left[\frac{\text{CF2}}{(1 + \text{IRR})^2}\right] + \left[\frac{\text{CF3}}{(1 + \text{IRR})^3}\right]$$

Plugging in the values, we get:

$$\$150{,}000 = \left[\frac{\$50{,}000}{(1 + \text{IRR})^1}\right] + \left[\frac{\$100{,}000}{(1 + \text{IRR})^2}\right] + \left[\frac{\$40{,}000}{(1 + \text{IRR})^3}\right]$$

While it is theoretically possible to solve the above equation, it is much simpler to use the financial calculator.

Keystrokes	*Explanation*	*Display*
[2nd] [QUIT]	Return to standard mode	0
[CF] [2nd] [CLR WRK]	Clear CF Register	CF = 0
150 [+/-] [ENTER]	Initial outlay (in 000's)	CF0 = -150
[↓] 50 [ENTER]	Enter CF at t = 1	C01 = 50
[↓] [↓] 100 [ENTER]	Enter CF at t = 2	C02 = 100
[↓] [↓] 40 [ENTER]	Enter CF at t = 3	C03 = 40
[↓] [ÌRR] [CPT]	Compute IRR	13.11%

Money-Weighted Rate of Return

The money-weighted rate of return is the internal rate of return (IRR) of an investment.

Let us consider a simple example to illustrate this point. At time t = 0, an investor buys a share for $20.00. At the end of the Year 1, he receives a dividend of $0.50 and purchases another share for $22.50. At the end of the Year 2, he sells both shares for $23.50 each after receiving another dividend of $0.50 per share. What is the money-weighted return?

Since the money-weighted return is the IRR, we can use a financial calculator. The first step is to determine the net cash flows for every period. This is illustrated in the table below:

Time (end of period)	*Outflow (-)*	*Inflow (+)*	*Net cash flow*
0	-$20.00 To purchase the first share		-$20.00

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1	-$22.50 To purchase the second share	$0.50 Dividend received on first share	-$22.00
2		Dividend received = $0.50 x 2 shares = $1.00 From sales of 2 shares = $47	+48.00

After entering these cash flows(CF0 = -20, CF1 = -22, and CF3 = 48), use the calculator's IRR function to find the money-weighted rate of return as 9.39%.

Time-Weighted Rate of Return

The time-weighted rate of return measures the compound growth rate of $1 initially invested in the portfolio over a stated measurement period. The time-weighted return can be calculated using the following steps:

1. Break the overall evaluation period into sub-periods based on the dates of cash inflows and outflows.
2. Calculate the holding period return on the portfolio for each sub-period.
3. Link or compound holding period returns to obtain an annual rate of return for the year (the time-weighted rate of return for the year).
4. If the investment is for more than a year, take the geometric mean of the annual returns to obtain the time-weighted rate of return over that measurement period.

Consider the same example we discussed above with the following cash flows:

Time	*Outflow*	*Inflow*
0	$20.00 to purchase the first share	
1	$22.50 to purchase the second share	$0.50 dividend received on first share
2		$1.00 dividends ($0.50 x 2 shares); $47.00 from selling 2 shares @ $23.50 per share

Calculating the TWRR for this example is relatively simple because cash flows only occur at the start/end of every year. We will follow the steps mentioned earlier:

Steps 1: Break into evaluation period and value the portfolio at start/end of every period.

- Value of the portfolio at the start of Year 1 (t = 0) is $20.00.
- Value of portfolio at the end of Year 1 (t = 1) before the purchase of the new share is 22.50 + 0.50 = $23.00. Note that the dividend of $0.50 on the first share is received at the end of Year 1.
- Value of the portfolio at the start of Year 2 (t = 1) after the purchase of the second share is 22.50 + 22.50 = $45.00. The dividend of $0.50 from the first share is paid out and is not considered as part of the portfolio.
- Value of the portfolio at the end of Year 2 (t = 2) is 23.50 + 23.50 + 0.50 + 0.50 = $48.00. Both shares pay a dividend of $0.50 at the end of the second year.

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Step 2: Calculate the holding period return on the portfolio for each sub-period.

- In this question the cash flows are taking place at the start/end of each period. Hence there are no sub-periods. *Scenarios involving sub-periods will be covered in the next example.*

Step 3: Link or compound holding period returns to obtain an annual rate of return for the year.

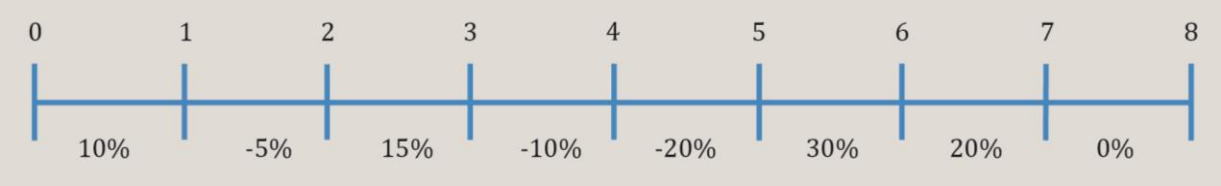
- The annual rate of return is based on the portfolio value at the start and end of each period.
- The portfolio value at the start of Year 1 was $20.00 and the value at the end of Year 1 was $23.00. Hence the holding period return was 15.00%.
- The portfolio value at the start of Year 2 was $45.00 and the value at the end of Year 2 was $48.00. Hence the holding period return was 6.67%.

Step 4: If the investment is for more than a year, take the geometric mean of the annual returns to obtain the time-weighted rate of return over that measurement period.

The TWRR is calculated as: $(1.15 * 1.067)^{\frac{1}{2}} - 1 = 0.1077 = 10.77\%$.

Example

Consider an investment where cash flows occur at the start/end of every quarter. Here each quarter is considered a sub-period. The return for each sub-period has already been calculated and is shown below:



Calculate the time-weighted return.

Solution:

Step 1 (break evaluation into sub-periods) and step 2 (calculate HPR for sub-periods) have been done for you.

Step 3: Link the quarterly returns to determine the return for Years 1 and 2 respectively.
For year 1: (1 + 0.10) (1 – 0.05) (1 + 0.15) (1 – 0.10) = 1.0816
For year 2: (1 – 0.20) (1 + 0.30) (1 + 0.20) (1 + 0) = 1.2480

Step 4: Determine the annualized return by taking the geometric mean.
TWRR = $(1.0816 \times 1.2480)^{1/2} - 1 = 0.1618 = 16.18\%$.

Money-weighted v/s time-weighted returns

- The money-weighted rate of return is impacted by the timing and amount of cash flows.

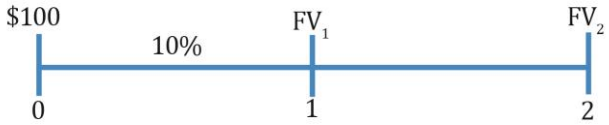
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- The time-weighted rate of return is not impacted by the timing and amount of cash flows.
- The time-weighted return is an appropriate performance measure if the portfolio manager does not control the timing and amount of investment.
- On the other hand, money-weighted return is an appropriate measure if the portfolio manager has control over the timing and amount of investment.

5. Annualized Return

Annual compounding

Let's understand this concept with a simple example.
Say present value (PV) = $100 and interest rate (r) = 10%.
What is the future value (FV) after one year?
What is the future value (FV) after two years?



The future value of a single cash flow with annual compounding can be computed using the following formula:

$$FV_N = PV\,(1 + r)^N$$

where:
FV_N = future value of the investment
N = number of periods
PV = present value of the investment
r = rate of interest

Therefore,

$$FV_1 = 100\,(1 + 0.1)^1 = \$110$$

$$FV_2 = 100\,(1 + 0.1)^2 = \$121$$

Notice that with compound interest, after two years we have $121. Whereas, with simple interest, after two years we would have $120. The difference between the two values ($1) represents the interest on interest component. In Year 2, we not only receive interest on the $100 principal, but we also receive interest on the $10 interest earned in Year 1 that has been reinvested.

Example

Cyndia Rojers deposits $5 million in her savings account. The account holders are entitled to a 5% interest. If Cyndia withdraws cash after 2.5 years, how much cash would she *most likely* be able to withdraw?

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Solution:

$FV_N = PV\,(1 + r)^N$

$FV_{2.5} = 5\,(1 + 0.05)^{2.5} = \$5.649$ million

FV Calculation using a Financial Calculator

You will often use the following keys on your TI BA II Plus calculator:
N = number of periods
I/Y = rate per period
PV = present value
FV = future value
PMT = payment
CPT = compute

One important point to note is the signs used for PV and FV. If the value for PV is negative "-", then the value for FV is positive "+". An inflow is often represented as a positive number, while outflows are denoted by negative numbers.
Before you begin, set the number of decimal points on your calculator to 9 to increase accuracy.

Keystrokes	Explanation	Display
[2nd] [FORMAT] [ENTER]	Get into format mode	DEC = 9
[2nd] [QUIT]	Return to standard calc mode	0

Question: You invest $100 today at 10% compounded annually. How much will you have in 5 years?

The key strokes to compute the future value of a single cash flow are illustrated below.

Keystrokes	Explanation	Display
[2nd] [QUIT]	Return to standard calc mode	0
[2nd] [CLR TVM]	Clears TVM Worksheet	0
5 [N]	Five years/periods	N = 5
10 [I/Y]	Set interest rate	I/Y = 10
100 [PV]	Set present value	PV = 100
0 [PMT]	Set payment	PMT = 0
[CPT] [FV]	Compute future value	FV = -161.05

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Non-annual Compounding

When our compounding frequency is not annual, we use the following formula to compute future value:

$$FV_N = PV\left(1 + \frac{r_s}{m}\right)^{mN}$$

where:
r_s = the stated annual interest rate in decimal format
m = the number of compounding periods per year
N = the number of years

Let's understand this concept using an example.

You invest $80,000 in a 3-year certificate of deposit. This CD offers a stated annual interest rate of 10% compounded quarterly. How much will you have at the end of three years?

Solution:
There are two methods to solve this question.
Formula Method
PV is $80,000.
The stated annual rate is 10%.
The number of compounding periods per year is 4. The total number of periods is 4 x 3 = 12.
Therefore, future value after 12 quarters (3 years) is

$$FV_{12} = \$80{,}000\left(1 + \frac{0.1}{4}\right)^{4\times3} = \$107{,}591$$

Calculator Method
You can also solve this problem using a financial calculator; the key strokes are given below:
N = 12, I/Y = 2.5%, PV = $80,000, PMT = 0, CPT FV = -$107,591
PMT is 0 because there are no intermediate payments in this example.

Example

Donald invested $3 million in an American bank that promises to pay 4% compounded daily. Which of the following is *closest* to the amount Donald receives at the end of the first year? Assume 365 days in a year.

A. $3.003 million
B. $3.122 million
C. $3.562 million

Solution
The correct answer is B.
Formula Method

$$FV_N = PV\left(1 + \frac{r_s}{m}\right)^{mN}$$

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$$FV_{1\ year} = 3\ million\left(1+\frac{0.04}{365}\right)^{365} = \$3.122\ million$$

Calculator Method

N = 365, I/Y = 4/365%, PV = $3 million, PMT = 0; CPT FV = -$3.122 million

Annualizing Returns

Annualized return converts the returns for periods that are shorter or longer than a year, to an annualized number for easy comparison.

$$\text{Annualized return} = \left(1+r_{period}\right)^{c}-1$$

Where c = number of periods in year

Example:

(*This is based on Example 13 from the curriculum.*)

An investor is evaluating the returns of three ETFs.

ETF	Time Since Inception	Return Since Inception (%)
1	146 days	4.61
2	5 weeks	1.10
3	15 months	14.35

Which ETF has the highest annualized rate of return?

Solution:

ETF 1 annualized return = $(1.0461^{365/146}) - 1 = 11.93\%$

ETF 2 annualized return = $(1.0110^{52/5}) - 1 = 12.05\%$

ETF 3 annualized return = $(1.1435^{12/15}) - 1 = 11.32\%$

ETF 2 has the highest annualized rate of return.

Continuously Compounded Returns

We covered examples with annual compounding, quarterly compounding and daily compounding. If we keep increasing the number of compounding periods until we have infinite number of compounding periods per year, then we can say that we have continuous compounding.

The continuously compounded return associated with a holding period return can be calculated as:

- Natural logarithm of one plus that holding period return, or
- Natural logarithm of the ending price over the beginning price

i.e. $R_c = \ln(1 + R_t) = \ln(P_t/P_0)$

Example:

1. If the one-week holding period return is 4%, then the equivalent continuously

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compounded return is: ln (1 + 0.04) = 3.922%

2. A stock was purchased at t = 0 for $30. One period later at t = 1, the stock has a value of $34.50. The continuously compounded return over the period can be calculated as: ln (34.50/30) = 13.976%

6. Other Major Return Measures and Their Applications

Gross Return

Gross return is the return earned by an asset manager prior to deducting management fees and taxes. It measures the investment skill of a manager.

Net Return

Net return is the return earned by the investor on an investment after all managerial and administrative expenses have been accounted for. This is the measure of return that should matter to an investor.

Assume an investment manager generates $120 for every $100, and charges a 2% fee for management and administrative expenses. The gross return, in this case, is 20% and the net return is 18%.

Pre-tax and After-tax Nominal Return

The returns we saw till now were pre-tax nominal returns, i.e., before deducting any taxes or any adjustments for inflation. This is the default, unless otherwise stated.

After-tax nominal return is the return after accounting for taxes. The actual return an investor earns should consider the tax implications as well.

In the example that we saw above for gross and net return, 18% was the pre-tax nominal return. If the tax rate for the investor is 33.33%, then the after-tax nominal return will be 18(1 - 0.3333) = 12.0006%.

Real Return

Real return is the return after deducting taxes and inflation.

$(1 + r) = (1 + r_{real})\ (1 + \pi)$

where:

r_{real} = real rate

π = rate of inflation

r = nominal rate

In the previous example, the after-tax nominal return was 12%. Assume the inflation rate for the period is 10%. What is the real rate of return?

Using the above formula, (1 + 0.12) = (1 + r) (1 + 0.1). Solving for r, we get 1.818%.

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Instructor's tip: If the answer choices are close to each other, use this formula to determine the correct answer. Else, you may use an approximation to solve for r quickly as nominal rate = real rate + inflation.

Leveraged Return

In cases, where an investor borrows money to invest in assets like bonds or real estate, the leveraged return is the return earned by the investor on his money after accounting for interest paid on borrowed money.

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Summary

LO: Interpret interest rates as required rates of return, discount rates, or opportunity costs and explain an interest rate as the sum of a real risk-free rate and premiums that compensate investors for bearing distinct types of risk.

An interest rate is the required rate of return. If you invest $100 today on the condition that you get $110 after one year, the required rate of return is 10%.
If the future value (FV) at the end of Year 1 is $110, you can discount at 10% to get the present value (PV) of $100. Hence, 10% can also be thought of as a discount rate.
Finally, if you spent $100 on taking your spouse out for dinner you gave up the opportunity to earn 10%. Thus, 10% can also be interpreted as an opportunity cost.

Interest rate = Real risk-free interest rate + Inflation premium + Default risk premium + Liquidity premium + Maturity premium.

Nominal risk free rate= real risk free rate + inflation premium

LO: Calculate and interpret different approaches to return measurement over time and describe their appropriate uses

Holding period return is the return earned on an asset during the period it was held.

$$\text{HPR single period} = \frac{P_T - P_0 + D_T}{P_0}$$

Arithmetic return is a simple arithmetic average of returns.

Geometric mean return is the compounded rate of return earned on an investment.

$$GM = [(1 + R_1) * (1 + R_2) * \ldots . * (1 + R_T)]^{\frac{1}{T}} - 1$$

The harmonic mean is a special type of weighted mean in which an observation's weight is inversely proportional to its magnitude. The harmonic mean is used to find average purchase price for equal periodic investments.

LO: Compare the money-weighted and time-weighted rates of return and evaluate the performance of portfolios based on these measures.

The money-weighted rate of return accounts for the timing and amount of all cash flows into and out of a portfolio. It is simply the internal rate of return.

The time-weighted rate of return measures the compound rate of growth of $1 initially invested in the portfolio over a stated measurement period.

Money-weighted v/s time-weighted returns

- The money-weighted rate of return is impacted by the timing and amount of cash flows.
- The time-weighted rate of return is not impacted by the timing and amount of cash flows.

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- The time-weighted return is an appropriate performance measure if the portfolio manager does not control the timing and amount of investment.
- On the other hand, money-weighted return is an appropriate measure if the portfolio manager has control over the timing and amount of investment.

LO: Calculate and interpret annualized return measures and continuously compounded returns, and describe their appropriate uses.

Annualized return converts the returns for periods that are shorter or longer than a year, to an annualized number for easy comparison.

The continuously compounded return associated with a holding period return can be calculated as:

- Natural logarithm of one plus that holding period return, or
- Natural logarithm of the ending price over the beginning price

LO: Calculate and interpret major return measures and describe their appropriate uses.

Gross return is the return earned by an asset manager prior to deducting management fees and taxes. It measures investment skill.

Net return accounts for all managerial and administrative expenses is what the investor is concerned with.

Pre-tax nominal return is the return before accounting for inflation and taxes; this is the default, unless otherwise stated.

After-tax nominal return is the return after accounting for taxes.

Real return is the return after accounting for taxes and inflation.

Leveraged return is the return earned by the investor on his money after accounting for interest paid on borrowed money

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LM02 The Time Value of Money in Finance

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1. Introduction

In this learning module we will apply time value of money principles to value financial assets. This learning module covers:

- Calculating the present value of fixed-income and equity instruments based on their expected cash flows.
- Calculating implied bond and stock returns given their current market prices.
- Applying the cash flow additivity and no-arbitrage principles to calculate implied forward interest rates, forward exchange rates, and option values.

2. Time Value of Money in Fixed Income and Equity

The relationship between present value (PV) and future value (FV) of a cash flow is:

$FV_t = PV(1 + r)^t$

where:
r = stated discount rate per period
t = number of compounding periods

With continuous compounding the relationship can be expressed as:

$FV_t = PVe^{rt}$

We can also rearrange the above equations and express present value in terms of future values as:

$$PV = \frac{FV_t}{(1 + r)^t}$$

$$PV = \frac{FV_t}{e^{rt}}$$

Fixed-Income Instruments and the Time Value of Money

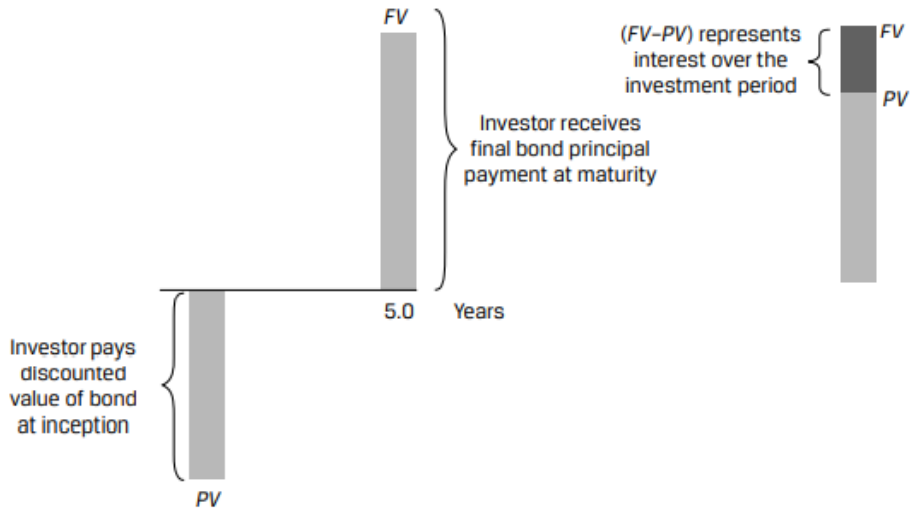
Fixed-income instruments are debt instruments such as a bond or a loan. Depending on the types of cash flows provided we can divide them into three types:

- Discount instruments
- Coupon instruments
- Annuity instruments

Discount instruments: A debt security that does not pay interest but is instead issued at a deep discount. At maturity the security is redeemed for its full face value. The difference between the face value and issue price represents interest earned over the life of the instrument. E.g., zero-coupon bonds.

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Exhibit 1 from the curriculum illustrates the cash flow pattern of a zero-coupon bond (aka discount bond).



The present value (PV) of a zero-coupon bond can be calculated as:

$$PV = \frac{FV_t}{(1 + r)^t}$$

where:
FV_t = principal amount received at maturity
r = market discount rate

Example:
(*This is based on Example 1 from the curriculum.*)

Consider a 20 year zero-coupon bond issued at a yield to maturity (YTM) of 6.70%. What should an investor expect to pay for this bond per INR100 of principal?

Solution:

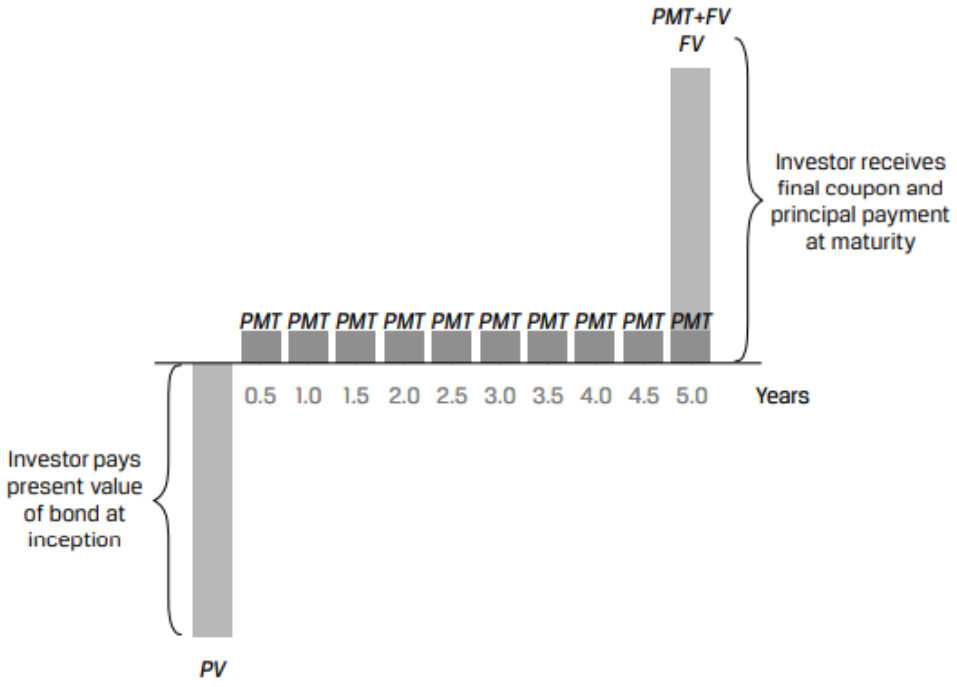
The YTM of a bond at issuance is the market discount rate at issuance. The PV of this bond can be calculated as:

$$PV = \frac{100}{(1 + 0.067)^{20}} = 27.33$$

Coupon instruments: These are regular bonds that make periodic coupon payments and repay principal at maturity. E.g., Coupon paying corporate bonds.

Exhibit 2 from the curriculum illustrates the cash flow pattern of a semi-annual coupon paying bond.

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The present value of a coupon paying bond can be calculated as:

$$\text{PV of bond} = \frac{\text{PMT}}{(1+\text{r})^1} + \frac{\text{PMT}}{(1+\text{r})^2} + \cdots + \frac{\text{PMT+FV}}{(1+\text{r})^{\text{N}}}$$

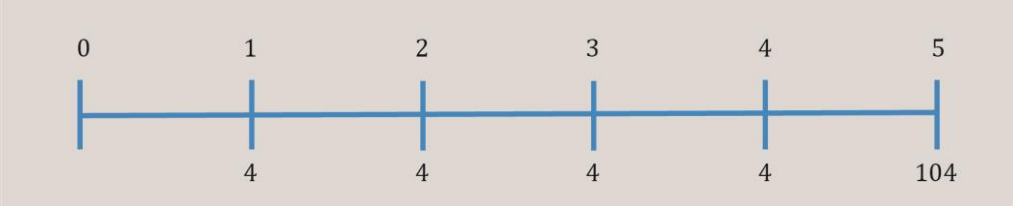
where:
PMT = coupon payment per period
FV= par value of the bond paid at maturity
r = market discount rate
N = number of periods until maturity

Example

The coupon rate on a bond is 4% and the payment is made once a year. The time to maturity is five years and the market discount rate is 6%. What is the bond price per 100 of par value?

Solution:

Start by drawing a timeline for the cash flows. The par value of the bond or principal is $100. A coupon payment of $4 is made every year. At maturity (at the end of five years), a payment of $104 (principal of $100 + coupon of $4) is made.



We are required to calculate the present value of bond at time t = 0. For that, we discount all the future cash flows at the market discount rate of 6%.

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End of year	Type of cash flow	Amount	Present value
1	Coupon	4	3.77
2	Coupon	4	3.55
3	Coupon	4	3.36
4	Coupon	4	3.17
5	Coupon + principal	104	77.71
			91.56

Using the formula, PV = $\frac{4}{1.06} + \frac{4}{1.06^2} + \frac{4}{1.06^3} + \frac{4}{1.06^4} + \frac{104}{1.06^5} = 91.575$

Note: The table and formula above were just for your understanding.

On the exam, solve this quickly using a financial calculator:

N = 5; I/Y = 6; FV = 100; PMT = 4; PV = ? PV = -91.575

Perpetual bonds

A perpetual bond is a special type of coupon bond that has no stated maturity, i.e. it is expected to keep paying coupons perpetually. Companies may issue perpetual bonds to obtain equity like financing.

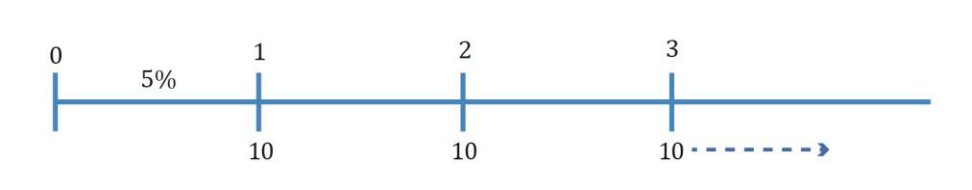
The present value of a perpetual bond can be calculated as:

$$\text{PV of perpetual bond} = \frac{\text{PMT}}{\text{r}}$$

Example:

A bond pays an annual coupon of $10 at the end of every year forever. Calculate the PV of this bond assuming a market discount rate of 5%.

Solution:



The PV of this perpetuity can be computed as:

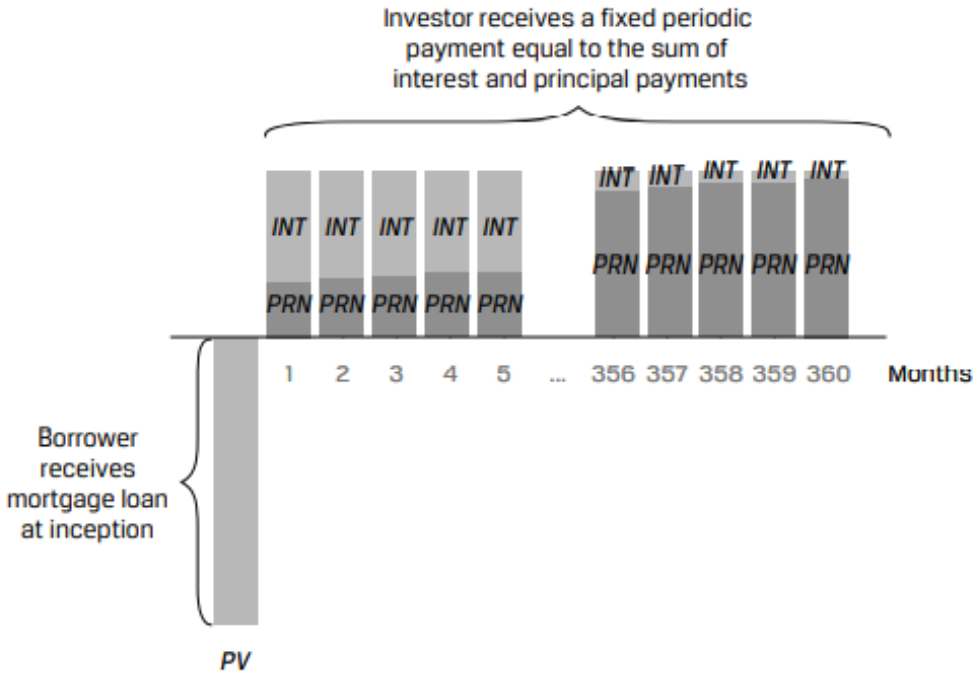
$$\text{PV} = \frac{10}{0.05} = \$200$$

Instructor's Note:
Keep in mind that the present value of $200 is one period before the first cash flow. Many students show the present value of $200 at the same time as the first cash flow, which is incorrect.

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Annuity instruments: They make periodic level (i.e. equal) payments consisting of both interest and principal. E.g. mortgages.

Exhibit 3 from the curriculum illustrates the cash flow pattern of a 30-year mortgage with monthly payments.



The periodic annuity cash flow (A) can be calculated as:

$$A = \frac{r(PV)}{1-(1+r)^{-t}}$$

Instructor's Note: Instead of remembering and using the above formula, we recommend that you use the TVM function of the calculator to solve for annuity payments. This is illustrated in the following example.

Example:

Freddie bought a car worth $42,000 today. He was required to make a 15% down payment. The remainder was to be paid as a monthly payment over the next 12 months with the first payment due at t = 1. Given that the interest rate is 8% per annum compounded monthly, what is the approximate monthly payment?

Loan amount = 85% of $42,000 = 0.85 x 42,000 = $35,700
PV = -$35,700
N = 12
I/Y = 8/12%
FV = 0 CPT PMT → PMT = $3,105.48

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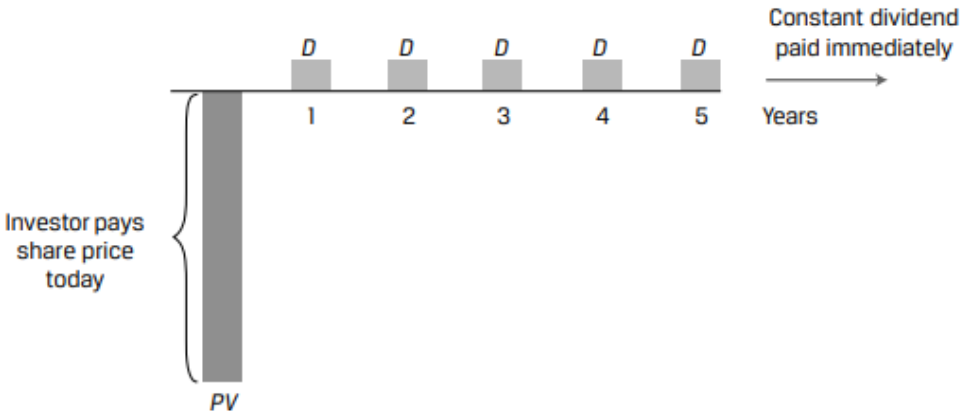
Equity Instruments and the Time Value of Money

One way of valuing equity instruments is as the present value of their expected dividends. Common assumptions used in dividend valuation models are:

- Constant dividend
- Constant dividend growth rate
- Changing dividend growth rate

Constant dividend: An investor pays an initial price (PV) for a share and receives a fixed periodic dividend (D) forever. Generally, preferred shares have such dividends.

Exhibit 4 from the curriculum illustrates the cash flow pattern of a stock with constant dividends.



The PV of a stock with constant dividend is similar to the valuation of a perpetual bond. The PV is calculated as:

$$PV = \frac{D}{r}$$

where: V_0 = present value of the perpetuity; D_o= dividend and r = rate of return

Example

A $100 par value, non-callable, non-convertible perpetual preferred stock pays a 5% dividend. The discount rate is 8%. Calculate the expected stock price.

Solution:

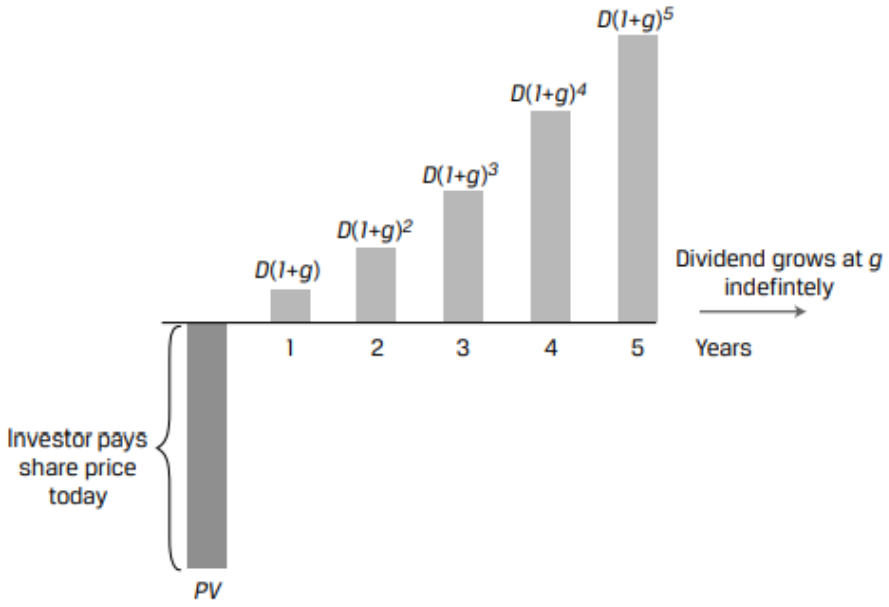
Expected annual dividend = 0.05 x 100 = 5

Value of the preferred share = 5.00/0.08 = 62.50

Constant dividend growth rate: An investor pays an initial price (PV) for a share and receives an initial dividend in one period (D_{t+1}). This dividend is expected to grow at a constant rate g forever.

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Exhibit 5 from the curriculum illustrates the cash flow pattern of a stock with constant dividend growth rate.



The present value of a stock with constant dividend growth rate can be found using the Gordon growth model. According to this model, the present value can be calculated as:

$$PV = \frac{D_1}{r - g}$$

where:
D_1 = next period's dividend
r = required rate of return
g = dividend growth rate

Instructor's Note: In the equation above, if the growth rate is zero, then the equation reduces to the present value of a perpetuity.

Example:

A company just paid a dividend of $2.00. The dividend is expected to grow at 5% into perpetuity. The required return is 10%. What is the estimated stock price?

Solution:

$$PV = \frac{D_1}{r-g} = \frac{D_0*(1+g)}{r-g} = \frac{2(1+0.05)}{0.10-0.05} = 42$$

Instructor's Note: Do not forget to use the next period's dividend i.e. $2 growing at 5% in the formula. The PV using the unadjusted current period dividend is commonly presented as one of the answer options as a trap.

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Changing dividend growth rate: It is an ideal situation to assume that all companies grow at a constant rate indefinitely and pay a constant dividend. The assumption is true to an extent only for stable companies. In reality, companies go through a finite rapid growth phase followed by an infinite period of sustainable growth.

A two-stage DDM can be used to calculate the value of such companies transitioning from growth to mature stage. The Gordon growth model may be used to calculate the terminal value at the beginning of the second stage which represents the present value of dividends during the sustainable growth phase.

$$PV = \sum_{t=1}^{n} \frac{D_0(1+g_s)^t}{(1+r)^t} + \frac{V_n}{(1+r)^n}$$

The first term is discounting the dividends during the high growth period. The second term is calculating the terminal value for the second sustainable growth period and then discounting it to the present value where V_n = terminal value at time n estimated using the Gordon growth model.

Example

Let us understand the concept better with the help of an example. The current dividend for a company is $4.00. The dividends are expected to grow at 20% a year for 4 years and then at 4% after that. The required rate of return is 18%. Estimate the intrinsic value.

Solution:

First draw a timeline.



We will use this formula:

$$PV = \sum_{t=1}^{n} \frac{D_0(1+g_s)^t}{(1+r)^t} + \frac{V_n}{(1+r)^n}$$

where n = 4 (high growth period)

Solve for the second term: $\frac{V_n}{(1+r)^n}$; $V_4 = \frac{D_5}{r-g} = \frac{D_4(1+g_L)}{r-g} = \frac{8.29 * 1.04}{0.18 - 0.04} = 61.58$

Using the financial calculator, we can calculate the present value of dividends and terminal value by entering the following values: $CF_0 = 0$; $CF_1 = 4.8$; $CF_2 = 5.76$; $CF_3 = 6.91$; $CF_4 = 8.29 + 61.58$; I = 18; NPV = 48.45

Note: while calculating V_4, you need to use 4% as growth rate since it is the long-term growth rate.

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3. Implied Return and Growth

Market participants may sometimes face situations where both the present value (current market price) and future cash flows of an instrument are known. In such cases, we can solve for the implied return or growth rate that the market has incorporated into instrument's current market price.

Implied Return for Fixed-Income Instruments

The yield-to-maturity (YTM) is a bond's internal rate of return based on the assumption that the bond is held to maturity and all future cash flows occur as promised.

We can calculate the YTM implied by a bond's current market price for both discount and coupon paying bonds as illustrated in the examples below.

Example:
(*This is based on Example 8 from the curriculum.*)

Continuing with our previous example on discount bonds. Consider a 20 year zero coupon bond that will pay a par value of INR100 at maturity. The bond was initially issued at a price of INR27.33 and offered a 6.70% YTM. Six years later, the bond trades at a price (PV) of INR40.

1. What is the initial investor's implied return on this bond over the six-year holding period?
2. What is the expected return of a new investor who purchases the discount bond now at a price of INR40 and holds it till maturity?

Solution to 1:

Here PV = -27.33, FV = 40, N = 6, PMT = 0; CPT→ I/Y = 6.55%

The investor received an annualized return of 6.55% which is slightly below the expected return of 6.70% at the time of issuance.

Instructor's note: The difference in returns is due to changes in market expectations about inflation and real interest rates. However, if the investor continues to hold the bond till maturity, he will receive exactly 6.70%, the YTM at issuance.

Solution to 2:

Here PV = -40, FV = 100, N = 14, PMT = 0, CPT → I/Y = 6.76%

The investor's expected return is 6.76%.

Example

The coupon rate on a bond is 4% and the payment is made once a year. The bond has a remaining maturity of 3 years and is currently trading in the market for $95. What is the YTM of this bond?

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Solution:

PV = -95, FV = 100, PMT = 4, N = 3, CPT→ I/Y = 5.86%

The expected return on this bond is 5.86%. Note that this YTM calculation assumes that all intermediate cash flows (the coupon payments of $4) can be reinvested at 5.86% through maturity.

Equity Instruments, Implied Return, and Implied Growth

The present value of an equity instrument reflects not only the required return but also the growth of cash flows. Under the assumption of constant growth of dividends, the PV of a stock was calculated as:

$$PV = \frac{D_1}{r - g}$$

Rearranging this equation, we get:

$$r - g = \frac{D_1}{PV}$$

$$r = \frac{D_1}{PV} + g$$

Thus, the implied return on a stock is equal to its expected dividend yield plus a constant growth rate.

We can also express the equation in terms of growth rate as:

$$g = r - \frac{D_1}{PV}$$

i.e., the implied growth rate of a stock can be calculated by deducting its expected dividend yield from its required return.

Example:
(*This is based on Example 10 from the curriculum.*)

Coca-Cola stocks currently trade at $63.00 per share. Its annualized expected dividend per share during the next year is $1.76.

1. If an analyst expects Coca-Cola's dividends to increase at a constant 4% per year indefinitely, calculate the required return expected by investors.

2. If the analyst believes that Coca-Cola investors should expect a return of 7%, calculate the implied dividend growth rate for Coca-Cola.

Solution to 1:

$$r = \frac{D_1}{PV} + g = \frac{1.76}{63} + 0.04 = 0.0679 = 6.79\%$$

Solution to 2:

$$g = r - \frac{D_1}{PV} = 0.07 - \frac{1.76}{63} = 0.0421 = 4.21\%$$

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Instructor's note: For the same expected dividend and current price, a higher expected return implies a higher dividend growth rate and vice versa.

Price to Earnings Ratio

Instead of comparing stock prices directly, it is common practice to compare their price-to-earnings ratio (P/E). The P/E is a relative valuation metric that is calculated as share price/earnings per share. A stock trading at a P/E of 20 implies that investors are willing to pay 20 time earnings per share, and this stock is more expensive than a stock trading at a P/E of 10. P/E ratios are not only calculated for individual stocks but also for stock indexes such as the S&P 500.

We can link this relative valuation metric to the Gordon growth model as follows.

$$PV = \frac{D_1}{r - g}$$

Dividing both sides by E_1 (the expected earnings over the next period) we get:

$$\frac{PV}{E_1} = \frac{D_1/E_1}{r - g}$$

The LHS of the equation is the forward price-to-earnings ratio, the ratio of the current stock price to its next period expected earnings. The numerator on the RHS, D_1/E_1, is the dividend payout ratio i.e., the proportion of earnings distributed to shareholders as dividends.

We can use these inputs to calculate the implied return and growth rate of a stock as illustrated in the examples below.

Example
(*This is based on Example 11 from the curriculum.*)

Suppose Coca-Cola stock trades at a forward P/E of 28 and its expected dividend payout is 70%.

1. If an analyst believes that Coca-Cola's dividends will grow at a constant rate of 4% per year, calculate Coca-Cola's required return.

2. If the analyst believes that Coca-Cola investors should expect a return of 7%, calculate the implied dividend growth rate for Coca-Cola.

Solution to 1:

$$\frac{PV}{E_1} = \frac{D_1/E_1}{r - g}$$

$$28 = \frac{0.7}{r - 0.04}$$

$$r = \frac{0.7}{28} + 0.04 = 0.065 = 6.5\%$$

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Solution to 2:

$$\frac{PV}{E_1} = \frac{D_1/E_1}{r - g}$$

$$28 = \frac{0.7}{0.07 - g}$$

$$g = 0.07 - \frac{0.7}{28} = 0.045 = 4.5\%$$

If we already know the required return and growth rate of the stock, we can also use the expression derived above to evaluate if a stock is overvalued or undervalued.

Example:
(*This is based on Example 11 from the curriculum.*)

Suppose Coca-Cola stock trades at a forward P/E of 28 and its expected dividend payout is 70%. Analysts believe that Coca-Cola stock should earn a 9% and that its dividend will grow by 4.5% per year indefinitely. Evaluate if the stock is overvalued or undervalued.

Solution:

$$\frac{PV}{E_1} = \frac{D_1/E_1}{r - g}$$

Using the above inputs, we can see that this results in an inequality.

$$28 > \frac{0.70}{0.09 - 0.045} = 15.56$$

Coca-Cola's forward price-to-earnings ratio of 28 is much greater than 15.56, which is computed from the equation. Therefore, the stock appears to be overvalued.

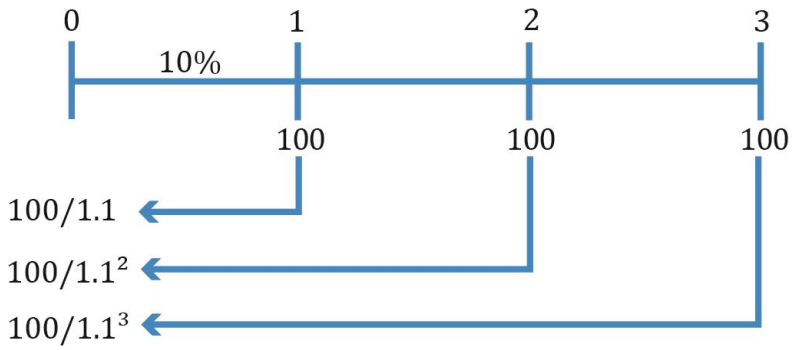
4. Cash Flow Additivity

According to the cash flow additivity principle, cash flows indexed at the same point in time are additive. For example, if you have the following cash flows:



You cannot simply add these three numbers. You have to take each of these numbers and bring them to a particular point in time. Let's say that we find the present values at time zero for each of these cash flows.

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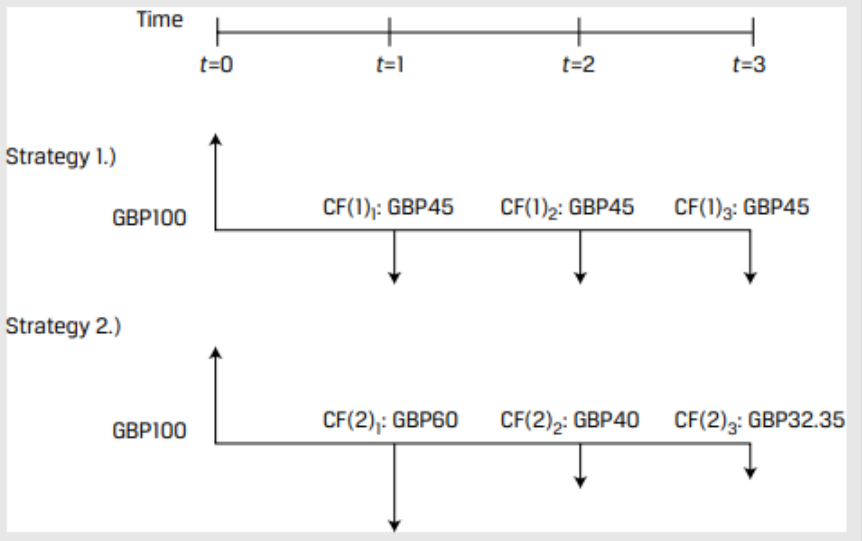


According to this principle, these present values that are all indexed to time zero can be added.

While valuing two cash flow streams, the cash flow additivity principle allows for the cash flow streams to be compared as long as the cash flows occur at the same point in time. This is illustrated in the example below:

Example:
(*This is based on Example 12 from the curriculum.*)

Assume that you have GBP100 to invest, and have two strategies to choose from with the following cash flow streams.

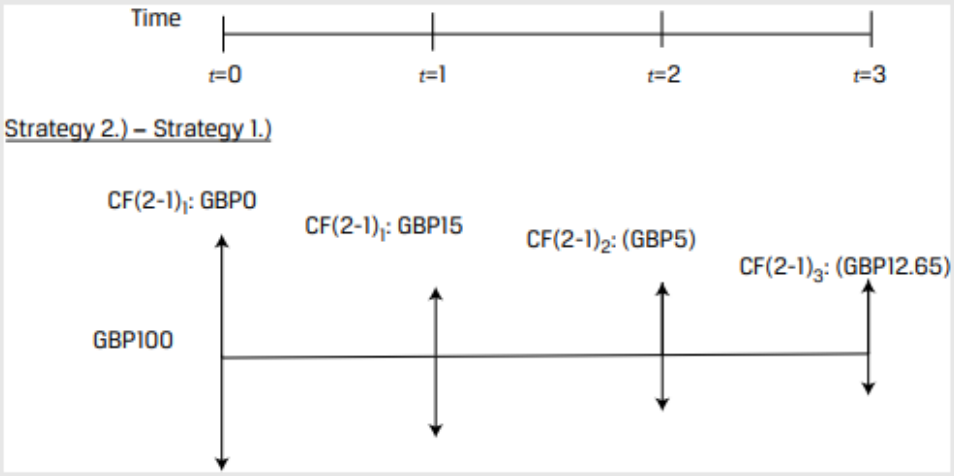


Your required return from both strategies is 10%. Recommend which investment strategy to choose.

Solution:

To compare the two strategies, we can calculate the difference between the cash flows at each time period and effectively create a new set of cash flows.

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We then calculate the present value of these cash flows:

$$\text{PV} = 0 + \frac{15}{1.10} + \frac{-5}{1.10^2} + \frac{-12.65}{1.10^3} = 0$$

Since the PV is 0 the two strategies are economically equivalent. You should not prefer one over the other.

Instructor's Note: We calculated the difference in cash flows as Strategy 2 – Strategy 1. If we got a positive PV instead of a zero, it would indicate that strategy 2 is superior. And if we got a negative PV, it would indicate that strategy 1 is superior.

The cash flow additivity principle helps ensure that market prices reflect the condition of no arbitrage. Asset prices for economically equivalent assets are the same even if the assets have differing cash flow streams. Several real-world applications of cash flow additivity are presented below to illustrate no-arbitrage pricing.

Implied Forward Rates Using Cash Flow Additivity

A forward rate is the interest rate that is determined today for a loan that will be initiated at a future time period. $F_{A,B}$ represents a forward rate for a loan initiated A years from today with further maturity of B years. For example, $F_{2,3}$ represents the forward rate for a loan which starts at the end of year 2 and matured at the end of year 5.

Given a set of zero rates we can calculate the 'implied forward rates'. Consider the zero-rates below:

Years to Maturity	Zero Rate
1	2.3960%
2	3.4197%
3	4.0005%

Over a two-year period, an investor faces the following investment choices:

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1. Invest USD100 for one year today at the zero rate (z_1) of 2.396% and reinvest the proceeds of USD102.40 at the one-year rate in one year's time, or the implied forward rate ($IFR_{1,1}$).
2. Invest USD100 for two years at the two-year zero rate (z_2) of 3.4197% to receive $USD100(1 + z_2)^2$, or USD106.96.

Based on the no-arbitrage principal both options should give the same results. Therefore,

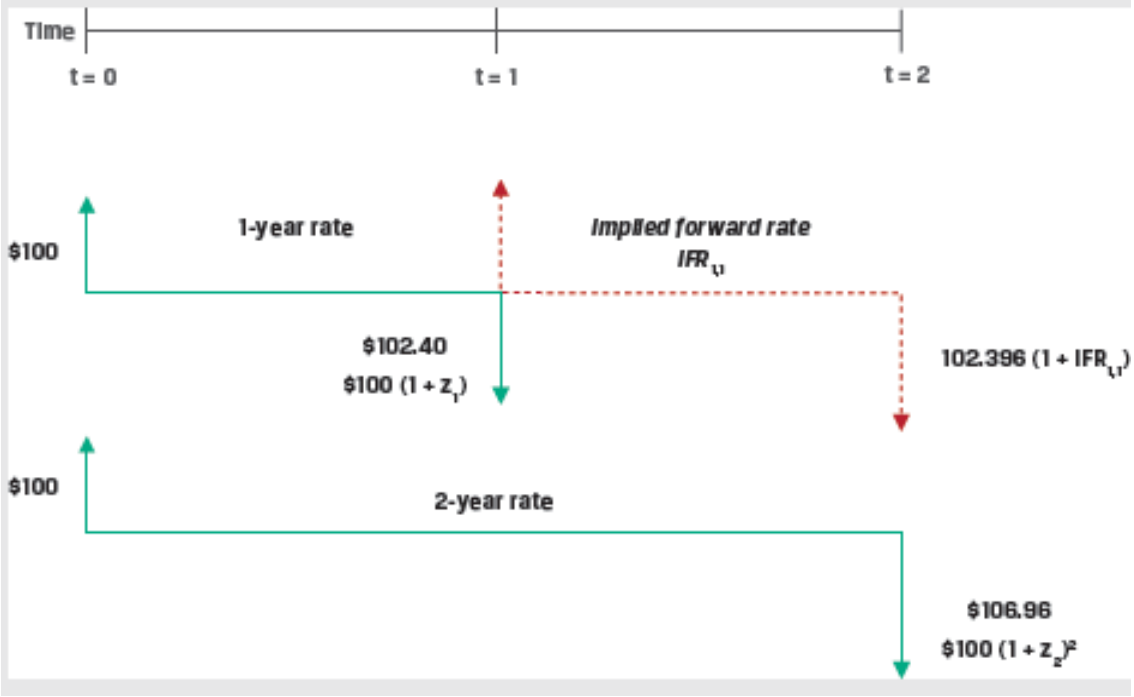
$USD100 \times (1 + z_1) \times (1 + IFR_{1,1}) = USD100 \times (1 + z_2)^2$

$(1 + z_1) \times (1 + IFR_{1,1}) = (1 + z_2)^2$

$(1.02396) \times (1 + IFR_{1,1}) = (1.034197)^2$

$IFR_{1,1} = 4.4536\%$

This is illustrated in the figure below.



We can also solve for the one-period rate in two periods ($IFR_{2,1}$) as follows:

$(1 + z_2)^2 \times (1 + IFR_{2,1})^1 = (1 + z_3)^3$

$(1.034197)^2 \times (1 + IFR_{2,1}) = (1.040005)^3$

$IFR_{2,1} = 5.1719\%$

Forward Exchange Rates Using No Arbitrage

An exchange rate is the price or cost of one currency expressed in terms of another currency. Stated otherwise, it is the number of units of the price currency needed to buy/sell one unit of the base currency.

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Consider an exchange rate quote of 1.4500 USD/EUR. The numerator currency (USD) is called as the price currency and the denominator currency (EUR) is called as the base currency. It implies that one EUR is exchangeable with 1.45 USD. So, 1.4500 USD/EUR is interpreted as $1.45 per euro. Here, USD is the currency used to express price per one unit of euro.

A spot rate is the rate that is in effect today. A forward rate is a fixed exchange rate that we lock in today based on which currencies will be exchanged at some future date.

The forward exchange rate depends on relative interest rates in the two countries. To derive the relationship between spot rates, forward rates, and interest rates, assume you have one unit of domestic currency to be invested for let us say, one year. There are two options you may consider:

The exchange rate convention used = $\frac{\text{Foreign currency}}{\text{Domestic currency}} = \frac{\text{Price currency}}{\text{Base currency}}$

Option 1**:** Invest one unit of domestic currency (base currency) at the domestic risk-free rate i_B for one period.

Amount at the end of the period = $1+ i_B$

Option 2**:** Convert one unit of domestic currency into foreign currency (price currency) today using the spot rate $\frac{S_P}{B}$. Invest this amount at the foreign risk-free rate i_P for one period. Determine the forward rate $F_{\frac{P}{B}}$ today at which the price currency will be converted back to base currency.

Amount of units of price currency at the end of the period **=** $\frac{S_P}{B}(1 + i_P)$

Amount in terms of base (domestic currency) at the end of the period = $\frac{S_P}{B}(1 + i_P) * \frac{1}{F_{\frac{P}{B}}}$

Note: By using forward rate, any foreign exchange risk was eliminated in this transaction.

Since the risk of these two investments is the same, they should generate equivalent returns. As an investor, you should be indifferent between the two as it is a no-arbitrage relationship.

$$1 + i_B = \frac{S_P}{B}(1 + i_P) * \frac{1}{F_{\frac{P}{B}}}$$

The above equation can be rewritten as:

$$F_{\frac{P}{B}} = S_{\frac{P}{B}} * \frac{1 + i_P}{1 + i_B}$$

where:

$F_{\frac{P}{B}} = \text{forward rate}$

$i_P = \text{risk} - \text{free rate of the price currency}$

$i_B = \text{risk} - \text{free rate of the base currency}$

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Let us use some values now to compute the forward rate. The spot rate for INR/USD $S_{\frac{INR}{USD}}$ = 100.

The risk-free rate for INR is 10% and the risk-free rate for dollar (base currency) is 1%.

Forward rate $F_{\frac{INR}{USD}} = S_{\frac{INR}{USD}} * \frac{1 + i_{INR}}{1 + i_{USD}} = 100 * \frac{1 + 0.1}{1 + 0.01} = 108.9108$

This relationship ensures that there is no arbitrage. The only forward rate that prevents arbitrage is 108.91. Otherwise, for any rate greater/lesser than 108.91 traders can exploit the mispricing by buying low and selling high.

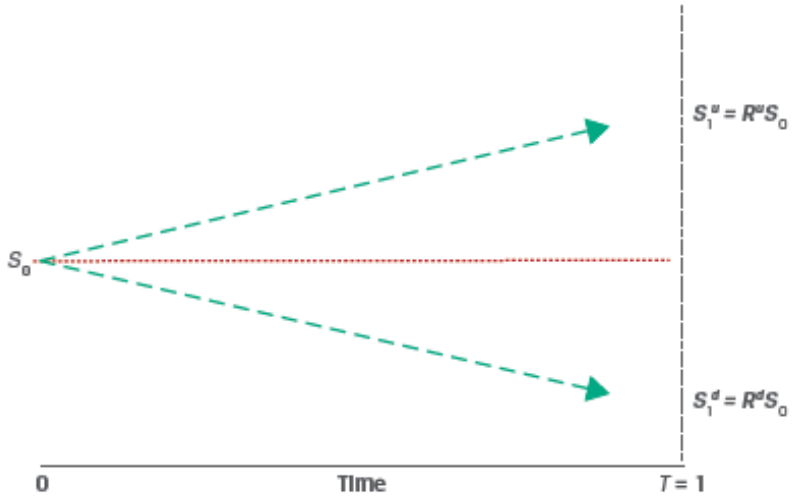
Option Pricing Using Cash Flow Additivity

Instructor's Note: This material is covered in greater depth later in the curriculum. At this point, only concentrate on gaining a basic understanding of the following example.

An option is a derivative contract that gives the buyer the right, but not the obligation, to buy or sell the underlying asset by a certain date (expiration date) at a specified price (strike price).

There are two kinds of options: Call options and put options. A call option confers the right to buy a stock at the strike price before the agreement expires. A put option gives the holder the right to sell a stock at a specific price.

Options can be valued using the binomial model. The binomial model assumes that over a given time period, there are only two possible outcomes - the stock's price will either go up to S_1^u or down to S_1^d. The following exhibit from the curriculum illustrates the binomial model.



Pricing a call option:

Consider a one-year European call option with an exercise price (X) of 100.

The underlying spot price (S_0) is 80.

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The two possible stock prices after 1 year are: S_1^d = 60 and S_1^u=110

At *t = 0*, the call option value is c_0.

At *t = 1*, the call option value is either c_1^u (if the price goes up) or c_1^d(if the price goes down). We can compute the values c_1^u and c_1^d as:

Up move: The call option ends in-the-money. The owner will choose to buy the stock at 100 and sell it in the market at 110.

$$c_1^u = 110 - 100 = 10$$

Down move: The call option ends out-of-the-money. The owner will not choose to buy the stock at 100 when the market price is only 60. So the option will expire worthless and have a value of 0.

$$c_1^d = 0$$

To compute c_0 we can use replication and no-arbitrage pricing. Assume that at *t = 0* we sell the call option at a price of c_0 and purchase *h* units of the underlying asset. V_0 represents the initial investment in the portfolio, and V_1^u and V_1^d represent the portfolio value if the underlying price moves up or down, respectively.

$$V_0 = hS_0 - c_0$$

$$V_1^u = hS_1^u - c_1^u$$

$$V_1^d = hS_1^d - c_1^d$$

We solve for h, such that the portfolio payoff in both the up and down scenarios are equal, i.e.:

$$V_1^u = V_1^d$$

$$hS_1^u - c_1^u = hS_1^d - c_1^d$$

Solving for h, we get:

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d}$$

This equation gives us the hedge ratio, or the proportion of the underlying that will offset the risk associated with an option.

The hedge ratio for our example is:

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d} = \frac{10 - 0}{110 - 60} = 0.20$$

i.e. for each call option unit sold, we must buy 0.2 units of the underlying asset.

The portfolio values at t=1 for the up and down scenarios are:

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$$V_1^u = hS_1^u - c_1^u = 0.20 \times 110 - 10 = 12$$

$$V_1^d = hS_1^d - c_1^d = 0.20 \times 60 - 0 = 12$$

As expected, the portfolio values are the same.

To prevent arbitrage, the portfolio value at *t =1* should be discounted at the risk-free rate so that:

$$V_0 = \frac{V_1}{1 + r}$$

For our example, V_1 =12. Assume that the annual risk-free rate is 5%. Therefore,

$$V_0 = \frac{12}{1.05} = 11.42$$

Replacing V_0 by the hedged portfolio we get:

$$hS_0 - c_0 = 11.42$$

$$c_0 = hS_0 - 11.42 = 0.2 \times 80 - 11.42 = 4.58$$

The value of the call option at *t = 0* is 4.58.

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Summary

LO: Calculate and interpret the present value (PV) of fixed-income and equity instruments based on expected future cash flows.

Fixed Income:

The present value (PV) of a zero-coupon bond can be calculated as:

$$PV = \frac{FV_t}{(1+r)^t}$$

The present value of a coupon paying bond can be calculated as:

$$\text{PV of bond} = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \cdots + \frac{PMT+FV}{(1+r)^N}$$

The present value of a perpetual bond can be calculated as:

$$\text{PV of perpetual bond} = \frac{PMT}{r}$$

Equity:

The PV of a stock with constant dividend is similar to the valuation of a perpetual bond. The PV is calculated as:

$$PV = \frac{D}{r}$$

The present value of a stock with constant dividend growth rate can be found using the Gordon growth model:

$$PV = \frac{D_1}{r-g}$$

The present value of a stock with changing dividend growth rate can be found using a two-stage model:

$$PV = \sum_{t=1}^{n} \frac{D_0(1+g_s)^t}{(1+r)^t} + \frac{V_n}{(1+r)^n}$$

LO: Calculate and interpret the implied return of fixed-income instruments and required return and implied growth of equity instruments given the present value (PV) and cash flows.

Fixed income: The yield-to-maturity (YTM) is a bond's internal rate of return based on the assumption that the bond is held to maturity and all future cash flows occur as promised.

We can calculate the YTM implied by a bond's current market price for both discount and coupon paying bonds by using the TVM function of a calculator.

Equity: The implied return on a stock is equal to its expected dividend yield plus a constant growth rate.

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$$r = \frac{D_1}{PV} + g$$

We can also express this relationship in terms of P/E ratios:

$$\frac{PV}{E_1} = \frac{D_1/E_1}{r - g}$$

LO: Explain the cash flow additivity principle, its importance for the no-arbitrage condition, and its use in calculating implied forward interest rates, forward exchange rates, and option values.

According to the cash flow additivity principle, cash flows indexed at the same point in time are additive.

While valuing two cash flow streams, the cash flow additivity principle allows for the cash flow streams to be compared as long as the cash flows occur at the same point in time.

The cash flow additivity principle helps ensure that market prices reflect the condition of no arbitrage. Asset prices for economically equivalent assets are the same even if the assets have differing cash flow streams.

Implied forward rate: Given the one year and two year spot rates, the implied 1 year forward rate can be found as:

$$(1 + z_1) \times (1 + IFR_{1,1}) = (1 + z_2)^2$$

Forward exchange rate: Given the spot exchange rate and the interest rates in the price and base currencies, the forward exchange rate can be found as:

$$F_{\frac{P}{B}} = S_{\frac{P}{B}} * \frac{1 + i_P}{1 + i_B}$$

Option values: The binomial model assumes that over a given time period, there are only two possible outcomes - the asset's price will either go up to S_1^u or down to S_1^d.

The hedge ratio for a call option is calculated as:

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d}$$

To compute c_0 we can use replication and no-arbitrage pricing. Assume that at $t = 0$ we sell the call option at a price of c_0 and purchase h units of the underlying asset.

$$V_0 = hS_0 - c_0$$

$$V_1^u = hS_1^u - c_1^u$$

$$V_1^d = hS_1^d - c_1^d$$

To prevent arbitrage, the portfolio value at $t = 1$ should be discounted at the risk-free rate so that:

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$$V_0 = \frac{V_1}{1 + r}$$

or

$$hS_0 - c_0 = \frac{V_1}{1 + r}$$

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LM03 Statistical Measures of Asset Returns

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1. Introduction

In this learning module we will learn how to summarize and analyze important aspects of financial returns. This learning module covers:

- Measures of central tendency and location
- Measures of dispersion
- Measures of the shape of return distributions
- Covariance and correlation between two variables

2. Measures of Central Tendency and Location

A 'population' is defined as all members of a specified group. A 'parameter' describes the characteristics of a population.

A 'sample' is a subset drawn from a population. A 'sample statistic' describes the characteristic of a sample.

For example, all stocks listed on a country's exchange refers to a population. If 30 stocks are selected from the listed stocks, then this refers to a sample.

Sample statistics—such as measures of central tendency, measures of dispersion, skewness, and kurtosis—help make probabilistic statements about investment returns.

'Measures of central tendency' specify where data are centered.

'Measures of location' include not only measures of central tendency but other measures that explain the location or distribution of data.

Measures of Central Tendency

The Arithmetic Mean

The arithmetic mean is the sum of the observations divided by the number of observations. It is the most frequently used measure of the middle or center of data.

The Sample Mean

The sample mean is the arithmetic mean calculated for a sample. It is expressed as:

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

where: n is the number of observations in the sample.

If the sample data is: 2, 4, 4, 6, 10, 10, 12, 12, and 12 the sample mean can be calculated as:

$$\overline{X} = \frac{2 + 4 + 4 + 6 + 10 + 10 + 12 + 12 + 12}{9} = 8$$

A drawback of the arithmetic mean is that it is sensitive to extreme values (outliers). It can be pulled sharply upward or downward by extremely large or small observations,

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respectively.

The Median

The median is the midpoint of a data set that has been sorted into ascending or descending order.

For odd number of observations: 2,5,7,11,14 → Median = 7

For even number of observations: 3, 9, 10, 20 → Median = (9 + 10)/2 = 9.5

As compared to a mean, a median is less affected by extreme values (outliers).

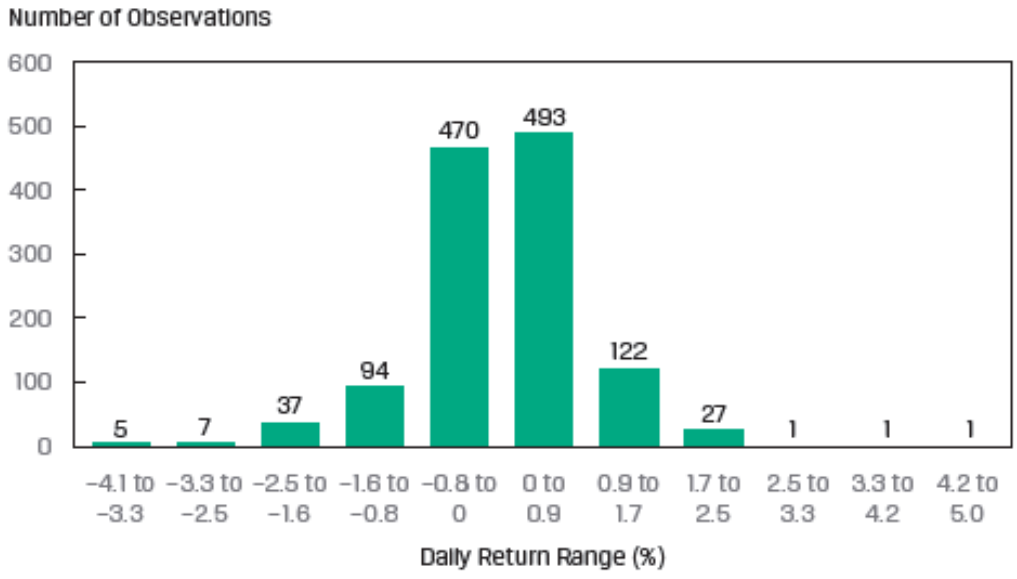
The Mode

The mode is the most frequently occurring value in a distribution.

For the following data set: 2, 4, 5, 5, 7, 8, 8, 8, 10, 12 → Mode = 8

A distribution can have more than one mode, or even no mode. When a distribution has one mode it is said to be unimodal. If a distribution has two or three modes, it is called bimodal or trimodal respectively.

When working with continuous data such as stock returns, 'modal interval' is often used instead of a mode. The data is divided into bins and the bin with the highest frequency is considered the modal interval. The following exhibit demonstrates this concept by plotting a histogram of the daily returns on an index. The highest bar in the histogram '0.0 to 0.9%' is the modal interval.



Dealing with Outliers

When data contains outliers, there are three options to deal with the extreme values:

Option 1: Do nothing; use the data without any adjustment.

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Option 2: Delete all the outliers.

Option 3: Replace the outliers with another value.

Option 1 is appropriate in cases when the extreme values are genuine.

Option 2 excludes extreme observations. A **trimmed mean** excludes a stated percentage of the lowest and highest values and then calculates the arithmetic mean of the remaining values. For example, a 5% trimmed mean discards the lowest 2.5% and the highest 2.5% of values and computes the mean of the remaining 95% of values.

Option 3 replaces extreme observations with observations closest to them. A **winsorized mean** assigns a stated percentage of the lowest values equal to one specified low value and a stated percentage of the highest values equal to one specified high value, and then computes a mean from the restated data. For example, a 95% winsorized mean sets the bottom 2.5% of values equal to the value at or below which 2.5% of all the values lie (the "2.5th percentile" value) and the top 2.5% of values equal to the value at or below which 97.5% of all the values lie (the "97.5th percentile" value).

Measures of Location

Quartiles, Quintiles, Deciles, and Percentiles

A quantile is a value at or below which a stated fraction of the data lies. Some examples of quantiles include:

- Quartiles: The distribution is divided into quarters.
- Quintiles: The distribution is divided into fifths.
- Deciles: The distribution is divided into tenths.
- Percentile: The distribution is divided into hundredths.

The formula for the position of a percentile in a data set with n observations sorted in ascending order is:

$$L_y = \frac{(n + 1)y}{100}$$

where:
y is the percentage point at which we are dividing the distribution.
n is the number of observations.
L_y is the location (L) of the percentile (P_y) in an array sorted in ascending order.

Some important points to remember are:

- When the location, L_y, is a whole number, the location corresponds to an actual observation.
- When L_y is not a whole number or integer, L_y lies between the two closest integer numbers (one above and one below) and we use linear interpolation between those two places to determine P_y.
- Interquartile range is the difference between the third and the first quartiles.

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Example

Consider the data set:

47 35 37 32 40 39 36 34 35 31 44

1. Find the 75th percentile point
2. Find the 1st quartile and 3rd quartile
3. Calculate the interquartile range
4. Find the 5th decile point
5. Find the 6th decile point.

Solution to 1:

First arrange the data in ascending order:

31, 32, 34, 35, 35, 36, 37, 39, 40, 44, 47

Location of the 75th percentile is the:

L_{75} = (11 + 1) (75/100) = 9th value. i.e. P_{75} = 40

With a small data set, such as this one, the location and the value is approximate. As the data set becomes larger, the location and percentile value estimates become more precise.

Solution to 2:

Location of the 1st quartile is:

L_{25} = (11 + 1) (25/100) = 3rd value. i.e. P_{25} = 34

Location of the 3rd quartile is:

L_{75} = (11 + 1) (75/100) = 9th value. i.e. P_{75} = 40

Solution to 3:

The interquartile range is the difference between the third and first quartiles, 40 – 34 = 6

Solution to 4:

Location of the 5th decile is:

L_{50} = (11 + 1) (50/100) = 6th value. i.e. P_{50} = 36

Solution to 5:

L_{60} = (11 + 1) (60/100) = 7.2

Use linear interpolation, which estimates an unknown value on the basis of two known values that surround it.

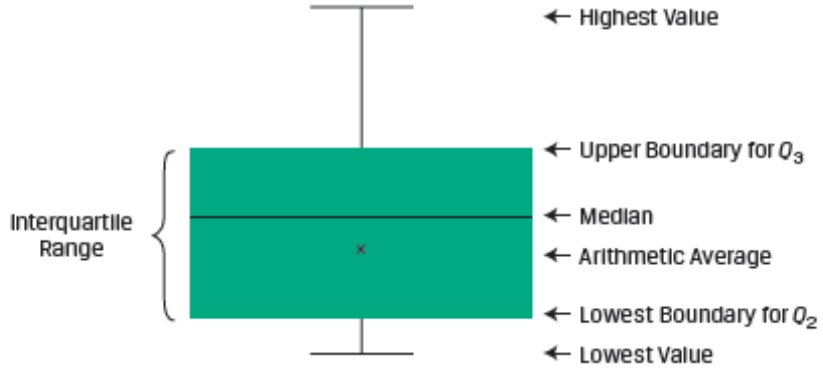
In this case, the 7th value is 37 and the 8th value is 39. The 6th decile is: P_{60} = 37+ 0.4 (0.2 times the linear distance between 37 and 39). P_{60} = 37.4

Box and Whiskers Plot

A box and whiskers plot is used to visualize the dispersion of data across quartiles. The box represents the interquartile range. The whiskers represent the highest and lowest values of

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the distribution. Exhibit 5 shows a sample box and whisker plot.



There are several variations of the box and whiskers plot. Sometimes the whiskers may be a function of the interquartile range instead of the highest and lowest values.

Quantiles in Investment Practice

Quantiles are used in:

- Portfolio performance evaluation: The performance of investment managers is often evaluated in terms of the percentile or quartile in which they fall relative to the performance of their peers.
- Investment research: For example, companies can be ranked based on their market capitalization and sorted into deciles. The first decile contains companies with smallest market values and the tenth decile contains companies with the largest market values. Such a classification allows analysts to compare the performance of small companies with large ones.

3. Measures of Dispersion

Measures of central tendency tell us where the investment results (expected returns) are centered. However, to evaluate an investment we also need to know how returns are dispersed around the mean. Measures of dispersion describe the variability of outcomes around the mean.

The Range

The range is the difference between the maximum and minimum values in a data set. It is expressed as:

Range = Max value – Min Value

If the annual returns data is: 10%, -5%, 10%, 25%. What is the range?

Here the maximum return is 25% and the minimum return is -5%. The range is 25% – (-5%) = 30%.

Another way to specify the range is to mention the actual minimum and maximum values. For example, for the above data the range is “from -5% to 25%”.

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The range is easy to compute; however, it does not tell us much about how the data is distributed.

Mean Absolute Deviations

It is the average of the absolute values of deviations from the mean. It is expressed as:

$$\text{MAD} = \left[\sum_{i=1}^{n} |X_i - \overline{X}|\right] / n$$

where: $\overline{X}$ is the sample mean and n is the number of observations in the sample.

Example

Consider the following data set: 8, 12, 10, 8 and 5. Calculate the mean absolute deviation.

Solution:

$\overline{X}$= (8 + 12 + 10 + 8 + 5) / 5 = 8.6

$$\text{MAD} = \frac{|8 - 8.6| + |12 - 8.6| + |10 - 8.6| + |8 - 8.6| + |5 - 8.6|}{5}$$

$$\text{MAD} = \frac{0.6 + 3.4 + 1.4 + 0.6 + 3.6}{5} = 1.92$$

Sample Variance and Sample Standard Deviation

'Variance' is defined as the average of the squared deviations around the mean. 'Standard deviation' is the positive square root of the variance.
'Sample variance' applies when we are dealing with a subset, or sample, of the total population. It is expressed as:

$$s^2 = \sum_{i=0}^{n} (X_i - \overline{X})^2 / (n - 1)$$

where: $\overline{X}$ is the sample mean and n is the number of observations in the sample.

'Sample standard deviation' is defined as the positive square root of the sample variance.

Example

Calculate the sample variance for the following data set: 8, 12, 10, 8 and 5.

Solution:

$$s^2 = \frac{[(8 - 8.6)^2 + (12 - 8.6)^2 + (10 - 8.6)^2 + (8 - 8.6)^2 + (5 - 8.6)^2]}{5 - 1}$$

$s^2 = 6.80\%$

The sample standard deviation is the positive square root of the sample variance. For the sample data given above, $s = \sqrt{6.80}$ = 2.61%

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Using a financial calculator to calculate variance and standard deviations

The sample standard deviation can easily be computed using a financial calculator. Assume the following data set: 10%, -5%, 10%, 25%, the calculator key strokes are shown below:

Keystrokes	*Description*	*Display*
[2nd] [DATA]	Enters data entry mode	
[2nd] [CLR WRK]	Clears data register	X01
10 [ENTER]		X01 = 10
[↓] [↓] 5+/- [ENTER]		X02 = -5
[↓] [↓] 10 [ENTER]		X03 = 10
[↓] [↓] 25 [ENTER]		X04 = 25
[2nd] [STAT] [ENTER]	Puts calculator into stats mode	
[2nd] [SET]	Press repeatedly till you see →	1-V
[↓]	Number of data points	N = 4
[↓]	Mean	X = 10
[↓]	Sample standard deviation	Sx = 12.25
[↓]	Population standard deviation	σx = 10.61

Notice that the calculator gives both the sample and the population standard deviation. On the exam we will have to determine whether we are dealing with population or sample data and choose the appropriate value.

Downside Deviation and Coefficient of Variation

Variance and standard deviation of returns take account of returns above and below the mean, but often investors are concerned only with downside risk, for example returns below the mean.

The target downside deviation, or target semideviation, is a measure of the risk of being below a given target. It is calculated as the square root of the average squared deviations from the target, but it includes only those observations below the target (B).
The sample target semideivation can be calculated as:

$$s_{Target} = \sqrt{\sum_{\text{for all } X_i \le B}^{n} \frac{(X_i - B)^2}{n-1}}$$

Example:
Suppose the monthly returns on a portfolio are as shown:

Month	**Return (%)**
Jan	6
Feb	4
Mar	-2
Apr	-5

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May	5
Jun	2
Jul	1
Aug	0
Sep	4
Oct	3
Nov	0
Dec	2

Calculate the target downside deviation when the target return is 4%.

Solution:

Month	**Observation**	**Deviation from the 4% target**	**Deviation below the target**	**Squared deviations below the target**
Jan	6	2	-	-
Feb	4	0	-	-
Mar	-2	-6	-6	36
Apr	-5	-9	-9	81
May	5	1	-	-
Jun	2	-2	-2	4
Jul	1	-3	-3	9
Aug	0	-4	-4	16
Sep	4	0	-	-
Oct	3	-1	-1	1
Nov	0	-4	-4	16
Dec	2	-2	-2	4
Sum				**167**

$$\text{Target semideviation} = \sqrt{\frac{167}{11}} = 3.8964\%$$

The target downside deviation will be less than the standard deviation, because deviations above the target are ignored. As the target is increased, the target downside deviation will increase.

Coefficient of Variation

Coefficient of variation expresses how much dispersion exists relative to the mean of a distribution and allows for direct comparison of dispersion across different data sets, even if the means are drastically different from one another. It is used in investment analysis to compare relative risks. When evaluating investments, a lower value is better. Coefficient of variation is expressed as:

$$CV = \frac{s}{\bar{X}}$$

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where: s = sample standard deviation of a set of observations and $\overline{X}$ = sample mean

Example
Investment A has a mean return of 7% and a standard deviation of 5%. Investment B has a mean return of 12% and a standard deviation of 7%. Calculate the coefficients of variation.

Solution
The coefficients of variation can be calculated as follows:

$$CV_A = \frac{5\%}{7\%} = 0.71$$

$$CV_B = \frac{7\%}{12\%} = 0.58$$

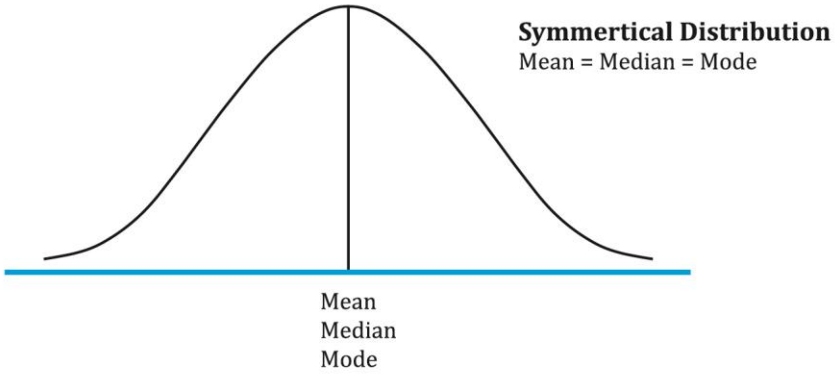
This metric shows that Investment A is riskier than Investment B.

4. Measures of Shape of a Distribution

Mean and variance may not adequately describe an investment's distribution of returns. To reveal other important characteristics of the distribution, we must look beyond measures of central tendency, location, and dispersion. One such characteristic is the degree of symmetry in return distributions.

Symmetrical distribution
A distribution is said to be symmetrical when the distribution on either side of the mean is a mirror image of the other.

In a normal distribution, mean = median = mode.



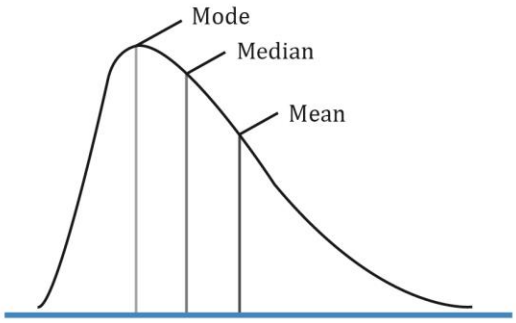
If a distribution is non-symmetrical, it is said to be skewed. Skewness is a measure of the asymmetry of the probability distribution. Skewness can be negative or positive.

Positively skewed distribution
A positively skewed distribution has a long tail on the right side, which means that there will be limited but frequent downside returns and unlimited but less frequent upside returns.

Here the mean > median > mode. The extreme values affect the mean the most which is

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pulled to the right. They affect the mode the least.

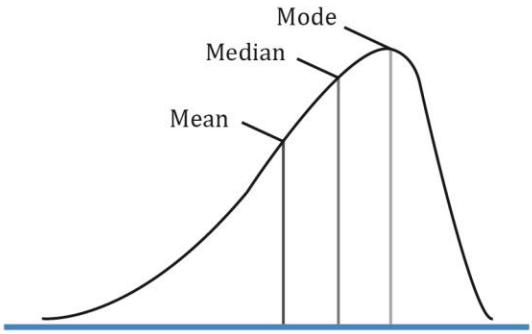


Right - Skewed (Positive Skewness)

Negatively skewed distribution

A negatively skewed distribution has a long tail on the left side, which means that there will be limited but frequent upside returns and unlimited but less frequent downside returns.

Here the mean < median< mode. The extreme values affect the mean the most which is pulled to the left. They affect the mode the least.



Left - Skewed (Negative Skewness)

Instructor's Note: Investors prefer positive skewness because it has a higher chance of very large returns and also because it has a higher mean return.

Example:

Which of the following distribution is most likely characterized by frequent small losses and a few extreme gains?

A. Normal distribution
B. Negatively skewed
C. Positively skewed

Solution:

C is correct. A positively skewed distribution is characterized by frequent small losses and a

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few extreme gains.

Example:

Which of the following is most likely to be true for a negatively skewed distribution?

A. Mean < Median < Mode
B. Mode < Median < Mean
C. Median < Mean < Mode

Solution:

A is correct. In a negatively skewed distribution, the mean < median < mode.

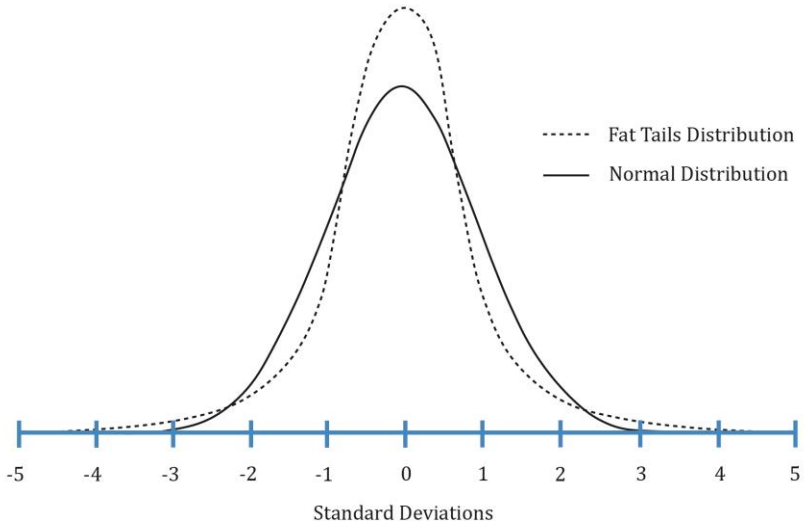
Kurtosis

Kurtosis is a measure of the combined weight of the tails of a distribution relative to the rest of the distribution.

Excess kurtosis = kurtosis - 3. An excess kurtosis with an absolute value greater than one is considered significant.

- A 'leptokurtic' distribution has fatter tails than a normal distribution. It has an excess kurtosis greater than 0.
- A 'platykurtic' distribution has thinner tails than a normal distribution. It has an excess kurtosis less than 0.
- A 'mesokurtic' distribution is identical to a normal distribution. It has an excess kurtosis equal to 0.

The following figure shows a leptokurtic distribution. As compared to a normal distribution, a leptokurtic distribution is more likely to generate observations in the tail region. It is also more likely to generate observations near the mean. However, to have the total probabilities sum to 1, it will generate fewer observations in the remaining regions (i.e. regions between the central and the two tail regions)



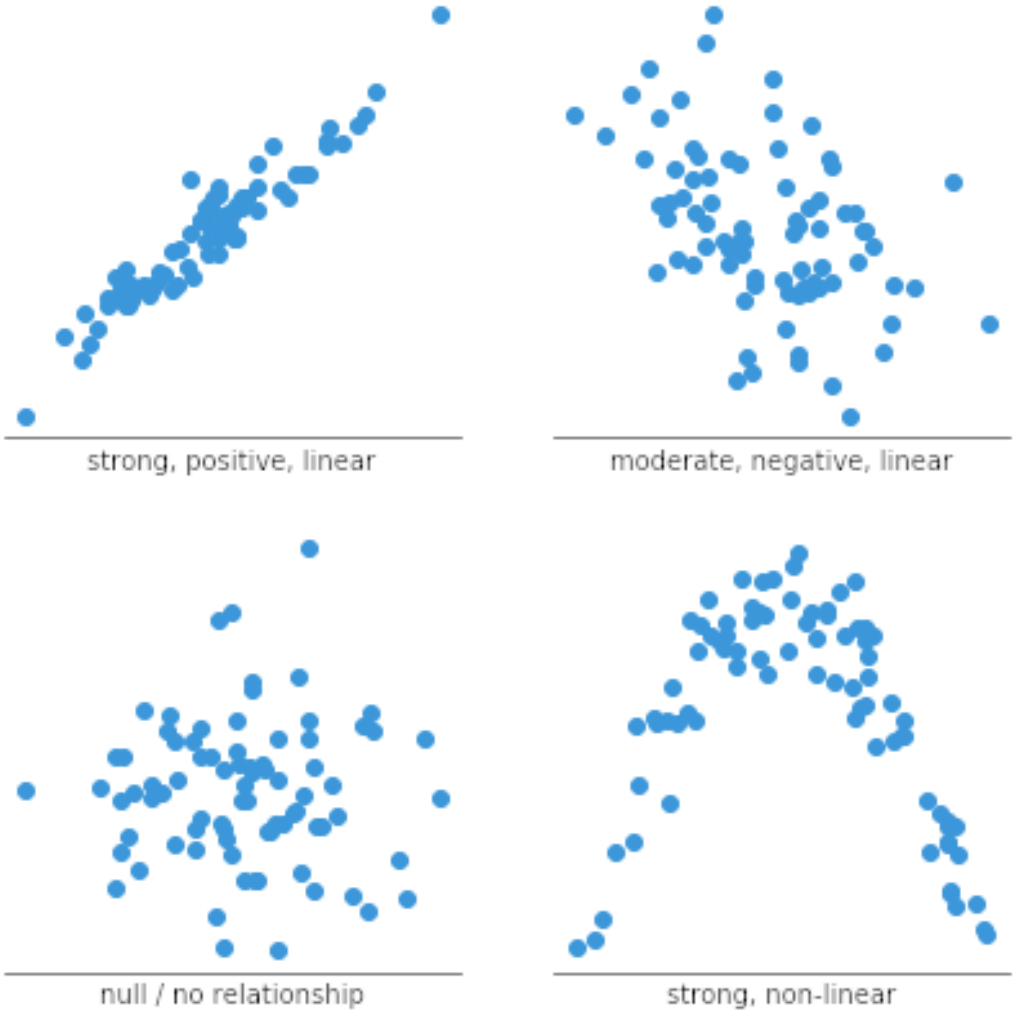
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5. Correlation Between Two Variables

Scatter Plot

A scatter plot is a type of graph used to visualize the joint variation in two numerical variables. It is constructed with the x-axis representing one variable and the y-axis representing the other variable. Dots are drawn to indicate the values of the two variables at different points in time.

The pattern of a scatter plot may indicate no relationship, linear relationship or a non-linear relationship between the two variables. In case of a linear relationship, a positive slope indicates that the variables move in the same direction; whereas a negative slope indicates that the variables move in opposite directions.



Covariance and Correlation

Covariance

Covariance is a measure of how two variables move together. The formula for computing the

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'sample covariance' of X and Y is:

$$s_{XY} = \frac{\sum_{i=1}^{N}(X_i - \overline{X})(Y_i - \overline{Y})}{n - 1}$$

The problem with covariance is that it can vary from negative infinity to positive infinity which makes it difficult to interpret. To address this problem, we use another measure called correlation.

Correlation

Correlation is a standardized measure of the linear relationship between two variables with values ranging between -1 and +1.

The 'sample correlation coefficient' can be calculated as:

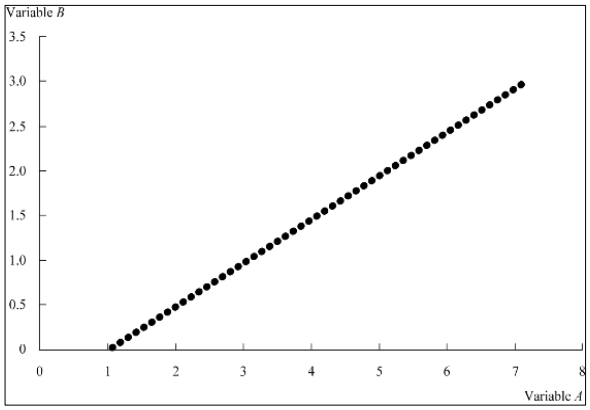
$$r_{XY} = \frac{s_{XY}}{s_x * s_y}$$

Properties of Correlation

- Correlation ranges from -1 and +1.
- A correlation of 0 (uncorrelated variables) indicates an absence of any linear (straight-line) relationship between the variables.
- A correlation of +1 indicates a perfect positive relationship.
- A correlation of -1 indicates a perfect negative relationship.

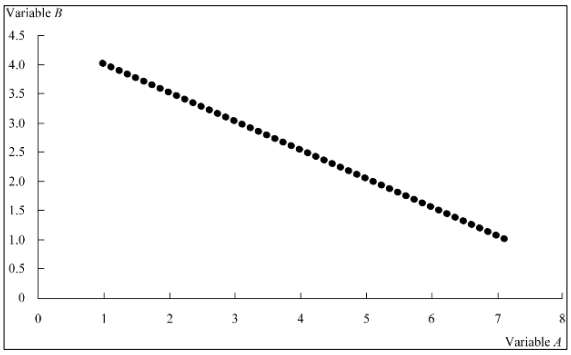
The three scatter plots below show a positive linear, negative linear, and no linear relation between two variables A and B. They have correlation coefficients of +1, -1 and 0 respectively.

Variables with a correlation of 1.

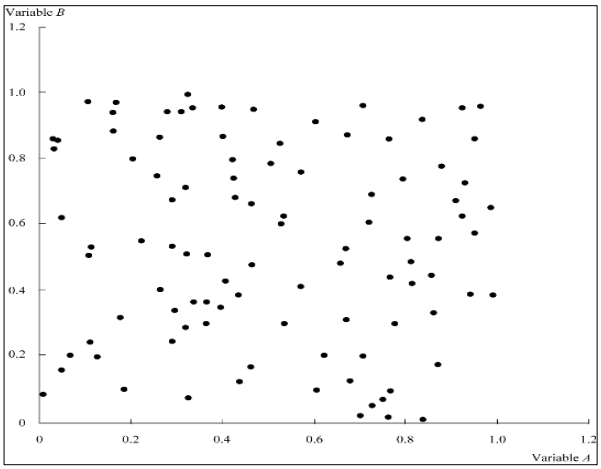


Variables with a correlation of -1.

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Variables with a correlation of 0.



Limitations of Correlation Analysis

The correlation analysis has certain limitations:

- Two variables can have a strong non-linear relation and still have a very low correlation.
- The correlation can be unreliable when outliers are present.
- The correlation may be spurious. 'Spurious correlation' refers to the following situations:
  - The correlation between two variables that reflects chance relationships in a particular data set.
  - The correlation induced by a calculation that mixes each of two variables with a third variable.
  - The correlation between two variables arising not from a direct relation between them, but from their relation to a third variable. Ex: shoe size and vocabulary of school children. The third variable is age here. Older shoe sizes simply imply that they belong to older children who have a better vocabulary.

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Summary

LO: Calculate, interpret, and evaluate measures of central tendency and location to address an investment problem.

Measures of central tendency specify where data are centered.

The arithmetic mean is the sum of the observations divided by the number of observations. It is the most frequently used measure of the middle or center of data.

The median is the midpoint of a data set that has been sorted into ascending or descending order.

The mode is the most frequently occurring value in a distribution.

Measures of location

A quantile is a value at or below which a stated fraction of the data lies. Some examples of quantiles include:

- Quartiles: The distribution is divided into quarters.
- Quintiles: The distribution is divided into fifths.
- Deciles: The distribution is divided into tenths.
- Percentile: The distribution is divided into hundredths.

The formula for the position of a percentile in a data set with n observations sorted in ascending order is:

$$L_y = \frac{(n + 1)y}{100}$$

A box and whiskers plot is used to visualize the dispersion of data across quartiles. The box represents the interquartile range. The whiskers represent the highest and lowest values of the distribution. There are several variations of the box and whiskers plot. Sometimes the whiskers may be a function of the interquartile range instead of the highest and lowest values.

LO: Calculate, interpret, and evaluate measures of dispersion to address an investment problem.

Measures of dispersion describe the variability of outcomes around the mean.

The range is the difference between the maximum and minimum values in a data set.

MAD is the average of the absolute values of deviations from the mean.

Variance is defined as the average of the squared deviations around the mean. Standard deviation is the positive square root of the variance.

Coefficient of variation expresses how much dispersion exists relative to the mean of a distribution and allows for direct comparison of dispersion across different data sets, even if the means are drastically different from one another.

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The target downside deviation, or target semideviation, is a measure of the risk of being below a given target. It is calculated as the square root of the average squared deviations from the target, but it includes only those observations below the target (B).

LO: Interpret and evaluate measures of skewness and kurtosis to address an investment problem.

Skewness is a measure of the asymmetry of the probability distribution. If a distribution is non-symmetrical, it is said to be skewed. Skewness can be negative or positive.

A positively skewed distribution has a long tail on the right side, which means that there will be frequent small losses and few large gains.

A negatively skewed distribution has a long tail on the left side, which means that there will be frequent small gains and few large losses.

Kurtosis is a measure of the combined weight of the tails of a distribution relative to the rest of the distribution.

Excess kurtosis = kurtosis - 3. An excess kurtosis with an absolute value greater than one is considered significant.

- A leptokurtic distribution has fatter tails than a normal distribution. It has an excess kurtosis greater than 0.
- A platykurtic distribution has thinner tails than a normal distribution. It has an excess kurtosis less than 0.
- A mesokurtic distribution is identical to a normal distribution. It has an excess kurtosis equal to 0.

LO: Interpret correlation between two variables to address an investment problem.

Correlation is a statistic that measures the degree to which two variables move in relation to each other.

- Correlation ranges from -1 and +1.
- A correlation of 0 (uncorrelated variables) indicates an absence of any linear (straight-line) relationship between the variables.
- A correlation of +1 indicates a perfect positive relationship.
- A correlation of -1 indicates a perfect negative relationship.

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LM04 Probability Trees and Conditional Expectations

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3. Probability Trees and Conditional Expectations....6
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1. Introduction

Since many investment decisions are made in an environment of uncertainty, it is essential for portfolio managers and investment managers to have a fundamental grasp of probability concepts. In this learning module, we will focus on:

- Calculation of the expected value, variance, and standard deviation for a random variable.
- Using probability trees to visualize conditional expectations and the total probabilities for expected value.
- Using Bayes' formula to adjust probabilities with the arrival of new information.

Instructor's Note: Before we get into the actual concepts covered in the curriculum, some prerequisite fundamental concepts are presented below.

Fundamental Concepts

A *random variable* is an uncertain quantity/number. For example, when you roll a die, the result is a random variable.

An *outcome* is the observed value of a random variable. For example, if you roll a 2, it is an outcome.

An *event* can be a single outcome or a set of outcomes. For example, you can define an event as rolling a 2 or rolling an even number.

Mutually exclusive events are events that cannot happen at the same time. For example, rolling a 2 and rolling a 3 are examples of mutually exclusive events. They cannot happen at the same time.

Exhaustive events are those that cover all possible outcomes. For example, 'rolling an even number' or' rolling an odd number' are exhaustive events. They cover all possible outcomes.

The *two defining properties of probability* are:

- The probability of any event has to be between 0 and 1.
- The sum of the probabilities of mutually exclusive and exhaustive events is equal to 1.

Conditional v/s Unconditional probabilities

Unconditional probability is the probability of an event occurring irrespective of the occurrence of other events. It is denoted as P(A). Unconditional probability is also called 'marginal' probability.

Conditional probability is the probability of an event occurring given that another event has occurred. It is denoted as P(A|B), which is the probability of event A given that event B has occurred.

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Joint Probability and Multiplication Rule

Multiplication rule is used to determine the joint probability of two events. It is expressed as:

$$P(AB) = P(A|B)\,P(B)$$

Rearranging the equation, we get the formula for computing conditional probabilities:

$$P(A|B) = P(AB) / P(B)$$

Example

P(interest rates will decrease) = P(D) = 40%
P(stock price increases) = P(S)
P(stock price will increase given interest rates decrease) = P(S|D) = 70%

Compute probability of a stock price increase **and** an interest rate decrease.

Solution:

$P(SD) = P(S|D) \times P(D) = 0.7 \times 0.4 = 0.28 = 28\%$

Addition Rule for Probabilities

Addition rule is used to determine the probability that at least one of the events will occur. It is expressed as:

$$P(A \text{ or } B) = P(A) + P(B) - P(AB)$$

P(AB) represents the joint probability that both A and B will occur. It is subtracted from the sum of the unconditional probabilities: P(A) + P(B), to avoid double counting.

If the two events are mutually exclusive, the joint probability: P(AB) is zero and the probability that either A or B will occur is simply the sum of the unconditional probabilities for each event:

$$P(A \text{ or } B) = P(A) + P(B)$$

Example

P(price of A increases) = P(A) = 0.5
P(price of B increases) = P(B) = 0.7
P(price of A and B increases) = P(AB) = 0.3

Compute the probability that the price of stock A **or** the price of stock B increases.

Solution

$P(A \text{ or } B) = 0.5 + 0.7 - 0.3 = 0.9$

Independent and Dependent Events

If the occurrence of one event does not influence the occurrence of the other event, then the two events are called *independent events.*

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i.e. P(A|B) = P(A) or P(B|A) = P(B)

Multiplication rule for independent events: P(AB) = P(A) P(B)

Addition rule for independent events: P(A or B) = P(A) + P(B) – P(AB). (The addition rule does not change.)

If the probability of an event is affected by the occurrence of another event, then it is called a *dependent event.*

Total Probability Rule

The total probability rule is used to calculate the unconditional probability of an event, given conditional probabilities.

In investment analysis, we often formulate a set of mutually exclusive and exhaustive scenarios and then estimate the probability of a particular event. For example, let's say that we have two scenarios S and non-S that are mutually exclusive and exhaustive.

According to the total probability rule, the probability of any event P(A) can be expressed as:

$$P(A) = P(AS) + P(AS^C)$$

Using the multiplication rule we get,

$$P(A) = P(A|S)\,P(S) + P(A|S^C)\,P(S^C)$$

If we have more than two scenarios, we can generalize this equation to:

$$P(A) = P(AS_1) + P(AS_2) + \ldots + P(AS_n) = P(A|S_1)\,P(S_1) + P(A|S_2)\,P(S_2) + \ldots + P(A|S_n)\,P(S_n)$$

2. Expected Value and Variance

Expected Value of a Random Variable

The expected value of a random variable can be defined as the probability-weighted average of the possible outcomes of the random variable. For a random variable X, the expected value of X is denoted as E(X) and is calculated as:

$$E(X) = \sum_{i=1}^{n} P(X_i)\,X_i$$

where:
X_i = One of n possible outcomes of the random variable X
$P(X_i)$ = Probability of X_i

Variance of a Random Variable

The expected value is our forecast, but we cannot count on the individual forecast being realized. This is why we need to measure the risk we face. Variance and standard deviation are examples of how we can measure this risk.

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The variance of a random variable is the expected value (the probability-weighted average) of squared deviations from the random variable's expected value:

$\sigma^2(X) = E[X - E(X)]^2$.

It can also be expressed as:

$$\sigma^2(X) = \sum_{i=1}^{n} P(X)[X - E(X)]^2$$

Variance is a number greater than or equal to 0 because it is the sum of squared terms. If variance is 0, there is no dispersion or risk. The outcome is certain and the quantity X is not random at all.

Standard deviation is the positive square root of variance. Like variance, standard deviation also measures dispersion, but it is measured in the same units as the variable.

We can calculate the expected value and variance of a random variable using a financial calculator as shown below:

Example

A project's cash flow for the upcoming year depends on the state of the economy, as shown in the table below. What is the expected value of the cash flow? What is the variance of the cash flow? What is the standard deviation?

State of Economy	Probability	Cash Flow
Good	0.3	50
Average	0.5	40
Weak	0.2	20

Solution:

Using a financial calculator:

Keystrokes	*Explanation*	*Display*
[2nd] [DATA]	Enters data entry mode	
[2nd] [CLR WRK]	Clears data register	X01
50 [ENTER]	1st possible value of random variable	X01 = 50
[↓] 30 [ENTER]	Probability of 30% for X01	Y01 = 30
[↓] 40 [ENTER]	2nd possible value of random variable	X02 = 40
[↓] 50 [ENTER]	Probability of 50% for X02	Y02 = 50
[↓] 20 [ENTER]	3rd possible value of random variable	X03 = 20

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[↓] 20 [ENTER]	Probability of 20% for X03	Y03 = 20
[2nd] [STAT]	Puts calculator into stats mode	
[2nd] [SET]	Press repeatedly till you see →	1-V
[↓]	Total number of entries	N = 100
[↓]	Expected value of random variable	X = 39
[↓]	Sample standard deviation	Sx = 10.49
[↓]	Population standard deviation	σx = 10.44

We can then square the population standard deviation of 10.44 to get the variance i.e. $10.44^2 = 109.00$

3. Probability Trees and Conditional Expectations

Total Probability Rule for Expected Value

Just like the total probability rule states unconditional probabilities in terms of conditional probabilities, the total probability rule for expected values states unconditional expected values in terms of conditional expected values.

$$E(X|S) = P(X_1|S)\,X_1 + P(X_2|S)\,X_2 + \ldots + P(X_n|S)\,X_n$$

Instructor's Note

Notice that this formula is exactly similar to the total probability rule formula.

$$P(A) = P(A|S_1)\,P(S_1) + P(A|S_2)\,P(S_2) + \ldots + P(A|S_n)\,P(S_n)$$

A probability tree is a means of illustrating the results of two or more independent events. The following example illustrates how we can use probability trees and the total probability rule to calculate the expected value of a random variable.

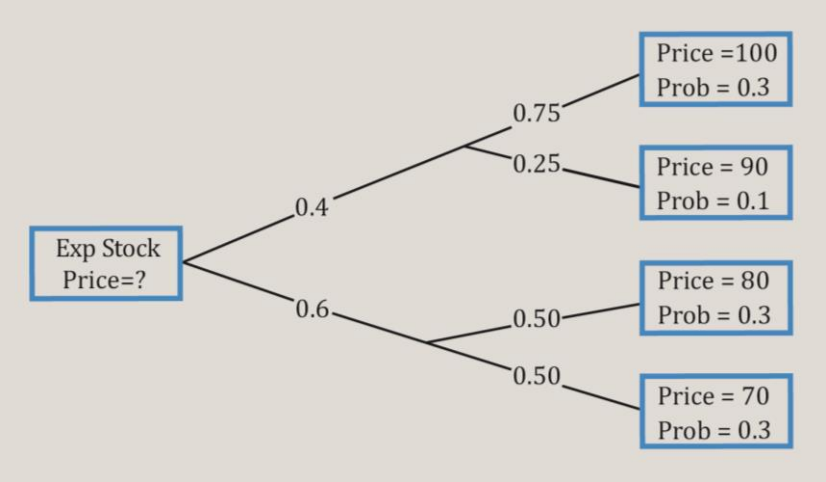
Example

What is the expected price of a stock at the end of the current period given the following information: probability that interest rates will decline = 0.4. If interest rates decline there is a 75% chance that stock price will be $100 versus a 25% chance that the stock price will be $90. If interest rates do not decline there is a 50% chance that the stock price will be $80 versus a 50% chance that stock price will be $70.

Solution:

We can plot the probabilities using a tree diagram.

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Consider the first node (top right). It refers to the probability that the stock price will be $100 given a decline in interest rates. We can calculate the probability of that happening by multiplying the probability of a decline in interest rates (0.4) by the probability of the stock price being $100 if that happens (0.75). This gives us a conditional probability of 0.30. In short, it is the joint probability of the stock price being $100 given a decline in interest rates. Similarly, probabilities are calculated for each of the other three nodes. We can then calculate:

E(Price | decline in interest rates) = 0.75 ($100) + 0.25 ($90) = $97.50

E(Price | no decline in interest rates) = 0.50 ($80) + 0.50 ($70) = $75.00

Now we use the total probability rule for expected value of stock price at the end of the current period:

E(Price) = E(Price | decline in interest rates) P(decline in interest rates) + E(Price | no decline in interest rates) P(no decline in interest rates)

E(Price) = $97.50 (0.40) + $75.00 (0.60)
E(Price) = $84.00

4. Bayes' Formula and Updating Probability Estimates

Bayes' formula is a rational method for updating or adjusting the probability of an event based on new information. According to Bayes' formula, the updated probability of an event given new information is:

$$P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Information})} \times P(\text{Event})$$

Example

Consider a factory that has three assembly lines. The percentage of output produced at each assembly line is as follows: Line A = 45%, Line B = 35%, Line C = 20%. The output defective from each line is estimated to be 3%, 5%, and 4%, respectively. Given that the product is defective, what is the probability that it came from Line C?

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Solution:

When dealing with questions related to Bayes' formula, the first step is to reproduce the information in probability notation:

P(Line A) = 0.45; P(Not Line A) = 0.55

P(Line B) = 0.35; P(Not Line B) = 0.65

P(Line C) = 0.20; P(Not Line C) = 0.80

P(Defective | Line A) = 0.03, P(Defective | Line B) = 0.05, P(Defective | Line C) = 0.04

P(Defective) = 0.45 x 0.03 + 0.35 x 0.05 + 0.20 x 0.04 = 0.039

Next write down the Bayes formula:

$$P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Information})} \text{ x } P(\text{Event})$$

We then have to distinguish between the event and the information and plug the relevant values into the formula. In this case, the information is that the product is defective. Hence, the formula can be written as:

$$P(\text{Line C} \mid \text{Defective}) = \frac{[P(\text{Defective} \mid \text{Line C}) * P(\text{Line C})]}{P(\text{Defective})} = \frac{0.04 * 0.20}{0.039} = 20.51\%$$

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Summary

LO: Calculate expected values, variances, and standard deviations and demonstrate their application to investment problems.

The expected value of a random variable can be defined as the probability-weighted average of the possible outcomes of the random variable.

$$E(X) = \sum_{i=1}^{n} P(X_i)\, X_i$$

The variance of a random variable is the expected value (the probability-weighted average) of squared deviations from the random variable's expected value:

$\sigma^2(X) = E\,[X - E(X)]^2$.

The square root of variance will give us the standard deviation.

LO: Formulate an investment problem as a probability tree and explain the use of conditional expectations in investment application.

A probability tree is a means of illustrating the results of two or more independent events.

The total probability rule for expected values states unconditional expected values in terms of conditional expected values.

$$E(X|S) = P(X_1|S)\, X_1 + P(X_2|S)\, X_2 + \ldots + P(X_n|S)\, X_n$$

LO: Calculate and interpret an updated probability in an investment setting using Bayes' formula.

Bayes' formula is a rational method for updating or adjusting the probability of an event based on new information. According to Bayes' formula, the updated probability of an event given new information is:

$$P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Information})} \times P(\text{Event})$$

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LM05 Portfolio Mathematics

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2. Portfolio Expected Return and Variance of Return....2
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Ver 1.0

1. Introduction

This learning module covers:

- Calculating the expected return and variance of a portfolio
- Calculating covariance and correlation of portfolio returns using a joint probability function
- Portfolio risk measures: Roy's safety-first ratio

2. Portfolio Expected Return and Variance of Return

Expected Return

A portfolio's expected return can be calculated as:

$$E(R_P) = w_1E(R_1) + w_2E(R_2) + \ldots + w_nE(R_n)$$

where:
w_n = portfolio weight of nth security in the portfolio
R_n = expected return of nth security in the portfolio
n = number of securities in the portfolio

We will discuss portfolio expected return and variance of return using a two-stock portfolio.

Example

40% of the portfolio is invested in Stock A and 60% is invested in Stock B. As shown in the table below, the expected return of each stock depends on the economic scenario.

Scenario	P(Scenario)	Expected returns of A	Expected returns of B
Recession	0.25	2%	4%
Normal	0.50	8%	10%
Boom	0.25	12%	16%

Calculate the expected return of A and B.

Solution:

Given the data presented above:

The expected return of A is: 0.25 x 2 + 0.50 x 8 + 0.25 x 12 = 7.5%.

The expected return of B is: 0.25 x 4 + 0.50 x 10 + 0.25 x 16 = 10%.

Expected return of the portfolio = weight of A in the portfolio x expected return of A + weight of stock B in the portfolio x expected return of B = 0.4 x 7.5 + 0.6 x 10 = 9%

The expected portfolio return is 9%. As the term implies, this is the expected return. The actual return will vary around 9%. The amount of variability is measured by the variance. In order to determine the variance of return, we must first calculate the covariance.

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Covariance

Covariance tells us how movements in a random variable vary with movements in another random variable, whereas variance tells us how a random variable varies with itself. Assume there are two random variables R_i and R_j. The covariance between R_i and R_j (used to measure how they move together) is given by:

$$\text{Cov}(R_i, R_j) = E([R_i - ER_i][R_j - ER_j])$$

where:
ER_i = expected return for variable R_i
ER_j = expected return for variable R_j

Example

Continuing with our previous example, calculate the covariance of returns between A and B.

Solution:

Say R_i represents the return on A and R_j represents the return on B, we have already calculated the expected returns of A and B as 7.5% and 10% respectively. The covariance of returns is:

$E[(Ri - 7.5)(Rj - 10)]$

$= 0.25(2\% - 7.5\%)(4\% - 10\%) + 0.5(8\% - 7.5\%)(10\% - 10\%) + 0.25(12\% - 7.5\%)(16\% - 10\%)$

$= 0.000825 + 0 + 0.000675 = 0.0015$

We can interpret the sign of covariance as:

- Covariance of returns is negative if, when the return on one asset is above its expected value, the return on the other asset tends to be below its expected value.
- Covariance of returns is 0 if the returns on the assets are unrelated.
- Covariance of returns is positive when the returns on both assets tend to be on the same side (above or below) their expected values at the same time.

Correlation

The problem with covariance is that it can vary from negative infinity to positive infinity which makes it difficult to interpret. To address this problem, we use another measure called correlation. Correlation is a standardized measure of the linear relationship between two variables with values ranging between -1 and +1.

- A correlation of 0 (uncorrelated variables) indicates an absence of any linear (straight-line) relationship between the variables.
- A correlation of +1 indicates a perfect positive relationship.
- A correlation of -1 indicates a perfect negative relationship.

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It is computed as:

$$\rho(R_i, R_j) = Cov(R_i, R_j)/\sigma(R_i)\,\sigma(R_j)$$

We will now apply this formula to calculate the correlation between the returns of A and B from our example. We have already shown that the covariance of returns is 0.0015. In order to calculate the correlation, we need the standard deviation of A and B. Using a financial calculator, we can determine that the standard deviation of A is 0.0357 and the standard deviation of B is 0.0424. The correlation, $\rho(A, B) = \frac{Cov(A,B)}{\sigma(A)\sigma(B)}$ = 0.0015/(0.0357 x 0.0424) = 0.99.

Instructor's Note: The keystrokes for calculating the standard deviation of A are shown below.

Keystrokes	***Explanation***	***Display***
[2nd] [DATA]	Enters data entry mode	
[2nd] [CLR WRK]	Clears data register	X01
0.02 [ENTER]	1st possible value of random variable	X01 = 0.02
[↓] 25 [ENTER]	Probability of 25% for X01	Y01 = 25
[↓] 0.08 [ENTER]	2nd possible value of random variable	X02 = 0.08
[↓] 50 [ENTER]	Probability of 50% for X02	Y02 = 50
[↓] 0.12 [ENTER]	3rd possible value of random variable	X03 = 0.12
[↓] 25 [ENTER]	Probability of 25% for X03	Y03 = 25
[2nd] [STAT]	Puts calculator into stats mode	
[2nd] [SET]	Press repeatedly till you see →	1-V
[↓]	Total number of entries	N = 100
[↓]	Expected value of random variable	X = 0.075
[↓]	Sample standard deviation	Sx = 0.0359
[↓]	Population standard deviation	σx = 0.0357

The correlation of 0.99 (almost 1) implies a very strong positive relationship between the returns of A and B. This is more meaningful than the covariance number of 0.0015 which tells us that there is a positive relationship between the returns of A and B but does not give a sense for the strength of the relationship.

Variance of returns

Once we know the covariance, we can calculate the variance of a portfolio using this formula:

$$\sigma^2(R_P) = w_1^2\sigma_1^2(R_1) + w_2^2\sigma_2^2(R_2) + 2w_1w_2Cov(R_1R_2)$$

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Example

Continuing with our example, the variance of the portfolio is

- Weight of the first asset, $w_1 = 0.40$
- Weight of the second asset, $w_2 = 0.60$
- Standard deviation of first asset = 0.0357
- Standard deviation of second asset = 0.0424
- Covariance between the two assets = 0.0015

Variance of the portfolio = $0.4^2 \times 0.0357^2 + 0.6^2 \times 0.0424^2 + 2 \times 0.4 \times 0.6 \times 0.0015 = 0.00157$

Standard deviation of the portfolio = $\sqrt{0.00157} = 0.0396$

The portfolio variance for a three-asset portfolio can be calculated as:

$$\sigma^2(R_P) = w_1^2\sigma_1^2(R_1) + w_2^2\sigma_2^2(R_2) + w_3^2\sigma_3^2(R_3) + 2w_1w_2Cov(R_1R_2) + 2w_1w_3Cov(R_1R_3) + 2w_2w_3Cov(R_2R_3)$$

Sometimes we may be provided a variance-covariance matrix. We can use this matrix to calculate the portfolio standard deviation as illustrated in the following example:

Example:

(*This is based on Question 9 from the curriculum.*)

An analyst has designed the following three asset portfolio:

	Asset 1	**Asset 2**	**Asset 3**
Expected return	5%	6%	7%
Portfolio weight	0.20	0.30	0.50

Variance-Covariance Matrix

	Asset 1	**Asset 2**	**Asset 3**
Asset 1	196	105	140
Asset 2	105	225	150
Asset 3	140	150	400

Instructor's Note: A variance covariance matrix is defined as a square matrix where the diagonal elements represent the variance and off-diagonal elements represent the covariance. In the above matrix, the variance of Asset 1, Asset 2 and Asset 3 are 196, 225 and 400 respectively. The covariance between Asset 1 and Asset 2 is 105. The covariance between Asset 1 and Asset 3 is 140. The covariance between Asset 2 and Asset 3 is 150.

Calculate the expected return and standard deviation of the portfolio.

Solution:

Expected return = $w_1E(R_1) + w_2E(R_2) + w_3E(R_3) = 0.20(5\%) + 0.30(6\%) + 0.50(7\%) = 6.3\%$

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Portfolio variance = $w_1^2\sigma_1^2\ (R_1) + w_2^2\sigma_2^2\ (R_2) + w_3^2\sigma_3^2\ (R_3) + 2w_1w_2\text{Cov}\ (R_1R_2) + 2w_1w_3\text{Cov}\ (R_1R_3) + 2w_2w_3\text{Cov}\ (R_2R_3)$

$= (0.20)^2(196) + (0.30)^2(225) + (0.50)^2(400) + 2(0.20)(0.30)(105) + 2(0.20)(0.50)(140) + 2(0.30)(0.50)(150) = 213.69$

Standard deviation is the square root of variance. Therefore, portfolio standard deviation = sqrt (213.69) = 14.62%

A variance covariance matrix can also be used to calculate correlation as illustrated in the following example.

Example:
(*This is based on Question 4 from the curriculum.*)

An analyst develops the following covariance matrix of returns.

	Hedge Fund	**Market Index**
Hedge Fund	256	110
Market Index	110	81

Calculate the correlation of returns between the hedge fund and the market index.

Solution:

$\rho(R_i, R_j) = \text{Cov}(R_i, R_j)/\sigma(R_i)\ \sigma(R_j)$

$$\rho(R_i, R_j) = \frac{110}{\sqrt{256} \times \sqrt{81}} = \frac{110}{16 \times 9} = 0.764$$

3. Forecasting Correlation of Returns: Covariance Given a Joint Probability Function

The same information we saw in section 2 can also be presented in the form of a joint probability function.

Example:

Scenario	**P(Scenario)**	**Expected returns of A**	**Expected returns of B**
Recession	0.25	2%	4%
Normal	0.50	8%	10%
Boom	0.25	12%	16%

This information can also be presented as a joint probability function of A's and B's returns:

	R_B = 4%	**R_B = 10%**	**R_B = 16%**
R_A = 2%	0.25	0	0
R_A = 8%	0	0.50	0
R_A = 12%	0	0	0.25

Row 1 and Column 1 represent the returns of A and B respectively. The other cells contain probabilities.

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We can calculate the covariance in the same way that we did in section 2.

The expected return of A is: 0.25 x 2 + 0.50 x 8 + 0.25 x 12 = 7.5%.

The expected return of B is: 0.25 x 4 + 0.50 x 10 + 0.25 x 16 = 10%.

Covariance of returns = E [(Ri – 7.5) (Rj – 10)]

= 0.25(2% - 7.5%) (4% - 10%) + 0.5(8% - 7.5%) (10% - 10%) + 0.25(12% - 7.5%) (16% - 10%)

= 0.000825 + 0 + 0.000675 = 0.0015

Instructor's Note: Do the above calculations by only looking at the data in the joint probability table. On the exam you may be presented information in either format (normal table or joint probability table).

4. Portfolio Risk Measures: Applications of The Normal Distribution

Shortfall risk

Shortfall risk is the risk that portfolio's return will fall below a specified minimum level of return over a given period of time.

Safety first ratio

Safety first ratio is used to measure shortfall risk. It is calculated as:

$$\text{SF Ratio} = \frac{[R_P - R_L]}{\sigma_P}$$

where:
R_p = Expected portfolio return
R_L = Threshold level
σp = Standard deviation of portfolio returns

The portfolio with the highest SF-Ratio is preferred, as it has the lowest probability of falling below the target return.

Roy's safety-first criteria

It states that an optimal portfolio minimizes the probability that the actual portfolio return will fall below the target return.

Example

An investor is considering two portfolios A and B. Portfolio A has an expected return of 10% and a standard deviation of 2%. Portfolio B has an expected return of 15% and a standard deviation of 10%. The minimum acceptable return for the investor is 8%. According to Roy's safety-first criteria, which portfolio should the investor select?

Solution:

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$$SF_A = \frac{10 - 8}{2} = 1$$

$$SF_B = \frac{15 - 8}{10} = 0.7$$

Since A has a higher safety-first ratio, the investor should select portfolio A.

If the risk-free rate is set as the threshold level R_L, the safety-first ratio becomes the Sharpe ratio.

$$\text{Sharpe Ratio} = \frac{[R_P - R_f]}{\sigma_P}$$

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Summary

LO: Calculate and interpret the expected value, variance, standard deviation, covariances, and correlations of portfolio returns.

Expected return: $E(R_P) = w_1E(R_1) + w_2E(R_2) + \ldots + w_nE(R_n)$

Covariance: $Cov(R_i, R_j) = E([R_i - ER_i][R_j - ER_j])$

Correlation: $\rho(R_i, R_j) = Cov(R_i, R_j)/\sigma(R_i)\,\sigma(R_j)$

Variance of a 2-asset portfolio: $\sigma^2(R_P) = w_1^2\sigma_1^2(R_1) + w_2^2\sigma_2^2(R_2) + 2w_1w_2Cov(R_1R_2)$

Variance of a 3-asset portfolio: $\sigma^2(R_P) = w_1^2\sigma_1^2(R_1) + w_2^2\sigma_2^2(R_2) + w_3^2\sigma_3^2(R_3) + 2w_1w_2Cov(R_1R_2) + 2w_1w_3Cov(R_1R_3) + 2w_2w_3Cov(R_2R_3)$

LO: Calculate and interpret the covariance and correlation of portfolio returns using a joint probability function for returns.

The covariance of portfolio returns can also be estimated using a joint probability function. This value is derived by summing all possible deviation cross-products weighted by the appropriate joint probability.

LO: Define shortfall risk, calculate the safety-first ratio, and identify an optimal portfolio using Roy's safety-first criterion.

Shortfall risk is the risk that portfolio's return will fall below a specified minimum level of return over a given period of time.

Safety first ratio is used to measure shortfall risk. It is calculated as:

$$\text{SF Ratio} = \frac{[R_P - R_L]}{\sigma_P}$$

According to Roy's safety-first criterion, the portfolio with the highest SF-Ratio is preferred, as it has the lowest probability of falling below the target return.

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LM06 Simulation Methods

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1. Introduction

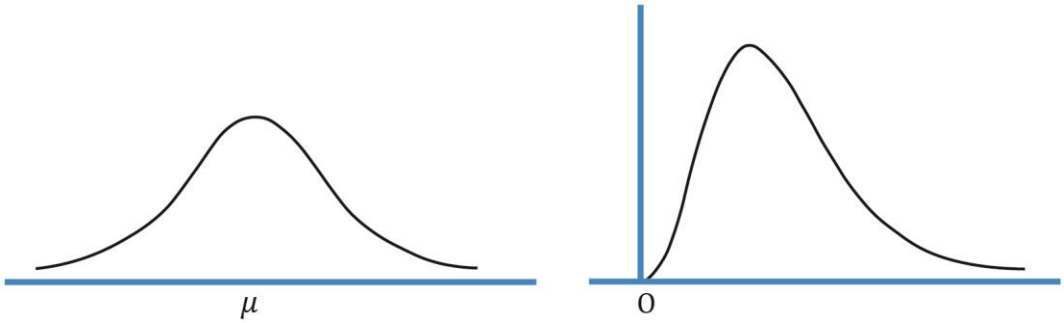
This learning module covers:

- Lognormal distribution and continuously compounded asset return
- Monte Carlo simulation
- Bootstrapping

2. Lognormal Distribution and Continuous Compounding

The Lognormal Distribution

If x is a random variable that is normally distributed, then to create a lognormal distribution of x we take e^x and plot the values on a graph.



Normal Distribution Lognormal Distribution

Instructor's Note: It is called a 'lognormal' distribution because 'the log is normal'.

The properties of a lognormal distribution are:

- It cannot be negative. It is bounded to the left by '0'.
- The upper end of its range extends to infinity.
- It is skewed to the right i.e. it has a long right tail.

Both normal and lognormal distributions are important for finance professionals.

A normal distribution is typically used to model asset returns, because returns generally vary about the mean, with a high probability of returns being close to the mean. Returns can also sometimes be negative.

However, asset prices do not vary equally about a mean price, since the probability of extreme changes in price decreases as the price approaches zero. This means asset prices will not form a symmetrical graph like that of the normal distribution. Instead, asset prices follow a lognormal distribution, which is skewed to the right and cannot be negative.

Continuously Compounded Rates of Return

The continuously compounded rate of return can be calculated as the natural logarithm of the ending price over the beginning price.

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$$R_c = \ln(P_t/P_0)$$

A property of the continuously compounded rate of return is that they are additive. The continuously compounded return from period 0 to period T is the sum of the incremental one-period returns between 0 and T.

$$r_{0,T} = r_{T-1,T} + r_{T-2,T-1} + \cdots + r_{0,1}$$

This is illustrated in the below example.

Example:

(*This is based on Practice Problem 1 from the curriculum.*)

The weekly closing prices of a share are as follows:

Date	Closing Price
1 August	112
8 August	160
15 August	120

Calculate the continuously compounded return of this share for the period August 1 to August 15.

Solution:

The continuously compounded return of an asset over a period is equal to the natural log of the asset's price change during the period. In this case, ln(120/112) = 6.90%.

The continuously compounded return from period 0 to period T is also equal to the sum of the incremental one-period continuously compounded returns, which in this case are weekly returns.

Week 1 return: ln(160/112) = 35.67%.

Week 2 return: ln(120/160) = –28.77%.

Continuously compounded return = 35.67% + –28.77% = 6.90%

Continuously compounded returns are used in many asset pricing models, as well as in risk management.

Many investment applications make the assumption that returns are **independently and identically distributed (i.i.d.**). The concept of independence captures the idea that investors cannot predict future returns based on past returns. The concept of identical distribution captures the assumption of stationarity, which implies that the mean and variance of return do not change form period to period.

Calculating Volatility

The volatility (measured in terms of standard deviation) of the continuously compounded returns on an asset is typically stated as an annualized number. However, in practice volatility is often estimated using a historical series of continuously compounded daily

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return. To convert daily volatility into annual volatility, we annualize based on 250 days in a year (the approximate number of business days that financial markets are open for trading). For example, if daily volatility is 0.01, the annual volatility will be $0.01\sqrt{250} = 0.1581$

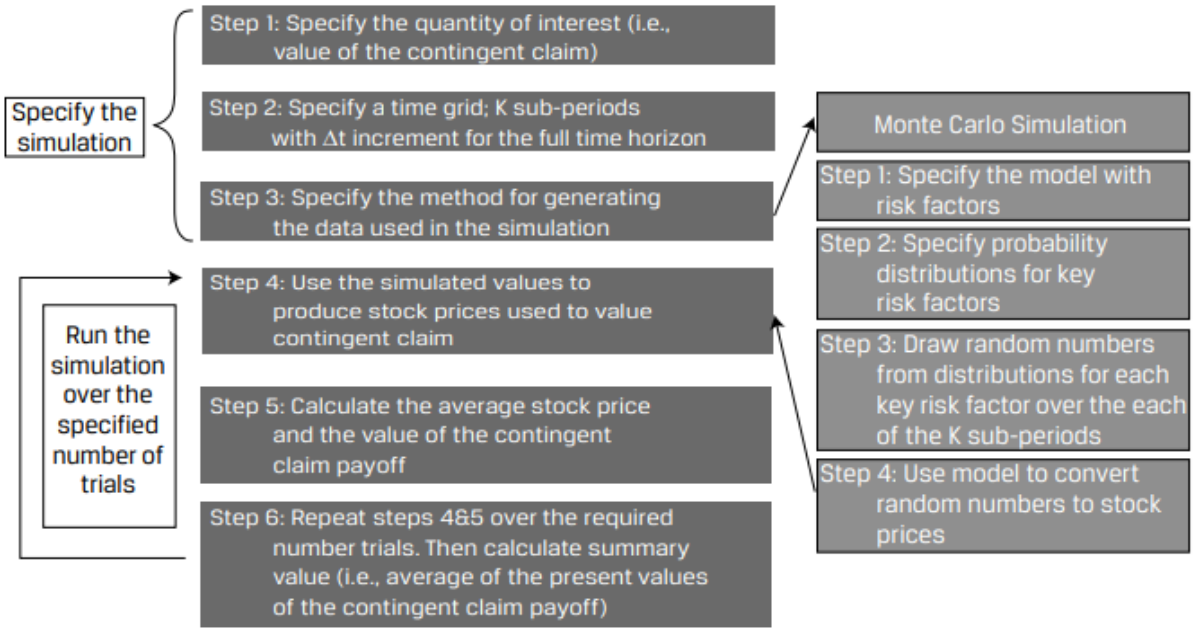
3. Monte Carlo Simulation

Monte Carlo simulation is a computer simulation used to simulate possible security prices based on risk factors. As input, it uses randomly generated values for risk factors based on their assumed probability distributions. It processes this information as per the specified model and runs thousands of iterations. Then it gives the distribution of the expected value of the security as output.

Monte Carlo simulation is widely used to estimate risk and return in investment applications. It is also used as a tool for valuing complex securities with embedded options where no analytic pricing formula is available (e.g. Mortgage backed securities).

A limitation of Monte Carlos simulation is that it is fairly complex and will provide answers that are no better than the assumptions. Also, simulation is not an analytical method but a statistical one. It cannot provide more insight into cause-and-effect relationships like analytical methods.

Exhibit 5 from the curriculum shows the steps for implementing a Monte Carlo simulation for valuing a contingent claim (an option).



4. Bootstrapping

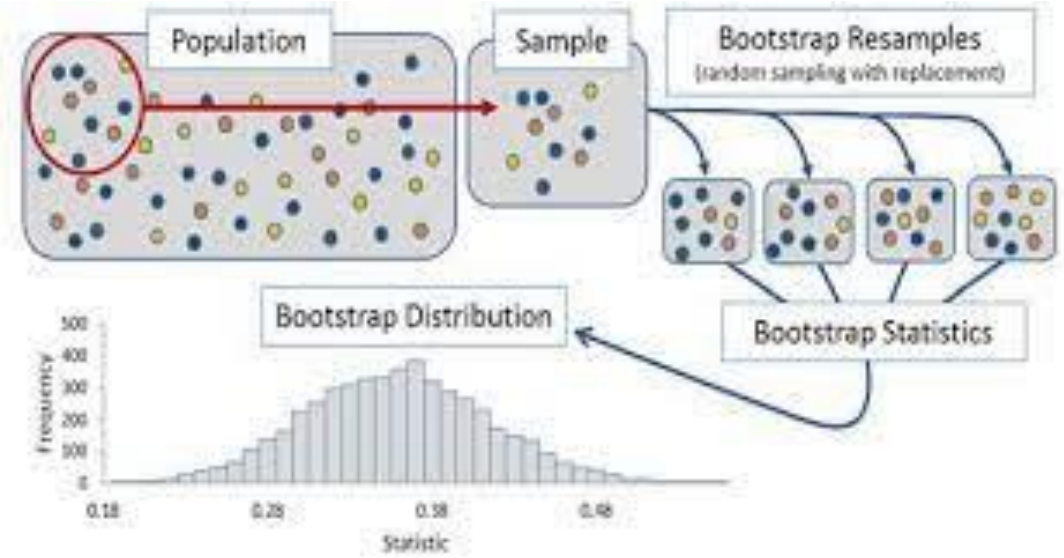
In an earlier lesson, we learnt how to find the standard error of the sample mean based on the central limit theorem. We will now cover 'resampling': a computational tool in which we

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repeatedly draw samples from the original observed data sample for the statistical inference of population parameters. A popular resampling method is bootstrap.

Bootstrap

The bootstrap technique is illustrated in the figure below:



The technique is used when we do not know what the actual population looks like. We simply have a sample of size n drawn from the population. Since the random sample is a good representation of the population, we can simulate sampling from the population by sampling from the observed sample i.e., we treat the randomly drawn sample as if it were the actual population.

Under this technique, samples are constructed by drawing observations from the large sample (of size n) one at a time and returning them to the data sample after they have been chosen. This allows a given observation to be included in a given small sample more than once. This sampling approach is called sampling with replacement.

If we want to calculate the standard error of the sample mean, we take many resamples and calculate the mean of each resample. We then construct a sampling distribution with these resamples. The bootstrap sampling distribution will approximate the true sampling distribution and can be used to estimate the standard error of the sample mean. Similarly, the bootstrap technique can also be used to construct confidence intervals for the statistic or to find other population parameters, such as the median.

Bootstrap is a simple but powerful technique that is particularly useful when no analytical formula is available to estimate the distribution of estimators.

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Summary

LO: Explain the relationship between normal and lognormal distributions and why the lognormal distribution is used to model asset prices when using continuously compounded asset returns.

If x is a random variable that is normally distributed, then to create a lognormal distribution of x we take e^x and plot the values on a graph.

A lognormal distribution is often used to model asset prices because the asset prices need to be positive, they cannot be negative.

LO: Describe Monte Carlo simulation and explain how it can be used in investment applications.

Monte Carlo simulation is a computer simulation that produces a distribution of possible security prices using randomly generated values for risk factors, based on their assumed distributions.

Monte Carlo simulation is widely used to estimate risk and return in investment applications. It is also used as a tool for valuing complex securities with embedded options where no analytic pricing formula is available.

Limitations include:

- It is fairly complex and will provide answers that are no better than the assumptions.
- Simulation is not an analytical method but a statistical one.

LO: Describe the use of bootstrap resampling in conducting a simulation based on observed data in investment applications.

Bootstrap treats a randomly drawn sample as if it were the population. It constructs the sampling distribution of an estimator by repeatedly drawing samples from the original sample with replacement.

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LM07 Estimation and Inference

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1. Introduction

This learning module covers:

- Sampling methods – simple random, stratified random, cluster, convenience, and judgmental sampling
- Central limit theorem, and standard error of the sample mean
- Resampling – bootstrap and jackknife techniques

A sample is a subset of a population. We can study a sample to infer conclusions about the population itself. For example, if all the stocks trading in the US are considered a population, then indices such as the S&P 500 are samples. We can look at the performance of the S&P 500 and draw conclusions about how all stocks in the US are performing. This process is known as sampling and estimation.

2. Sampling Methods

There are various methods for obtaining information on a population through samples. The information we obtain usually concerns a **parameter**, a quantity used to describe a population. To estimate a parameter, we use sample statistics. A **statistic** is a quantity used to describe a sample.

There are two reasons why sampling is used:

- Time saving: In many cases it will be very time consuming to examine every member of the population.
- Monetary saving: In some cases, examining every member of the population becomes economically inefficient.

There are two types of sampling methods:

- Probability sampling: Every member of the population has an equal chance of being selected. Therefore, the sample created is representative of the population.
- Non-probability sampling: Every member of the population may not have an equal chance of being selected. This is because sampling depends on factors such as the sampler's judgement or the convenience to access data. Therefore, the sample created may not be representative of the population.

All else equal, the probability sampling method is more accurate and reliable as compared to the non-probability sampling method.

In the subsequent sections, we will discuss the following sampling methods:

- Probability sampling
  - Simple random sampling
  - Systematic sampling
  - Stratified random sampling

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  - Cluster sampling

- Non-probability sampling
  - Convenience sampling
  - Judgement sampling

Simple Random Sampling

Simple random sampling is the process of selecting a sample from a larger population in such a way that each member of the population has the same probability of being included in the sample.

Sampling distribution

If we draw samples of the same size several times and calculate the sample statistic, the sample statistic will be different each time. The distribution of values of the sample statistic is called a sampling distribution.

For example, say you select 100 stocks from a universe of 10,000 stocks and calculate the average annual returns of these 100 stocks. Let's say you get an average return of 15%. You repeat this process with a second sample of 100 stocks. This time, you get an average return of 14%. You keep repeating this process and each time you get a different average return. The distribution of these sample average returns is called a sampling distribution.

Sampling error

Sampling error is the difference between a sample statistic and the corresponding population parameter.

The sampling error of the mean is given by:

$$\text{Sampling error of the mean} = \bar{x} - \mu$$

For example, let's say you want to estimate the average returns of 10,000 stocks. You draw a sample of 100 stocks and calculate the average return of these 100 stocks as 15%. However, the actual average of the 10,000 stocks was 12%. Then the sampling error = 15% - 12% = 3%.

Systematic sampling: In this technique, we select every kth member of the population until we have a sample of the desired size. Samples created using this technique should be approximately random.

Instructor's Note: Researchers calculate the sampling interval 'k' by dividing the entire population size by the desired sample size.

Stratified Random Sampling

In stratified random sampling, the population is divided into subgroups based on one or more distinguishing characteristics. Samples are then drawn from each subgroup, with

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sample size proportional to the size of the subgroup relative to the population. Finally, samples from each subgroup are pooled together to form a stratified random sample.

The advantage of stratified random sampling is that the sample will have the same distribution of key characteristics as the overall population. This can help reduce the sampling error. Stratified random sampling therefore produces more precise parameter estimates than simple random sampling

For example, you divide the universe of 10,000 stocks as per their market capitalization such that you have 5,000 large cap stocks, 3,000 mid cap stocks, and 2,000 small cap stocks. In stratified random sampling, to select a total sample of 100 stocks, you will randomly select 50 large cap stocks, 30 mid cap stocks, and 20 small cap stocks and pool all these samples together to form a stratified random sample.

Example

Paul wants to categorize publicly listed stocks for his research project. He first divides the stocks into 15 industries. Then from each industry, he categorizes companies into three groups: small, medium, large. Finally, he divides these into value versus growth stocks. How many cells or strata does the sampling plan entail?

A. 20
B. 45
C. 90

Solution:

C is correct. This is an application of the multiplication rule of counting. The total number of cells is the product of 15, 3, and 2. Thus the answer is 90.

Cluster Sampling

Cluster sampling is similar to stratified random sampling as it also requires the population to be divided into subpopulation groups, called clusters. Each cluster is essentially a mini-representation of the entire population. Then some random clusters are chosen as a whole for sampling.

Instructor's Note: Clusters are generally based on natural groups separating the population. For example, you might be able to divide your data into natural groupings like city blocks, voting districts, or school districts.

The main difference between cluster sampling and stratified random sampling is that in cluster sampling, the whole cluster is selected; and not all clusters are included in the sample. In stratified random sampling, however, only a few members from each stratum are selected; but all strata are included in the sample.

The difference between simple random sampling, stratified random sampling, and cluster sampling is illustrated in the figure below:

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As compared to SRS and stratified sampling, cluster sampling is less accurate because the chosen sample may be less representative of the entire population. However, this method is the most time-efficient and cost-efficient amongst the three.

Non-Probability Sampling

The two major types of non-probability sampling methods are:

- Convenience sampling: In this method, the researcher selects members from a population based on how easy it is to access the member i.e., data is collected from a conveniently available pool of respondents. The disadvantage of this method is that the sample selected may not be representative of the entire population. The advantage is that data can be collected quickly and at a low cost. Hence this method is particularly suitable for small-scale pilot studies.
- Judgmental sampling: In this method, the researcher uses his knowledge and professional judgment to selectively handpick members from the population. The disadvantage of this method is that the sampling may be impacted by the researcher's bias and the results may be skewed. The advantage of this method is that it allows the researcher to directly go to the target population of interest. For example, when auditing financial statements, experienced auditors can use their professional judgment to select important accounts or transactions that can provide sufficient audit coverage.

Sampling from Different Distributions

Instructor's Note: This section does not contain testable concepts. The core point is presented below.

In addition to selecting an appropriate sampling method, researchers also need to be careful when sampling from a population that is not under one single distribution. In such cases, the larger population should be divided into smaller parts, and samples should be drawn from the smaller parts separately.

3. Central Limit Theorem and Inference

The sample mean is a random variable with a probability distribution known as the statistic's sampling distribution. To understand this concept, consider the following population: last year's returns on every stock traded in the United States. We are interested in the mean return of all stocks but do not have time to calculate the population mean.

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Hence, we draw a sample of 50 stocks and compute the sample mean. We then draw another sample of 50 stocks and compute the sample mean. This exercise can be repeated several times giving us a distribution of sample means. This distribution is called the statistic's sampling distribution. The central limit theorem, explained below, helps us understand the sampling distribution of the mean.

The Central Limit Theorem

According to the central limit theorem, if we draw a sample from a population with a mean μ and a variance σ^2, then the sampling distribution of the sample mean:

- will be normally distributed (irrespective of the type of distribution of the original population).
- will have a mean of μ.
- will have a variance of σ^2/n.

For example, suppose the average return of the universe of 10,000 stocks is 12% and its standard deviation is 10%. Through central limit theorem we can conclude that if we keep drawing samples of 100 stocks and plot their average returns, we will get a sampling distribution that will be normally distributed with mean = 12% and variance of $10^2/100$ = 1%.

Standard Error of the Sample Mean

The standard deviation of the distribution of the sample means is known as the standard error of the sample mean.

When we know the population standard deviation, the standard error of the sample mean can be calculated as:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

When we do not know the population standard deviation (σ) we can use the sample standard deviation (s) to estimate the standard error of the sample mean:

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Example

The mean of a population is 12 and the standard deviation is 3. Given that the population comprises of 64 observations, what is the standard error of the sample mean?
A. 0.375
B. 0.378
C. 0.667

Solution:

A is correct. Standard Error = $\sigma/\sqrt{n} = 3/\sqrt{64} = 0.375$

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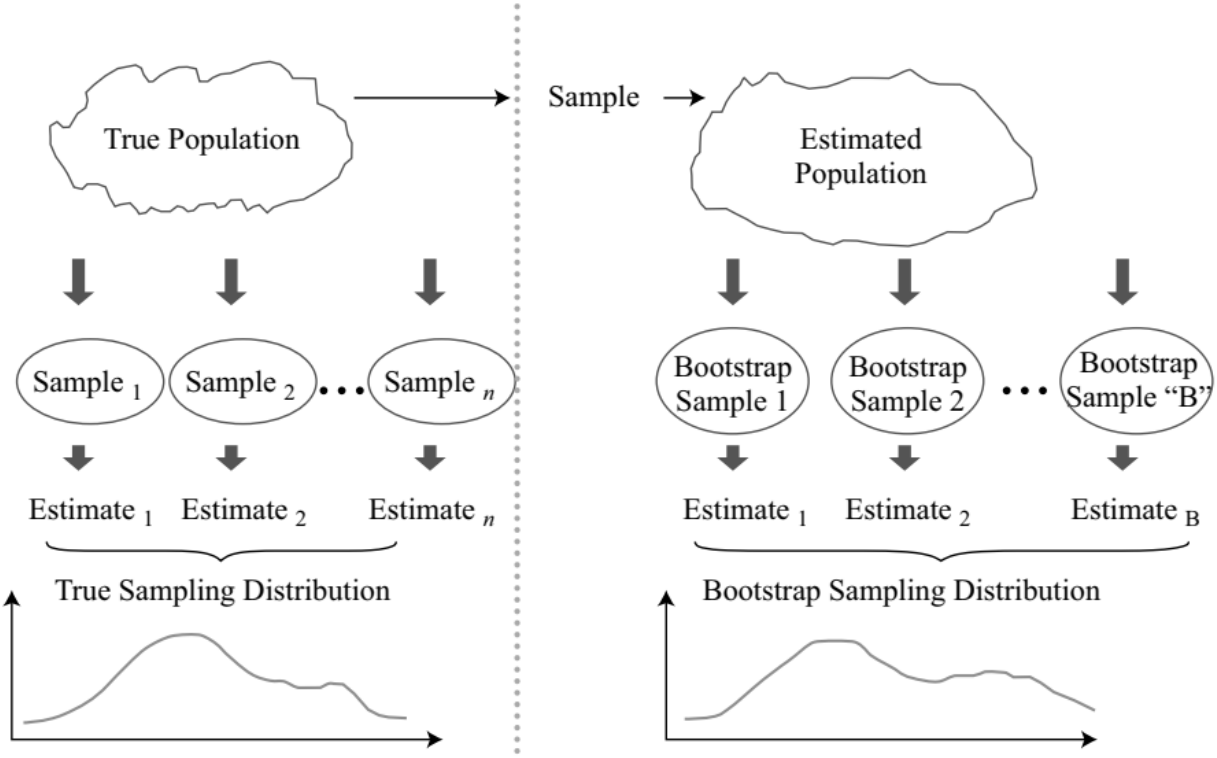
4. Bootstrapping and Empirical Sampling Distributions

Resampling is a computational tool in which we repeatedly draw samples from the original observed data sample for the statistical inference of population parameters. Two popular resampling methods are:

- Bootstrap
- Jackknife

Bootstrap

The bootstrap technique is illustrated in the figure below:



The technique is used when we do not know what the actual population looks like. We simply have a sample of size n drawn from the population. Since the random sample is a good representation of the population, we can simulate sampling from the population by sampling from the observed sample i.e., we treat the randomly drawn sample as if it were the actual population.

Under this technique, samples are constructed by drawing observations from the large sample (of size n) one at a time and returning them to the data sample after they have been chosen. This allows a given observation to be included in a given small sample more than once. This sampling approach is called sampling with replacement.

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If we want to calculate the standard error of the sample mean, we take many resamples and calculate the mean of each resample. We then construct a sampling distribution with these resamples. The bootstrap sampling distribution will approximate the true sampling distribution and can be used to estimate the standard error of the sample mean. Similarly, the bootstrap technique can also be used to construct confidence intervals for the statistic or to find other population parameters, such as the median.

Bootstrap is a simple but powerful technique that is particularly useful when no analytical formula is available to estimate the distribution of estimators.

Jackknife

In the Jackknife technique we start with the original observed data sample. Subsequent samples are then created by leaving out one observation at a time from the set (and not replacing it). Thus, for a sample of size n, jackknife usually requires n repetitions. The Jackknife method is frequently used to reduce the bias of an estimator.

Note that bootstrap differs from jackknife in two ways:

- Bootstrap repeatedly draws samples with full replacement.
- The researcher has to decide how many repetitions are appropriate.

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Summary

LO: Compare and contrast simple random, stratified random, cluster, convenience, and judgmental sampling and their implications for sampling error in an investment problem.

Probability sampling: Every member of the population has an equal chance of being selected. Therefore, the sample created is representative of the population.

Non-probability sampling: Every member of the population may not have an equal chance of being selected. This is because sampling depends on factors such as the sampler's judgement or the convenience to access data. Therefore, the sample created may not be representative of the population.

All else equal, the probability sampling method is more accurate and reliable as compared to the non-probability sampling method.

Simple random sampling is the process of selecting a sample from a larger population in such a way that each element of the population has the same probability of being included in the sample.

In stratified random sampling, the population is divided into subgroups based on one or more distinguishing characteristics. Samples are then drawn from each subgroup, with sample size proportional to the size of the subgroup relative to the population. Finally, samples from each subgroup are pooled together to form a stratified random sample.

Cluster sampling is similar to stratified random sampling as it also requires the population to be divided into subpopulation groups, called clusters. Each cluster is essentially a mini-representation of the entire population. Then some random clusters are chosen as a whole for sampling.

In convenience sampling, the researcher selects members from a population based on how easy it is to access the member i.e., data is collected from a conveniently available pool of respondents

In judgmental sampling, the researcher uses his knowledge and professional judgment to selectively handpick members from the population.

LO: Explain the central limit theorem and its importance for the distribution and standard error of the sample mean.

According to the central limit theorem, if we draw a sample from a population with a mean μ and a variance σ^2, then the sampling distribution of the sample mean:

- will be normally distributed (irrespective of the type of distribution of the original population).
- will have a mean of μ.
- will have a variance of σ^2/n.

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The standard deviation of the distribution of the sample means is known as the standard error of the sample mean.

When we know the population standard deviation, the standard error of the sample mean can be calculated as: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$

When we do not know the population standard deviation (σ) we can use the sample standard deviation (s) to estimate the standard error of the sample mean $s_{\bar{X}} = \frac{s}{\sqrt{n}}$

LO: Describe the use of resampling (bootstrap, jackknife) to estimate the sampling distribution of a statistic.

Resampling is a computational tool in which we repeatedly draw samples from the original observed data sample for the statistical inference of population parameters. Two popular resampling methods are:

- Bootstrap: Constructs the sampling distribution of an estimator by repeatedly drawing samples from the original sample with replacement.
- Jackknife: Draws repeated samples while leaving out one observation at a time from the set, without replacing it.

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LM08 Hypothesis Testing

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1. Introduction

This learning module covers:

- Hypothesis testing process
- Impact of errors in the hypothesis testing process
- Parametric and nonparametric tests

2. Hypothesis Tests for Finance

Hypothesis testing is the process of making judgments about a larger group (a population) on the basis of observing a smaller group (a sample). The results of such a test then help us evaluate whether our hypothesis is true or false.

For example, let's say you are a researcher and you believe that the average return on all Asian stocks was greater than 2%. To test this belief, you can draw samples from a population of all Asian stocks and employ hypothesis testing procedures. The results of this test can tell you if your belief is statistically valid.

In this learning module we will look at hypothesis tests concerning the mean and variance.

The Process of Hypothesis Testing

A hypothesis is defined as a statement about one or more populations. In order to test a hypothesis, we follow these steps:

1. State the hypothesis.
2. Identify the appropriate test statistic and its probability distribution.
3. Specify the significance level.
4. State the decision rule.
5. Collect data and calculate the test statistic.
6. Make a decision.

1. Stating the Hypothesis

For each hypothesis test, we always state two hypotheses: the null hypothesis (H_0) and the alternative hypothesis (H_a).

Null hypothesis (H_0): It is the hypothesis that the researcher wants to reject.

Alternative hypothesis (H_a)**:** It is the hypothesis that the researcher wants to prove. If the null hypothesis is rejected then the alternative hypothesis is considered valid.

Suppose you are a researcher and believe that the average return on all Asian stocks was greater than 2%. In this case, you are making a statement about the population mean (μ) of all Asian stocks.

For this example, the null and alternative hypotheses are:

$H_0: \mu \leq 2\%$

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H_a: $\mu > 2\%$

(The value 2% is known as μ_0, the hypothesized value of the population mean.)

Instructor's Note:
An easy way to differentiate between the two hypotheses is to remember that the null hypothesis always contains some form of the equal sign.

Two-Sided vs. One-Sided Hypotheses

The alternative hypothesis can be one-sided or two-sided depending on the proposition being tested. A one-sided test is also called a one-tailed test, and a two-sided test is also called a two-tailed test.

If we want to determine whether the estimated value of a population parameter is less than (or greater than) a hypothesized value we use a one-tailed test. However, if we want to determine whether the estimated value of a population parameter is different than a hypothesized value, we use a two tailed test.

Two-sided test: Suppose we want to test if the population mean is equal to 2%. The null and alternative hypothesis can be expressed as:

H_0: $\mu = 2\%$

H_a: $\mu \neq 2\%$

Since the alternative hypothesis contains a ≠ sign this is a two-sided test.

One-sided test (right side): Suppose we want to test if the population mean is greater than 2%. The null and alternative hypothesis can be expressed as:

H_0: $\mu \leq 2\%$

H_a: $\mu > 2\%$

Since the alternative hypothesis contains a > sign this is a one-sided test, and we are interested in the right side.

Instructor's Note
The sign in the alternative hypothesis points to the direction of the tail that we should use in our test. Since in our example the alternative hypothesis has a '>' sign it points to the right, therefore we are interested in the right tail.

One-sided test (left side): Suppose we want to test if the population mean is less than 2%. The null and alternative hypothesis can be expressed as:

H_0: $\mu \geq 2\%$

H_a: $\mu < 2\%$

Since the alternative hypothesis contains a < sign this is a one-sided test, and we are interested in the left side.

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Selecting the Appropriate Hypothesis

The easiest approach is to specify the alternative hypothesis first and then specify the null. Using a '<' or '>' sign in the alternative hypothesis instead of a '≠' sign reflects that belief of the researcher more strongly. However, a researcher may sometimes select a two-sided test to depict neutrality in his beliefs.

2. Identify the Appropriate Test Statistic

A test statistic is calculated from sample data and is compared to a critical value to decide whether or not we can reject the null hypothesis. The test statistic that should be used depends on what we are testing. For example, the test statistic for the test of a population mean is calculated as:

$$\text{test statistic} = \frac{\text{sample statistic} - \text{value of the parameter under } H_0}{\text{standard error of the sample statistic}} = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Continuing with our Asian stocks example, suppose we want to test if the population mean is greater than a particular hypothesized value. We draw 36 observations and get a sample mean of 4. We are also told that the standard deviation of the population is 4. If the hypothesized value of the population mean is 2, the test statistic is calculated as:

$$\text{test statistic} = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{4 - 2}{\frac{4}{\sqrt{36}}} = 3$$

However, if the hypothesized value of the population mean is 0, then the test statistic is calculated as:

$$\text{test statistic} = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{4 - 0}{\frac{4}{\sqrt{36}}} = 6$$

Identifying the Distribution of the Test Statistic

Exhibit 2 from the curriculum shows which test-statistics should be used depending on what we want to test and their corresponding distributions.

Instructor's Note: You will understand this table better, after you finish reading the remaining sections.

What We Want to Test	Test Statistic	Probability Distribution of the Statistic	Degrees of Freedom
Test of a single mean	$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$	*t*-Distributed	$n - 1$

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Test of the difference in means	$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}}$	*t*-Distributed	$n_1 + n_2 - 2$
Test of the mean of differences	$t = \frac{\bar{d} - \mu_{d0}}{s_{\bar{d}}}$	*t*-Distributed	$n - 1$
Test of a single variance	$\chi^2 = \frac{s^2(n-1)}{\sigma_0^2}$	Chi-square distributed	$n - 1$
Test of the difference in variances	$F = \frac{s_1^2}{s_2^2}$	*F*-distributed	$n_1 - 1, n_2 - 1$
Test of a correlation	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	*t*-Distributed	$n - 2$
Test of independence (categorical data)	$\chi^2 = \sum_{i=1}^{m} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$	Chi-square distributed	$(r - 1)(c - 1)$

3. Specify the Level of Significance

In reaching a statistical decision, we can make two possible errors:

- Type I error: We may reject a true null hypothesis.
- Type II error: We may fail to reject a false null hypothesis.

The following table shows the possible outcomes of a test.

Decision	**True condition**	
	H_0 true	**H_0 false**
Do not reject H_0	Correct decision	Type II error
Reject H_0 (accept H_a)	Type I error	Correct decision

The probability of a Type I error is also known as the **level of significance** of a test and is denoted by 'α'. A related term, **confidence level** is calculated as (1 - α). For example, a level of significance of 5% for a test means that there is a 5% probability of rejecting a true null hypothesis and corresponds to the 95% confidence level.

Controlling the two types of errors involves a trade-off. If we decrease the probability of a Type I error by specifying a smaller significance level (for e.g., 1% instead of 5%), we increase the probability of a Type II error. The only way to reduce both types of error simultaneously is by increasing the sample size, n.

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The probability of a Type II error is denoted by 'β'. The **power of test** is calculated as (1 - β). It represents the probability of correctly rejecting the null when it is false.

The different probabilities associated with the hypothesis testing decisions are presented in the table below:

Decision	**True condition**	
	H_0 true	**H_0 false**
Do not reject H_0	Confidence level (1 - α)	β
Reject H_0 (accept H_a)	Level of significance (α)	Power of the test (1 - β)

The most commonly used levels of significance are: 10%, 5% and 1%.

4. State the Decision Rule

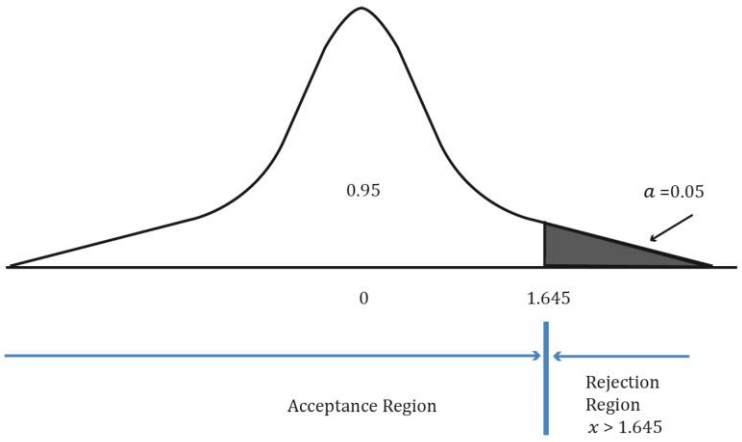
A decision rule involves determining the critical values based on the level of significance; and comparing the test statistic with the critical values to decide whether to reject or not reject the null hypothesis. When we reject the null hypothesis, the result is said to be statistically significant.

Determining Critical Values

<u>One-tailed test</u>:

Continuing with our Asian stocks example, suppose we want to test if the population mean is greater than 2%. Say we want to test our hypothesis at the 5% significance level. This is a one-tailed test and we are only interested in the right tail of the distribution. (If we were trying to assess whether the population mean is less than 2%, we would be interested in the left tail.)

The critical value is also known as the **rejection point** for the test statistic. Graphically, this point separates the acceptance and rejection regions for a set of values of the test statistic. This is shown below:



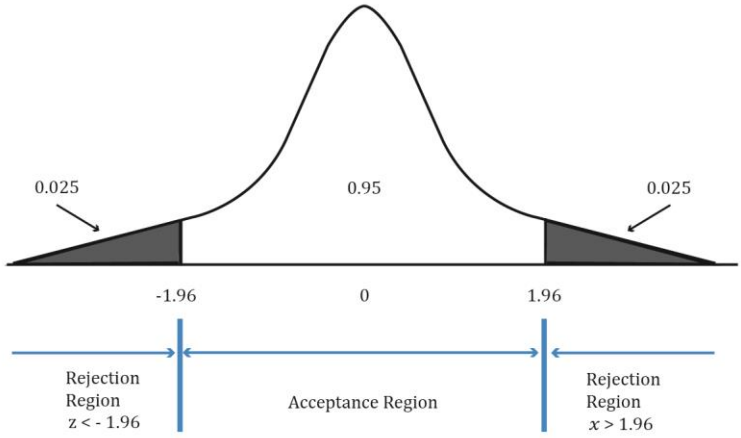
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The region to the left of the test statistic is the 'acceptance region'. This represents the set of values for which we do not reject (accept) the null hypothesis. The region to the right of the test statistic is known as the 'rejection region'.

Using the Z –table and 5% level of significance, the critical value = $Z_{0.05}$= 1.645

Two-tailed test:

In a two-tailed test, two critical values exist, one positive and one negative. For a two-sided test at the 5% level of significance, we split the level of significance equally between the left and right tail i.e. $\frac{0.05}{2}$ = 0.025 in each tail.
This corresponds to rejection points of +1.96 and -1.96. Therefore, we reject the null hypothesis if we find that the test statistic is less than -1.96 or greater than +1.96. We fail to reject the null hypothesis if -1.96 ≤ test statistic ≤ +1.96. Graphically, this can be shown as:



5. Collect Data and Calculate the Test Statistic

In this step we first ensure that the sampling procedure does not include biases, such as sample selection or time bias. Then, we cleanse the data by removing inaccuracies and other measurement errors in the data. Once we are convinced that the sample data is unbiased and accurate, we use it to calculate the appropriate test statistic.

6. Make a Decision

A statistical decision simply consists of rejecting or not rejecting the null hypothesis. If the test statistic lies in the rejection region, we will reject H_0. On the other hand, if the test statistic lies in the acceptance region, then we cannot reject H_0.

An economic decision takes into consideration all economic issues relevant to the decision, such as transaction costs, risk tolerance, and the impact on the existing portfolio. Sometimes a test may indicate a result that is statistically significant, but it may not be economically significant.

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The Role of p-Values

The p-value is the smallest level of significance at which the null hypothesis can be rejected. It can be used in the hypothesis testing framework as an alternative to using rejection points.

- If the p-value is lower than our specified level of significance, we reject the null hypothesis.
- If the p-value is greater than our specified level of significance, we do not reject the null hypothesis.

For example, if the p-value of a test is 4%, then the hypothesis can be rejected at the 5% level of significance, but not at the 1% level of significance.

Relationship between test-statistic and p-value

A high test-statistic implies a low p-value.

A low test-statistic implies a high p-value.

3. Tests of Return and Risk in Finance

Tests Concerning a Single Mean

One of the decisions we need to make in hypothesis testing is deciding which test statistic and which corresponding probability distribution to use. We use the following table to make this decision:

Sampling from		Small sample size (n<30)	Large sample size (n≥30)
Normal distribution	Variance known	z	z
	Variance unknown	t	t (or z)
Non –normal distribution	Variance known	NA	z
	Variance unknown	NA	t (or z)

If the population variance is known and our sample size is large, we can use the z-statistic and z-distribution to compute the critical value. However, if we do not know the population variance and we have a small sample size, then we have to use the t-statistic and t-distribution to compute the critical values.

Example

An analyst believes that the average return on all Asian stocks was *less* than 2%. The sample size is 36 observations with a sample mean of -3. The standard deviation of the population is 4. Will he reject the null hypothesis at the 5% level of significance?

Solution:

In this case, our null and alternative hypotheses are:

H_0: $\mu \geq 2$

H_a: $\mu < 2$

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The standard error of the sample is: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{36}} = 0.67$

The test statistic is:

$$\text{test statistic} = \frac{\text{sample statistic} - \text{value of the parameter under } H_0}{\text{standard error of the sample statistic}} = \frac{-3-2}{0.67} = -7.5$$

The critical values corresponding to a 5% level of significance is -1.645.

When we consider the left tail of the distribution, our decision rule is as follows: Reject the null hypothesis if the test statistic is less than the critical value and vice versa. Since our calculated test statistic of -7.5 is less than the critical value of -1.645, we reject the null hypothesis.

Example

Fund Alpha has been in existence for 20 months and has achieved a mean monthly return of 2% with a sample standard deviation of 5%. The expected monthly return for a fund of this nature is 1.60%. Assuming monthly returns are normally distributed, are the actual results consistent with an underlying population mean monthly return of 1.60%?

Solution:

The null and alternative hypotheses for this example will be:

H_0: $\mu = 1.60$ versus H_a: $\mu \neq 1.60$

$$\text{test statistic} = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{2 - 1.60}{\frac{5}{\sqrt{20}}} = 0.36$$

Using this formula, we see that the value of the test statistic is 0.36.

The critical values at a 0.05 level of significance can be calculated from the t-distribution table. Since this is a two-tailed test, we should look at a 0.05/2 = 0.025 level of significance with df = n - 1 = 20 – 1 = 19. This gives us two values of -2.1 and +2.1.

Since our test statistic of 0.35 lies between -2.1 and +2.1, i.e., the acceptance region, we do not reject the null hypothesis.

Test Concerning Difference Between Means with Independent Samples

Instructor's Note:

Focus on the basics of this topic, the probability of being tested on the details is low.

In this section, we will learn how to calculate the difference between the means of two independent and normally distributed populations. We perform this test by drawing a sample from each group. If it is reasonable to believe that the samples are normally distributed and also independent of each other, we can proceed with the test. We may also assume that the population variances are equal or unequal. However, the curriculum focuses on tests under the assumption that the population variances are equal.

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The test statistic is calculated as:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{(\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2})^{1/2}}$$

The term s_p^2 is known as the pooled estimator of the common variance. It is calculated by the following formula:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

The number of degrees of freedom is $n_1 + n_2 - 2$.

Example

(*This is based on Example 1 from the curriculum.*)

An analyst wants to test if the returns for an index are different for two different time periods. He gathers the following data:

	Period 1	**Period 2**
Mean	0.01775%	0.01134%
Standard deviation	0.31580%	0.38760%
Sample size	445 days	859 days

Note that these periods are of different lengths and the samples are independent; that is, there is no pairing of the days for the two periods.

Test whether there is a difference between the mean daily returns in Period 1 and in Period 2 using a 5% level of significance.

Solution:

The first step is to formulate the null and alternative hypotheses. Since we want to test whether the two means were equal or different, we define the hypotheses as:

H_0: $\mu_1 - \mu_2 = 0$

H_a: $\mu_1 - \mu_2 \neq 0$

We then calculate the test statistic:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(445 - 1)0.09973 + (859 - 1)0.15023}{445 + 859 - 2} = 0.1330$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{(\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2})^{1/2}} = \frac{(0.01775 - 0.01134) - 0}{(\frac{0.1330}{445} + \frac{0.1330}{859})^{1/2}} = 0.3099$$

For a 0.05 level of significance, we find the t-value for 0.05/2 = 0.025 using df = 445 + 859 - 2=1302. The critical t-values are ±1.962. Since our test statistic of 0.3099 lies in the acceptance region, we fail to reject the null hypothesis.

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We conclude that there is insufficient evidence to indicate that the returns are different for the two time periods.

Test Concerning Differences between Means with Dependent Samples

Instructor's Note:
Focus on the basics of this topic, the probability of being tested on the details is low.

In the previous section, in order to perform hypothesis tests on differences between means of two populations, we assumed that the samples were independent. What if the samples are not independent? For example, suppose you want to conduct tests on the mean monthly return on Toyota stock and mean monthly return on Honda stock. These two samples are believed to be dependent, as they are impacted by the same economic factors.

In such situations, we conduct a t-test that is based on data arranged in **paired observations**. Paired observations are observations that are dependent because they have something in common.

We will now discuss the process for conducting such a t-test.

Example:

Suppose that we gather data regarding the mean monthly returns on stocks of Toyota and Honda for the last 20 months, as shown in the table below:

Month	Mean return of Toyota stock	Mean monthly return of Honda stock	Difference in mean monthly returns (d_i)
1	0.5%	0.4%	0.1%
2	0.7%	1.0%	-0.3%
3	0.3%	0.7%	-0.4%
…	…	…	…
20	0.9%	0.6%	0.3%
Average	0.750%	0.600%	**0.075%**

Here is a simplified process for conducting the hypothesis test:

Step 1: Define the null and alternate hypotheses

We believe that the mean difference is not 0. Hence the null and alternate hypotheses are:

$H_0: \mu_d = \mu_{d0}$ versus $H_a: \mu_d \neq \mu_{d0}$

μ_d stands for the population mean difference and μ_{d0} stands for the hypothesized value for the population mean difference.

Step 2: Calculate the test-statistic

Determine the sample mean difference using:

$$\bar{d} = \frac{1}{n}\sum_{i=0}^{n} d_i$$

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For the data given, the sample mean difference is 0.075%.

Calculate the sample standard deviation. The process for calculating the sample standard deviation has been discussed in an earlier reading. The simplest method is to plug the numbers (0.1, -0.3, -0.4…0.3) into a financial calculator. The entire data set has not been provided. We'll take it as a given that the sample standard deviation is 0.150%.

Use this formula to calculate the standard error of the mean difference:

$$s_{\bar{d}} = \frac{s_d}{\sqrt{n}}$$

For our data this is 0.150 / $\sqrt{20}$ = 0.03354.

We now have the required data to calculate the test statistic using a t-test. This is calculated using the following formula using n - 1 degrees of freedom:

$$t = \frac{\bar{d} - \mu_{d0}}{s_{\bar{d}}}$$

For our data, the test statistic is $\frac{0.075 - 0}{0.03354} = 2.24$.

Step 3: Determine the critical value based on the level of significance

We will use a 5% level of significance. Since this is a two-tailed test we have a probability of 2.5% (0.025) in each tail. This critical value is determined from a t-table using a one-tailed probability of 0.025 and df = 20 – 1 = 19. This value is 2.093.

Step 4: Compare the test statistic with the critical value and make a decision

In our case, the test statistic (2.23) is greater than the critical value (2.093). Hence we will reject the null hypothesis.

Conclusion: The data seems to indicate that the mean difference is not 0.

The hypothesis test presented above is based on the belief that the population mean difference is not equal to 0. If μ_{d0} is the hypothesized value for the population mean difference, then we can formulate the following hypotheses:

1. If we believe the population mean difference is greater than 0:
   $H_0: \mu_d \leq \mu_{d0}$ versus $H_a: \mu_d > \mu_{d0}$
2. If we believe the population mean difference is less than 0:
   $H_0: \mu_d \geq \mu_{d0}$ versus $H_a: \mu_d < \mu_{d0}$
3. If we believe the population mean difference is not 0:
   $H_0: \mu_d = \mu_{d0}$ versus $H_a: \mu_d \neq \mu_{d0}$

Test Concerning a Single Variance

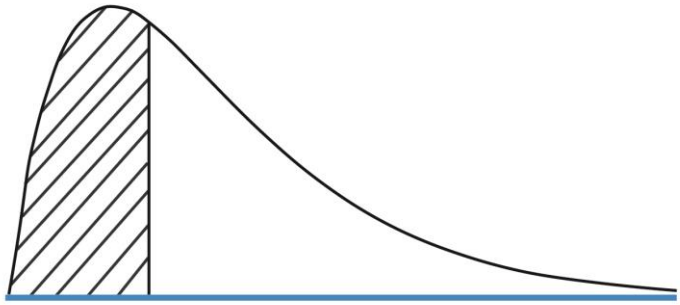
Instructor's Note:
Focus on the basics of this topic, the probability of being tested on the details is low.

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In tests concerning the variance of a single normally distributed population, we use the chi-square test statistic, denoted by χ^2.

Properties of the chi-square distribution

The chi-square distribution is asymmetrical and like the t-distribution, is a family of distributions. This means that a different distribution exists for each possible value of degrees of freedom, n - 1. Since the variance is a squared term, the minimum value can only be 0. Hence, the chi-square distribution is bounded below by 0. The graph below shows the shape of a chi-square distribution:



There are three hypotheses that can be formulated (σ^2 represents the true population variance and σ_0^2 represents the hypothesized variance):

1. $H_0: \sigma^2 = \sigma_0^2$ versus $H_a: \sigma^2 \neq \sigma_0^2$. This is used when we believe the population variance is not equal to 0, or it is different from the hypothesized variance. It is a two-tailed test.
2. $H_0: \sigma^2 \geq \sigma_0^2$ versus $H_a: \sigma^2 < \sigma_0^2$. This is used when we believe the population variance is less than the hypothesized variance. It is a one-tailed test.
3. $H_0: \sigma^2 \leq \sigma_0^2$ versus $H_a: \sigma^2 > \sigma_0^2$. This is used when we believe the population variance is greater than the hypothesized variance. It is a one-tailed test.

After drawing a random sample from a normally distributed population, we calculate the test statistic using the following formula using n - 1 degrees of freedom:

$$\chi^2 = \frac{(n-1)(s^2)}{\sigma_0^2}$$

where:
n = sample size
s = sample variance

We then determine the critical values using the level of significance and degrees of freedom. The chi-square distribution table is used to calculate the critical value.

Example

Consider Fund Alpha which we discussed in an earlier example. This fund has been in existence for 20 months. During this period the standard deviation of monthly returns was

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5%. You want to test a claim by the fund manager that the standard deviation of monthly returns is less than 6%.

Solution:
The null and alternate hypotheses are: H_0: $\sigma^2 \geq 36$ versus H_a: $\sigma^2 < 36$

Note that the standard deviation is 6%. Since we are dealing with population variance, we will square this number to arrive at a variance of 36%.

We then calculate the value of the chi-square test statistic:

$\chi^2 = (n - 1) s^2 / \sigma_0^2 = 19 \times 25/36 = 13.19$

Next, we determine the rejection point based on df = 19 and significance = 0.05. Using the chi-square table, we find that this number is 10.117.

Since the test statistic (13.19) is higher than the rejection point (10.117) we cannot reject H_0. In other words, the sample standard deviation is not small enough to validate the fund manager's claim that population standard deviation is less than 6%.

Test Concerning the Equality of Two Variances

Instructor's Note:
Focus on the basics of this topic, the probability of being tested on the details is low.

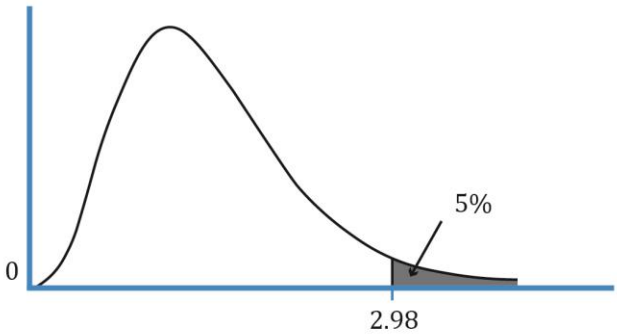
In order to test the equality or inequality of two variances, we use an F-test which is the ratio of sample variances.

The assumptions for a F-test to be valid are:

- The samples must be independent.
- The populations from which the samples are taken are normally distributed.

Properties of the F-distribution

The F-distribution, like the chi-square distribution, is a family of asymmetrical distributions bounded from below by 0. Each F-distribution is defined by two values of degrees of freedom, called the numerator and denominator degrees of freedom. As shown in the figure below, the F-distribution is skewed to the right and is truncated at zero on the left hand side.



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The rejection region is always in the right-side tail of the distribution.

When working with F-tests, there are three hypotheses that can be formulated:

1. $H_0: \sigma_1^2 = \sigma_2^2$ versus $H_a: \sigma_1^2 \neq \sigma_2^2$. This is used when we believe the two population variances are not equal.
2. $H_0: \sigma_1^2 \leq \sigma_2^2$ versus $H_a: \sigma_1^2 > \sigma_2^2$. This is used when we believe the variance of the first population is greater than the variance of the second population.
3. $H_0: \sigma_1^2 \geq \sigma_2^2$ versus $H_a: \sigma_1^2 < \sigma_2^2$. This is used when we believe the variance of the first population is less than the variance of the second population.

The term σ_1^2 represents the population variance of the first population and σ_2^2 represents the population variance of the second population.

The formula for the test statistic of the F-test is:

$$F = \frac{s_1^2}{s_2^2}$$

where:
s_1^2 = the sample variance of the first population with n observations
s_2^2 = the sample variance of the second population with n observations

A convention is to put the larger sample variance in the numerator and the smaller sample variance in the denominator.

$df_1 = n_1 - 1$ numerator degrees of freedom
$df_2 = n_2 - 1$ denominator degrees of freedom

The test statistic is then compared with the critical values found using the two degrees of freedom and the F-tables.

Finally, a decision is made whether to reject or not to reject the null hypothesis.

Example
You are investigating whether the population variance of the Indian equity market changed after the deregulation of 1991. You collect 120 months of data before and after deregulation. Variance of returns before deregulation was 13. Variance of returns after deregulation was 18. Check your hypothesis at a confidence level of 99%.

Solution:
Null and alternate hypothesis: $H_0: \sigma_1^2 = \sigma_2^2$ versus $H_A: \sigma_1^2 \neq \sigma_2^2$

F-statistic: $\frac{18}{13} = 1.4$
df = 119 for the numerator and denominator

$\alpha = 0.01$ which means 0.005 in each tail. From the F-table: critical value = 1.6

Since the F-stat is less than the critical value, do not reject the null hypothesis.

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4. Parametric Versus Nonparametric Tests

The hypothesis-testing procedures we have discussed so far have two characteristics in common:

- They are concerned with parameters, such as the mean and variance.
- Their validity depends on a set of assumptions.

Any procedure which has either of the two characteristics is known as a **parametric test**.

Nonparametric tests are not concerned with a parameter and/or make few assumptions about the population from which the sample are drawn. We use nonparametric procedures in three situations:

- Data does not meet distributional assumptions.
- Data has outliers
- Data are given in ranks. (Example: relative size of the company and use of derivatives.)
- The hypothesis does not concern a parameter. (Example: Is a sample random or not?)

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Summary

LO: Explain hypothesis testing and its components, including statistical significance, Type I and Type II errors, and the power of a test.

LO: Construct hypothesis tests and determine their statistical significance, the associated Type I and Type II errors, and power of the test given a significance level.

A hypothesis is a statement about the value of a population parameter developed for the purpose of testing a theory.

In order to test a hypothesis, we follow these steps:

1. State the hypothesis.
2. Identify the appropriate test statistic and its probability distribution.
3. Specify the significance level.
4. State the decision rule.
5. Collect data and calculate the test statistic.
6. Make a decision.

A test statistic is a quantity, calculated on the basis of a sample, and is used to decide whether to reject or not to reject the null hypothesis. The formula for computing the test statistic is:

$$\text{test statistic} = \frac{\text{sample statistic} - \text{value of the parameter under } H_0}{\text{standard error of the sample statistic}}$$

In reaching a statistical decision, we can make two possible errors: We may reject a true null hypothesis (a Type I error), or we may fail to reject a false null hypothesis (a Type II error).

The level of significance of a test is the probability of a Type I error. As α gets smaller the critical value gets larger and it becomes more difficult to reject the null hypothesis.

The power of a test is the probability of correctly rejecting the null (rejecting the null when it is false). It is expressed as:

Power of a test = 1 – P (Type II error)

LO: Compare and contrast parametric and nonparametric tests, and describe situations where each is the more appropriate type of test.

A parametric test is a hypothesis test concerning a parameter or a hypothesis test based on specific distributional assumptions. In contrast, a nonparametric test is either not concerned with a parameter or makes minimal assumptions about the population from which the sample is drawn.

A nonparametric test is primarily used in three situations: when data do not meet distributional assumptions, when data is given in ranks, or when the hypothesis we are addressing does not concern a parameter.

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LM09 Parametric and Non-Parametric Tests of Independence

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1. Introduction

This learning module covers:

- Parametric and non-parametric tests of correlation
- Tests of independence based on contingency table data

2. Tests Concerning Correlation

The strength of linear relationship between two variables is assessed through correlation coefficient. The significance of a correlation coefficient is tested by using hypothesis tests concerning correlation.

The most common way of setting up hypotheses concerning correlation is to check if the population correlation is not equal to 0. In this case the null and alternate hypothesis will be (ρ represents the population correlation coefficient):

- $H_0: \rho = 0$
- $H_a: \rho \neq 0$

It is a two-tailed test.

We can also set up hypothesis to check if the population correlation is positive or negative. The null and alternate hypothesis in these cases will be:

One sided (right side): $H_0: \rho \leq 0$ versus $H_a: \rho > 0$

One sided (left side): $H_0: \rho \geq 0$ versus $H_a: \rho < 0$

Parametric Test of a Correlation

As long as the two variables are distributed normally, we can use sample correlation, r for our hypothesis testing. The formula for the t-test is

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

where: n – 2 = degrees of freedom if H_0 is true.

The magnitude of r needed to reject the null hypothesis $H_0: \rho = 0$ decreases as sample size n increases due to the following:

i. As n increases, the number of degrees of freedom increases and the absolute value of the critical value t_c decreases.

ii. As n increases, the absolute value of the numerator increases, leading to larger-magnitude t-values.

In other words, as n increases, the probability of Type-II error decreases, all else equal.

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Example

The sample correlation between the oil prices and monthly returns of energy stocks in a Country A is 0.7986 for the period from January 2014 through December 2018. Can we reject a null hypothesis that the underlying or population correlation equals 0 at the 0.05 level of significance?

Solution:

H_0: $\rho = 0$ → true correlation in the population is 0.

H_a: $\rho \neq 0$ → correlation in the population is different from 0.

From January 2014 through December 2018, there are 60 months, so n = 60. We use the following statistic to test the above.

$$t = \frac{0.7986\sqrt{60-2}}{\sqrt{1-0.7986^2}} = \frac{6.0820}{0.6019} = 10.1052$$

At the 0.05 significance level, the critical level for this test statistic is 2.00 (n = 60, degrees of freedom = 58). When the test statistic is either larger than 2.00 or smaller than 2.00, we can reject the hypothesis that the correlation in the population is 0. The test statistic is 10.1052, so we can reject the null hypothesis.

Non-Parametric Test of Correlation: The Spearman Rank Correlation Coefficient

Instructor's Note:
Focus on the basics of this topic, the probability of being tested on the details is low.

If the two variables under consideration are not normally distributed, we can use a test based on the Spearman rank correlation coefficient, r_S. The Spearman rank correlation coefficient is equivalent to the usual correlation coefficient but is calculated on the *ranks* of two variables within their respective samples.

Steps for calculating r_S:

1. Sort the X observations from largest to smallest. Assign the number 1 to the largest value observation, the number 2 to the second largest value observation, and so on. In the event of a tie, assign the average of the ranks that the tied observations share to each tied observation. For example, if the third and fourth largest values are tied, we assign a rank of 3.5 to both observations (the average of 3 and 4). Repeat the procedure for the observations on Y.
2. Calculate the difference in ranks, d_i, for each pair of observations on X and Y, and then calculate d_i^2 (the squared difference in ranks).
3. For a sample size n, the spearman rank correlation is:

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$$

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Example:
(This is based on Question Set from the curriculum.)

You want to determine whether the alpha of a mutual fund is correlated with its expense ratio. You gather the following information about 9 mutual funds.

Mutual Fund	Alpha	Expense Ratio
1	-0.52	1.34
2	-0.13	0.40
3	-0.50	1.90
4	-1.01	1.50
5	-0.26	1.35
6	-0.89	0.50
7	-0.42	1.00
8	-0.23	1.50
9	-0.60	1.45

Perform a Spearman rank correlation test based on this sample data. Determine whether to reject the null hypothesis at the 0.05 level of significance if the critical values are ±2.306.

Solution:

The null and alternate hypotheses are:

H_0: $\rho = 0$ versus H_a: $\rho \neq 0$

To calculate the Spearman rank correlation, we first construct the following table:

Mutual Fund	Alpha	Expense Ratio	Rank by Alpha	Rank by Expense Ratio	Difference in Rank	Difference Squared
1	-0.52	1.34	6	6	0	0
2	-0.13	0.40	1	9	-8	64
3	-0.50	1.90	5	1	4	16
4	-1.01	1.50	9	2.5	6.5	42.25
5	-0.26	1.35	3	5	-2	4
6	-0.89	0.50	8	8	0	0
7	-0.42	1.00	4	7	-3	9
8	-0.23	1.50	2	2.5	-0.5	0.25
9	-0.60	1.45	7	4	3	9
					Sum	**144.5**

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2 - 1)}$$

$$r_s = 1 - \frac{6(144.5)}{9(9^2 - 1)} = -0.20416$$

The test statistic is calculated as:

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$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$t = \frac{-0.20416\sqrt{9-2}}{\sqrt{1-0.20416^2}} = -0.55177$$

Since this value falls within the range of ±2.306, we fail to reject the null hypothesis that the Spearman rank correlation coefficient is zero.

3. Tests of Independence Using Contingency Table Data

Instructor's Note:
Focus on the basics of this topic, the probability of being tested on the details is low.

When dealing with categorical or discrete data presented in the form of a contingency table, we use a chi-squared distributed test statistic.

Consider the following contingency table which classifies 1,594 ETFs based on two dimensions: size and investment type.

Investment Type	**Size Based on Market Capitalization**			
	Small	**Medium**	**Large**	**Total**
Value	50	110	343	503
Growth	42	122	202	366
Blend	56	149	520	725
Total	148	381	1,065	1,594

Suppose we want to test whether a relationship exists between the size and investment type, we can perform a test of independence using a chi-squared distributed test statistic. This nonparametric test compares actual observed frequencies with those expected on the basis of independence.

The test statistic is calculated as:

$$\chi^2 = \sum_{i=1}^{m} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}},$$

where:
m = number of cells in the table
Oij = observed frequency
Eij = expected frequency calculated as: $E_{ij} = \frac{(\text{Total row } i) \times (\text{Total column } j)}{\text{Overall total}}$.

This test statistic has degrees of freedom of (r − 1)(c − 1), where r is the number of categories for the first variable and c is the number of categories of the second variable.

The expected frequency for one cell – small cap value can be calculated as:

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$$E_{ij} = \frac{503 \times 148}{1{,}594} = 46.703$$

In the same way we can calculate the expected frequency for all 9 cells to get the following table.

Expected Frequency of ETFs by Size and Investment Type

Investment Type	Size Based on Market Capitalization		
	Small	**Medium**	**Large**
Value	46.703	120.228	336.070
Growth	33.982	87.482	244.536
Blend	67.315	173.290	484.395
Total	148.000	381.000	1,065.000

We then calculate the squared difference between observed and expected frequencies scaled by the expected frequency for each cell.

$$\frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

For the small cap value cell, this calculation is:

$$\frac{(50 - 46.703)^2}{46.703} = 0.2327$$

We perform the same calculations for other cells and get the following table:

Scaled Squared Deviation for Each Combination of Size and Investment Type

Investment Type	Size Based on Market Capitalization		
	Small	**Medium**	**Large**
Value	0.233	0.870	0.143
Growth	1.892	13.620	7.399
Blend	1.902	3.405	2.617

Finally, we sum all the above values to get a chi-squared test statistic as 32.08025.

Testing Our Hypothesis

The null and alternate hypothesis are:

H_0: ETF size and investment type are not related, so these classifications are independent;

H_a : ETF size and investment type are related, so these classifications are not independent.

With a (3-1) x (3 -1) = 4 degrees of freedom and a one-sided test with a 5% level of significance, the critical value is 9.4877.

Since the calculated chi-squared test statistic (32.08025) is greater than 9.4877 we reject the null hypothesis of independence and conclude that ETF size and investment type are related.

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Summary

LO: Explain parametric and nonparametric tests of the hypothesis that the population correlation coefficient equals zero, and determine whether the hypothesis is rejected at a given level of significance.

Parametric test: The significance of a correlation coefficient is tested by using hypothesis tests concerning correlation. The formula for the t-test is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

where: n – 2 = degrees of freedom

Non-parametric test: The Spearman rank correlation coefficient is equivalent to the usual correlation coefficient but is calculated on the *ranks* of two variables within their respective samples.

LO: Explain tests of independence based on contingency table data.

A chi-square distributed test statistic is used to test for independence of two categorical variables. This nonparametric test compares actual frequencies with those expected on the basis of independence.

This test statistic has degrees of freedom of $(r-1)(c-1)$, where r is the number of categories for the first variable and c is the number of categories of the second variable.

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LM10 Simple Linear Regression

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1 & 2. Introduction and Estimation of the Simple Linear Regression Model

Financial analysts often need to find whether one variable X can be used to explain another variable Y. Linear regression allows us to examine this relationship.

Suppose an analyst is evaluating the return on assets (ROA) for an industry. He gathers data for six companies in that industry.

Company	ROA (%)
A	6
B	4
C	15
D	20
E	10
F	20

Range = 4% to 20%

Average value = 12.5%

Let Y represents the variable we are trying to explain (in this case ROA), Y_i represents a particular observation, and $\overline{Y}$ represent the mean value. Then the variation of Y can be expressed as:

$$\text{Variation of Y} = \sum_{i=1}^{N} (Y_i - \overline{Y})^2$$

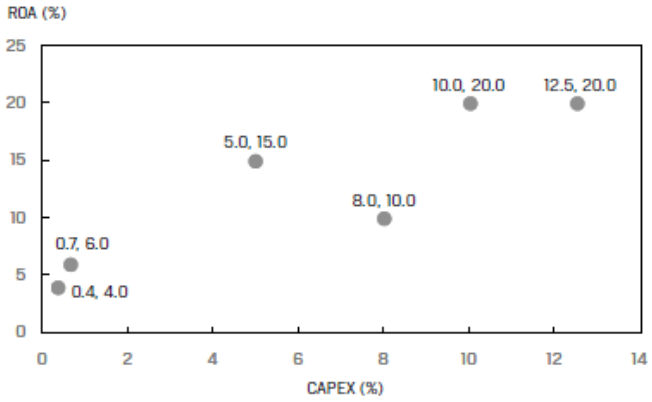
The variation of Y is also called **sum of squares total (SST)** or the total sum of squares. Our aim is to understand what explains the variation of Y.

The analyst now wants to check if another variable – CAPEX can be used to explain the variation of ROA. The analyst defines CAPEX as: capital expenditures in the previous period, scaled by the prior period's beginning property, plant, and equipment. He gathers the following data for the six companies:

Company	ROA (%)	CAPEX(%)
A	6	0.7
B	4	0.4
C	15	5.0
D	20	10.0
E	10	8.0
F	20	12.5
Arithmetic mean	12.50	6.10

The relation between the two variables can be visualized using a scatter plot.

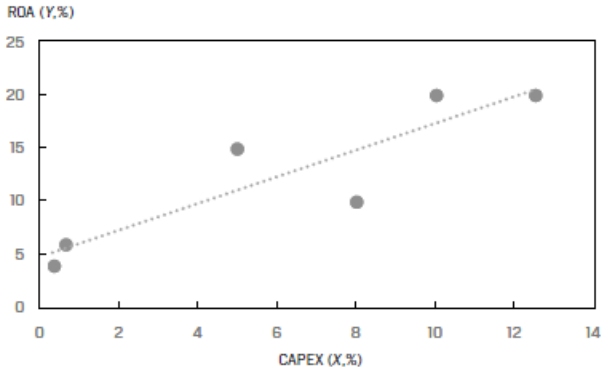
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The variable whose variation we want to explain, also called the **dependent variable** is presented on the vertical axis. It is typically denoted by Y. In our example ROA is the dependent variable.

The explanatory variable, also called the **independent variable** is presented on the horizontal axis. It is typically denoted by X. In our example CAPEX is the independent variable.

A linear regression model computes the best fit line through the scatter plot, which is the line with the smallest distance between itself and each point on the scatter plot. The regression line may pass through some points, but not through all of them.



Regression analysis with only one independent variable is called **simple linear regression (SLR)**. Regression analysis with more than one independent variable is called multiple regression. In this reading we focus on single independent variable, i.e., simple linear regression.

Estimating the Parameters of a Simple Linear Regression

The Basics of Simple Linear Regression

Linear regression assumes a linear relationship between the dependent variable (Y) and independent variable (X). The regression equation is expressed as follows:

$$Y_i = b_0 + b_1 X_i + \varepsilon_i$$

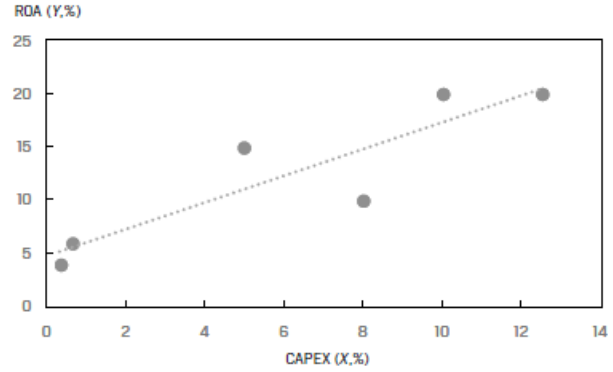
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where:
i = 1, …, n
Y = dependent variable
b_0 = intercept
b_1 = slope coefficient
X = independent variable
ε = error term
b_0 and b_1 are called the **regression coefficients.**

The equation shows how much Y changes when X changes by one unit.

Estimating the Regression Line

Linear regression chooses the estimated values for intercept $\widehat{b_0}$ and slope $\widehat{b_1}$ such that the **sum of the squared errors (SSE)**, i.e., the vertical distances between the observations and the regression line is minimized.



This is represented by the following equation. The error terms are squared so that they don't cancel out each other. The objective of the model is that the sum of the squared error terms should be minimized.

$$\text{SSE} = \sum_{i=1}^{N} \left(Y_i - \hat{Y}_i\right)^2 = \sum_{i=1}^{N} \left[Y_i - (\hat{b}_0 + \hat{b}_1 X_i)\right]^2$$

The slope coefficient is calculated as:

$$\hat{b}_1 = \frac{\text{Covairance of Yand X}}{\text{Variance of X}} = \frac{\frac{\sum_{i=1}^{N}(Y_i - \bar{Y})(X_i - \bar{X})}{n-1}}{\frac{\sum_{i=1}^{N}(X_i - \bar{X})^2}{n-1}} = \frac{\sum_{i=1}^{N}(Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^{N}(X_i - \bar{X})^2}$$

Note: This formula is similar to correlation which is calculated as:

$$r = \frac{\text{Covairance of Yand X}}{(\text{Std dev of X})(\text{Std dev of Y})}$$

Once we calculate the slope, the intercept can be calculated as:

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$$\hat{b}_0 = \overline{Y} - \hat{b}_1\overline{X}$$

Note: On the exam you will most likely be given the values of b_1 and b_0. It is unlikely that you will be asked to calculate these values. Nevertheless, the following table shows how to calculate the slope and intercept.

Company	ROA(Y_i)	CAPEX (X_i)	$(Y_i - \overline{Y})^2$	$(X_i - \overline{X})^2$	$(Y_i - \overline{Y})(X_i - \overline{X})$
A	6.0	0.7	42.25	29.16	35.10
B	4.0	0.4	72.25	32.49	48.45
C	15.0	5.0	6.25	1.21	-2.75
D	20.0	10.0	56.25	15.21	29.25
E	10.0	8.0	6.25	3.61	-4.75
F	20.0	12.5	56.25	40.96	48.00
Sum	75.0	36.6	239.50	122.64	153.30
Mean	12.5	6.100			

Slope coefficient: $\hat{b}_1 = \frac{153.30}{122.64} = 1.25$

Intercept: $\hat{b}_0 = 12.5 - (1.25 \text{ x } 6.10) = 4.875$

The regression model can thus be expressed as:

$$Y_i = 4.875 + 1.25X_i + \varepsilon_i$$

Interpreting the Regression Coefficients

The intercept is the value of the dependent variable when the independent variable is zero. In our example, if a company makes no capital expenditures, its expected ROA is 4.875%.

The slope measures the change in the dependent variable for a one-unit change in the independent variable. If the slope is positive, the two variables move in the same direction. If the slope is negative, the two variables move in opposite directions. In our example, if CAPEX increases by one unit, ROA increases by 1.25%.

Cross-Sectional vs. Time-Series Regressions

Regression analysis can be used for two types of data:

- Cross sectional data: Many observations for different companies, asset classes, investment funds etc. for the same time period.
- Time-series data: Many observations from different time periods for the same company, asset class, investment funds etc.

3. Assumptions of the Simple Linear Regression Model

The simple linear regression model is based on the following four assumptions:

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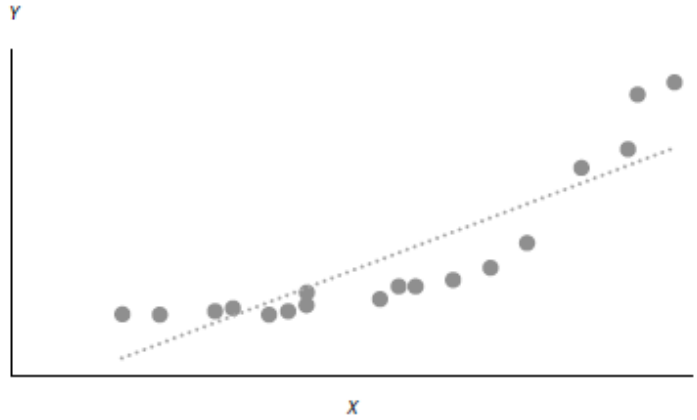
1. Linearity: The relationship between the dependent variable, Y, and the independent variable, X, is linear.
2. Homoskedasticity: The variance of the regression residuals is the same for all observations.
3. Independence: The observations, pairs of Ys and Xs, are independent of one another. This implies the regression residuals are uncorrelated across observations.
4. Normality: The regression residuals are normally distributed.

Assumption 1: Linearity

Since we are fitting a straight line through a scatter plot, we are implicitly assuming that the true underlying relationship between the two variables is linear.

If the relationship between the variables is non-linear, then using a simple linear regression model will produce invalid results.

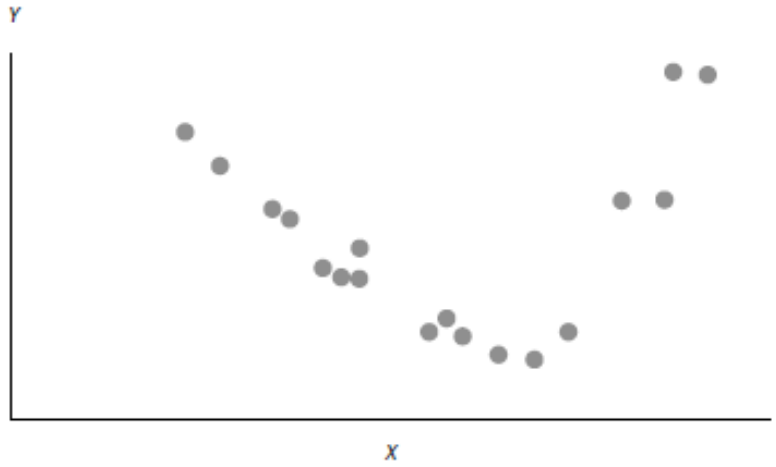
Exhibit 10 from the curriculum shows two variables that have an exponential relationship. A linear regression line does not fit this relationship well. At lower and higher values of X, the model underestimates Y. Whereas, at middle values of X, the model overestimates Y.



Another point related to this assumption is that the independent variable X should not be random. If X is random, there would be no linear relationship between the two variables.

Also, the residuals of the model should be random. They should not exhibit a pattern when plotted against the independent variable. As shown in Exhibit 11, the residuals from the linear regression model in Exhibit 10 do not appear random.

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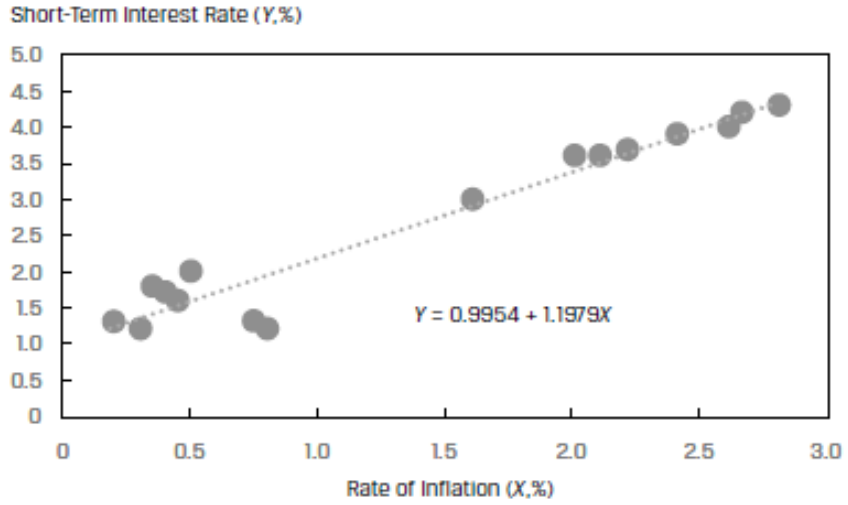


Hence, a linear regression model should not be used for these two variables.

Assumption 2: Homoskedasticity

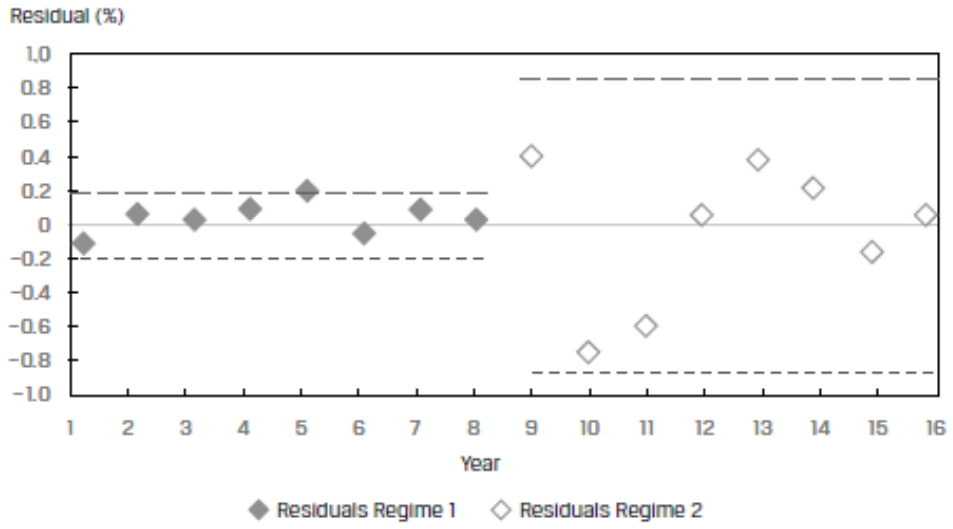
Assumption#2 is that the variance of the residuals is constant for all observations. This condition is called homoskedasticity. If the variance of the error term is not constant, then it is called heteroskedasticity.

Exhibit 12 shows a scatter plot of short-term interest rates versus inflation rate for a country. The data represents a total span of 16 years. We will refer to the first eight years of normal rates as Regime 1 and the second eight years of artificially low rates as Regime 2.



At first glance, the model seems to fit the data well. However, when we plot the residuals of the model against the years, we can see that the residuals for the two regimes appear different. This is shown in Exhibit 13 below. The variation in residuals for Regime 1 is much smaller than the variation in residuals for Regime 2. This is a clear violation of the homoskedasticity assumption and the two regimes should not be clubbed together in a single model.

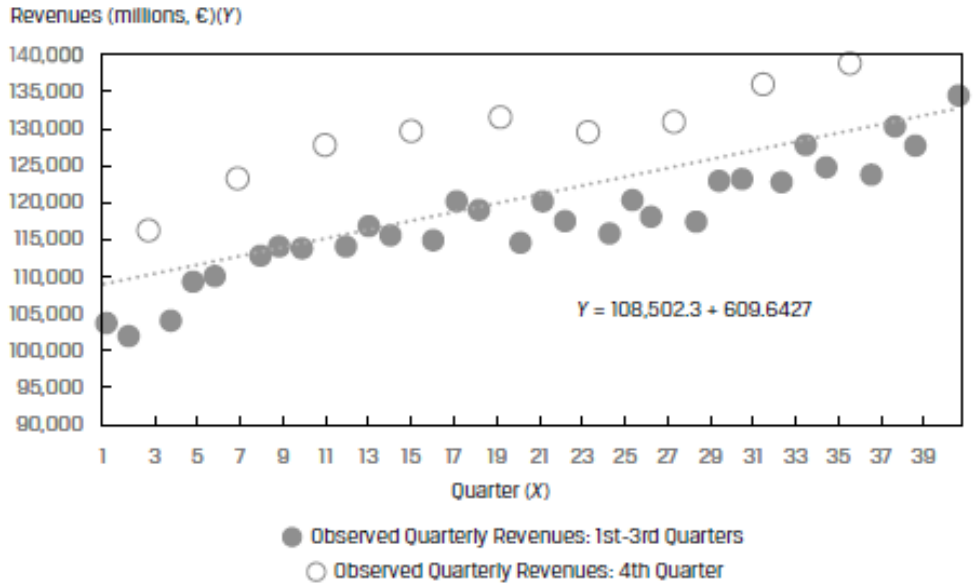
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Assumption 3: Independence

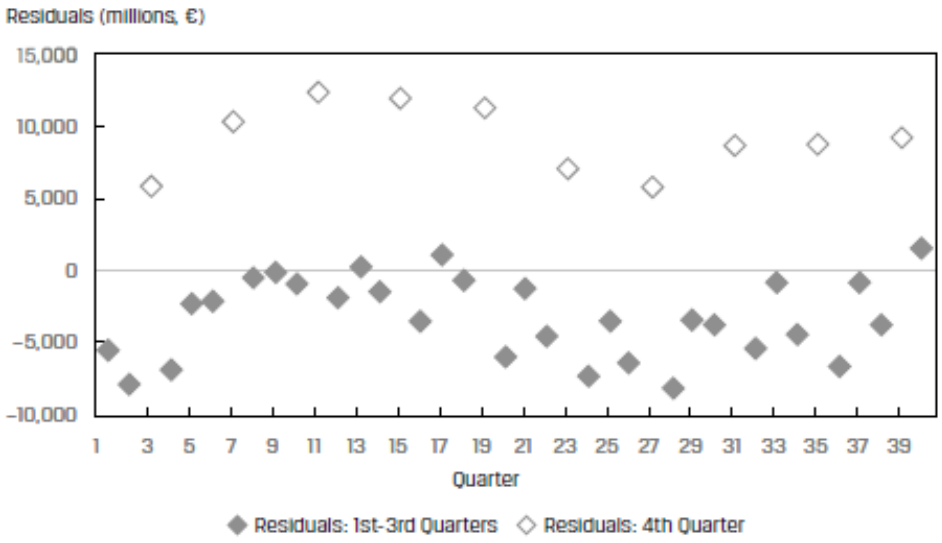
Assumption# 3 states that the residuals should be uncorrelated across observations. If the residuals exhibit a pattern, then this assumption will be violated.

For example, Exhibit 15 from the curriculum plots the quarterly revenues of a company over 40 quarters. The data displays a clear seasonal pattern. Quarter 4 revenues are considerably higher than the first 3 quarters.



A plot of the residuals from this model in Exhibit 16 also helps us see this pattern. The residuals are correlated – they are high in Quarter 4 and then fall back in the other quarters.

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Both exhibits show that the assumption of residual independence is violated and the model should not be used for this data.

Assumption 4: Normality

Assumption#4 states that the residuals should be normally distributed.

Instructor's Note: This assumption does not mean that X and Y should be normally distributed, it only means that the residuals from the model should be normally distributed.

4.1 Analysis of Variance

Breaking down the Sum of Squares Total into Its Components

To evaluate how well a linear regression model explains the variation of Y we can break down the total variation in Y (SST) into two components: Sum of square errors (SSE) and regression sum of squares (RSS).

Total sum of squares (**SST**), $\sum_{i=1}^{N} (Y_i - \overline{Y})^2$: This measures the total variation in the dependent variable. SST is equal to the sum of squared distances between the actual values of Y and the average value of Y.

The **regression sum of squares** (**RSS**), $\sum_{i=1}^{N} (\hat{Y}_i - \overline{Y})^2$. This is the amount of total variation in Y that is explained in the regression equation. RSS is the sum of squared distances between the predicted values of Y and the average value of Y.

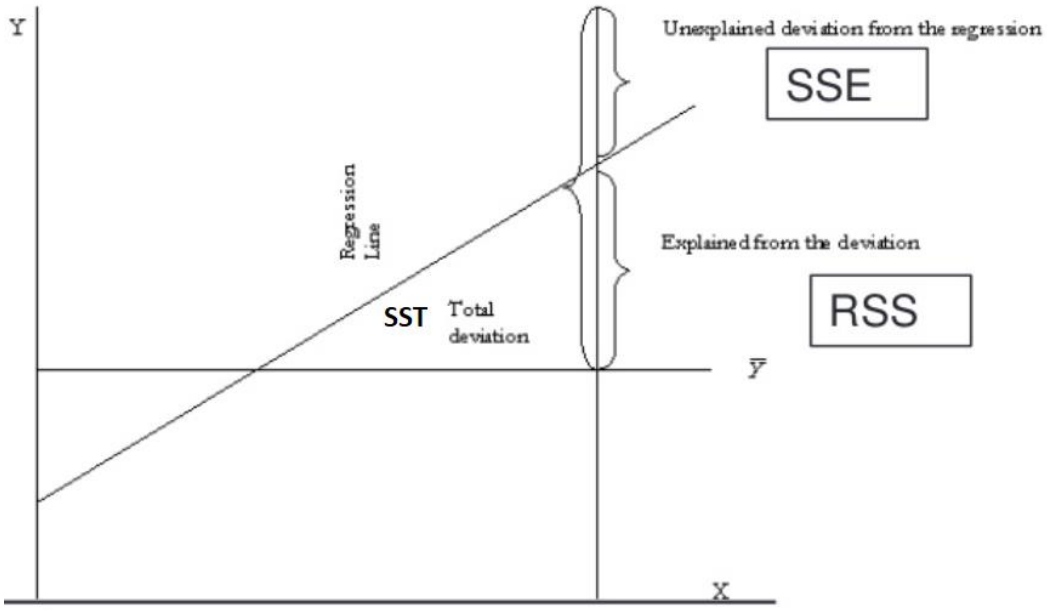
The **sum of squared errors** or residuals (**SSE**), $\sum_{i=1}^{N} (Y_i - \hat{Y}_i)^2$.This is also known as residual sum of squares. It measures the unexplained variation in the dependent variable. SSE is the sum of the vertical distances between the actual values of Y and the predicted values of Y on the regression line.

SST = RSS + SSE

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i.e., Total variation = explained variation + unexplained variation

These concepts are illustrated in the figure below:



Measures of Goodness of Fit

Goodness of fit indicates how well the regression model fits the data. Several measures are used to evaluate the goodness of fit.

Coefficient of determination: The coefficient of determination, denoted by R^2, measures the fraction of the total variation in the dependent variable that is explained by the independent variable.

$$R^2 = \frac{\text{Explained variation}}{\text{Total variation}} = \frac{\text{Regression sum of squares (RSS)}}{\text{Sum of squares total (SST)}}$$

Characteristics of coefficient of determination, R^2:

The higher the R^2, the more useful the model. R^2 has a value between 0 and 1. For example, if R^2 is 0.03, then this explains only 3% of the variation in the independent variable and the explanatory power of the model is low. However, if R^2 is 0.85, the model explains over 85% of the variation and the explanatory power of the model is high.

It tells us how much better the prediction is by using the regression equation rather than just $\bar{y}$ (average value) to predict y.

With only one independent variable, R^2 is the square of the correlation between X and Y.

The correlation, r, is also called the "multiple-R".

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F-test: For a meaningful regression model, the slope coefficients should be non-zero. This is determined through the F-test which is based on the F-statistic. The F-statistic tests whether all the slope coefficients in a linear regression are equal to 0. In a regression with one independent variable, this is a test of the null hypothesis H_0: $b_1 = 0$ against the alternative hypothesis H_a: $b_1 \neq 0$.

The F-statistic also measures how well the regression equation explains the variation in the dependent variable. The four values required to construct the F-statistic for null hypothesis testing are:

- The total number of observations (n)
- The total number of independent variables (k)
- The regression sum of squares (RSS)
- The sum of squared errors or residuals (SSE)

The F-statistic is calculated as:

$$F = \frac{\frac{RSS}{k}}{\frac{SSE}{n-(k+1)}} = \frac{\text{Mean square regression}}{\text{Mean square error}} = \frac{MSR}{MSE}$$

Interpretation of F-test statistic:

- The higher the F-statistic, the better.
- A high F-statistic implies that the regression model does a good job of explaining the variation in the dependent variable.
- A low F-statistic implies that the regression model does not do a good job of explaining the variation in the dependent variable.
- An F-statistic of 0 indicates that the independent variable does not explain variation in the dependent variable.

ANOVA and Standard Error of Estimate in Simple Linear Regression

Analysis of variance or ANOVA is a statistical procedure of dividing the total variability of a variable into components that can be attributed to different sources. We use ANOVA to determine the usefulness of the independent variable or variables in explaining variation in the dependent variable.

ANOVA table

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F-statistic
Regression (explained variation)	k	RSS	$MSR = \frac{RSS}{k}$	$F = \frac{MSR}{MSE}$

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Error (unexplained variation)	n - 2	SSE	$\text{MSE} = \frac{\text{SSE}}{\text{n} - \text{k} - 1}$	
Total variation	n – 1	SST		

n represents the number of observations and k represents the number of independent variables. With one independent variable, k = 1. Hence, MSR = RSS and MSE = SSE / (n - 2).

Information from the ANOVA table can be used to compute the standard error of estimate (SEE). The **standard error of estimate (SEE)** measures how well a given linear regression model captures the relationship between the dependent and independent variables. It is the standard deviation of the prediction errors. A low SEE implies an accurate forecast.

The formula for the standard error of estimate is given below:

$$\text{Standard error of estimate (SEE)} = \sqrt{\text{MSE}}$$

A low SEE implies that the error (or residual) terms are small and hence the linear regression model does a good job of capturing the relationship between dependent and independent variables.

The following example demonstrates how to interpret an ANOVA table.

Example: Using ANOVA Table Results to Evaluate a Simple Linear Regression

You are provided the following ANOVA table:

Source	Sum of Squares	Degrees of Freedom	Mean Square
Regression	576.1485	1	576.1485
Error	1,873.5615	98	19.1180
Total	2,449.7100		

1. What is the coefficient of determination for this regression model?
2. What is the standard error of the estimate for this regression model?
3. At a 5% level of significance, do we reject the null hypothesis of the slope coefficient equal to zero if the critical F-value is 3.938?
4. Based on your answers to the preceding questions, evaluate this simple linear regression model.

Solution to 1: Coefficient of determination (R^2) = RSS / SST = 576.148/2,449.71 = 23.52%.

Solution to 2: Standard error of estimate (SEE) $= \sqrt{\text{MSE}} = \sqrt{19.1180} = 4.3724$

Solution to 3:

$$\text{F} = \frac{\text{MSR}}{\text{MSE}} = \frac{576.1485}{19.1180} = 30.1364$$

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Since the calculated F-stat is higher than the critical value of 3.938, we can conclude that the slope coefficient is statistically different from 0.

Solution to 4: The coefficient of determination indicates that the model explains 23.52% of the variation in Y. Also, the F-stat confirms that the model's slope coefficient is statistically different from 0. Overall, the model seems to fit the data reasonably well.

4.2 Hypothesis Testing of Linear Regression Coefficients

Hypothesis Tests of the Slope Coefficient

In order to test whether an estimated slope coefficient is statistically significant, we use hypothesis testing.

Continuing with our previous example of a simple linear regression with ROA as the dependent variable and CAPEX as the independent variable. Suppose we want to test whether the slope coefficient of CAPEX is different from zero.

The steps are:

Step 1: State the hypothesis

$H_0: b_1 = 0$; $H_a: b_1 \neq 0$

Step 2: Identify the appropriate test statistic

To test the significance of individual slope coefficients we use the t-statistic. It is calculated by subtracting the hypothesized population slope (B_1) from the estimated slope coefficient ($\widehat{b_1}$) and then dividing this difference by the standard error of the slope coefficient, $s_{\widehat{b_1}}$:

$$t = \frac{\widehat{b_1} - B_1}{s_{\widehat{b_1}}} = \frac{\text{Estimated value} - \text{Hypothesized value}}{\text{standard error}}$$

with n – 2 = 6 – 2 = 4 degrees of freedom

Step 3: Specify the level of significance

Typically, a 5% level of significance is selected.

Step 4: State the decision rule

Since the alternate hypothesis contains a '≠' sign, this is a two tailed test. For a 5% level of significance, 4 degrees of freedom and a two tailed test the critical t-values are ±2.776.

We reject the null hypothesis if the calculated t-statistic is less than −2.776 or greater than +2.776.

Step 5: Calculate the test statistic

Instructor's Note: On the exam, the value of the standard error will most likely be given to you. You are unlikely to be asked to calculate this value.

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Let's say that the standard error given is 0.31

$$t = \frac{\widehat{b_1} - B_1}{s_{\widehat{b_1}}} = \frac{1.25 - 0}{0.31} = 4$$

Instructor's Note: In an extreme case if the value of the standard error is not given to you. It can be calculated as shown below:

The standard error of the slope coefficient is calculated as the ratio of the model's standard error of the estimate (SEE) to the square root of the variation of the independent variable:

$$s_{\widehat{b_1}} = \frac{SEE}{\sqrt{\sum_{i=1}^{n}(X_i - \overline{X})^2}}$$

The slope coefficient is 1.25.

The mean square error is 11.96875.

The variation of CAPEX is 122.640

SEE = $\sqrt{11.96875}$ = 3.459588

$$s_{\widehat{b_1}} = \frac{SEE}{\sqrt{\sum_{i=1}^{n}(X_i - \overline{X})^2}}$$

$$= \frac{3.459588}{\sqrt{122.640}} = 0.312398$$

Step 6: Make a decision

Reject the null hypothesis of a zero slope. There is sufficient evidence to indicate that the slope is different from zero.

What if we want to test if the slope coefficient is statistically different from 1.0?

Instead of using a hypothesized value of zero for the slope coefficient, what if we want to test if the slope coefficient is statistically different from 1.0?

The hypotheses become H_0: $b_1 = 1$ and H_a: $b_1 \neq 1$.

The calculated t-statistic is:

$$t = \frac{1.25 - 1}{0.31} = 0.8$$

This calculated test statistic falls within the critical values, ±2.776, so we fail to reject the null hypothesis: There is not sufficient evidence to indicate that the slope is different from 1.0.

What if we want to test if the slope coefficient is positive?

The hypotheses become H_0: $b_1 \leq 0$ and H_a: $b_1 > 0$.

The critical values change because this is a one tailed test. For a 5% level of significance, 4 degrees of freedom and a one tailed test the critical t-value is +2.132.

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All other steps stay the same.

The calculated test stat is greater than 2.132. Therefore, we can reject the null hypothesis. There is sufficient evidence to indicate that the slope is greater than zero.

Testing the correlation:

We can also use hypothesis testing to test the significance of the correlation between the two variables. The process is the same except that the hypothesis are written and the test statistic is calculated differently.

This is demonstrated using the ROA and CAPEX example. The regression software provided us an estimated correlation of 0.8945. Suppose we want to test if this correlation is statistically different from zero.

The steps are:

Step 1: State the hypothesis

Here the null and alternate hypothesis are: H_0: $\rho = 0$ versus H_a: $\rho \neq 0$.

Step 2: Identify the appropriate test statistic

The test-statistic is calculated as:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

with 6 – 2 = 4 degrees of freedom

Step 3: Specify the level of significance

$\alpha = 5\%$.

Step 4: State the decision rule

Critical t-values = ±2.776.

Reject the null if the calculated t-statistic is less than −2.776 or greater than +2.776.

Step 5: Calculate the test statistic

$$t = \frac{0.8945\sqrt{6-2}}{\sqrt{1-0.8001}} = 4.00131$$

Step 6: Make a decision

Reject the null hypothesis of no correlation. There is sufficient evidence to indicate that the correlation between ROA and CAPEX is different from zero.

In a simple linear regression, the t-statistic used to test the slope coefficient and the t-statistic used to test the correlation will have the same value. And we will get the same results using either hypothesis tests. This is demonstrated in the table below:

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	Test of the Slope	**Test of the Correlation**
Null and alternate hypothesis	H_0: $b_1 = 0$ versus H_a: $b_1 \neq 0$	H_0: $\rho = 0$ versus H_a: $\rho \neq 0$.
Critical values based on the level of significance and degrees of freedom	$\alpha = 5\%$ and dof =4: CV = ±2.776	$\alpha = 5\%$ and dof =4: CV = ±2.776
Test statistic	$t = \frac{\widehat{b_1} - B_1}{s_{\widehat{b_1}}} = \frac{1.25 - 0}{0.312398} = 4.00131$	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$ $= \frac{0.8945\sqrt{6-2}}{\sqrt{1-0.8001}} = 4.00131$
Decision	Since calculated t-stat is greater than the critical value, reject the null hypothesis of zero slope.	Since calculated t-stat is greater than the critical value, reject the null hypothesis of zero correlation.

Another feature of simple linear regression is that the F-stat is simply the square of the t-stat for the slope/correlation. For our example, the F-stat is $4.00131^2 = 16.0104$

Hypothesis Tests of the Intercept

For the ROA regression example, the intercept is 4.875%. Say you want to test if the intercept is statistically greater than 3%. This will be a one-tailed hypothesis test and the steps are:

Step 1: State the hypothesis

H_0: $b_0 \leq 3\%$ versus H_a: $b_0 > 3\%$

Step 2: Identify the appropriate test statistic

To test whether the population intercept is a specific value we can use the following t-stat:

$$t_{\text{intercept}} = \frac{\widehat{b_0} - B_0}{s_{\widehat{b_0}}}$$

with 6 – 2 = 4 degrees of freedom

The standard error of the intercept is calculated as:

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$$s_{\widehat{b_0}} = \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}}$$

Instructor's Note: On the exam, the value of the standard error will most likely be given to you. You are unlikely to be asked to calculate this value.

Step 3: Specify the level of significance

$\alpha = 5\%$.

Step 4: State the decision rule

Critical t-value = 2.132.

Reject the null if the calculated t-statistic is greater than 2.132.

Step 5: Calculate the test statistic

$$s_{\widehat{b_0}} = \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}} = \sqrt{\frac{1}{6} + \frac{6.1^2}{122.64}} = 0.68562$$

$$t_{intercept} = \frac{\widehat{b_0} - B_0}{s_{\widehat{b_0}}} = \frac{4.875 - 3.0}{0.68562} = 2.73475$$

Step 6: Make a decision

Since the calculated t-stat is greater than the critical value, we can reject the null hypothesis and conclude that the intercept is greater than 3%.

Hypothesis Tests of Slope When Independent Variable Is an Indicator Variable

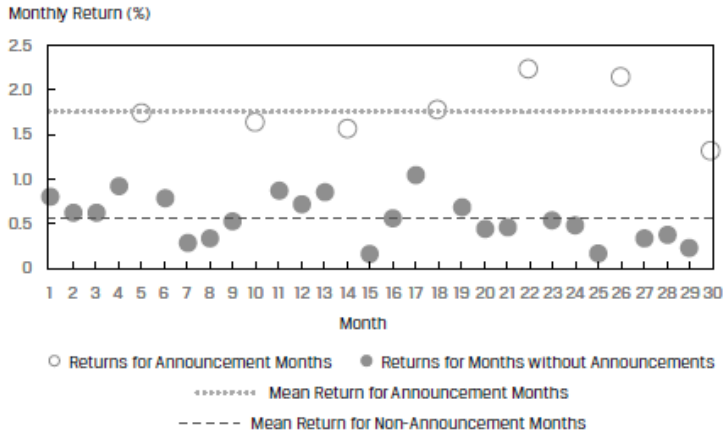
An indicator variable or a dummy variable can only take values of 0 or 1. An independent variable is set up as an indicator variable in specific cases. Say we want to evaluate if a company's quarterly earnings announcement influences its monthly stock returns. Here the monthly returns RET would be regressed on the indicator variable, EARN, that takes on a value of 0 if there is no earnings announcement that month and 1 if there is an earnings announcement.

The simple linear regression model can be expressed as:

$$RET_i = b_0 + b_1 EARN_i + \epsilon_i$$

Say we run the regression analysis over a 30-month period. The observations and regression results are shown in Exhibit 28.

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Clearly the returns for announcement months are different from the returns for months without announcement.

The results of the regression are given in Exhibit 29.

	Estimated Coefficients	**Standard Error of Coefficients**	**Calculated Test Statistic**
Intercept	0.5629	0.0560	10.0596
EARN	1.2098	0.1158	10.4435

We can draw the following inferences this table:

- The t-stats for both the intercept and the slope are high hence we can conclude that both the intercept and the slope are statistically significant.
- The intercept is the predicted value of Y when X = 0. Therefore, the intercept (0.5629) is the mean of the returns for non-earnings-announcement months.
- The slope coefficient (1.2098) is the difference in means of returns between earnings-announcement and non-announcement months.

Test of Hypotheses: Level of Significance and p-Values

p-value: At times financial analysts report the *p*-value or probability value for a particular hypothesis. The p-value is the smallest level of significance at which the null hypothesis can be rejected. It allows the reader to interpret the results rather than be told that a certain hypothesis has been rejected or accepted. In most regression software packages, the p-values printed for regression coefficients apply to a test of the null hypothesis that the true parameter is equal to 0 against the alternative that the parameter is not equal to 0, given the estimated coefficient and the standard error for that coefficient.

Here are a few important points connecting t-statistic and p-value:

- Higher the t-statistic, smaller the p-value.
- The smaller the p-value, the stronger the evidence to reject the null hypothesis.

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- Given a p-value, if p-value ≤ α, then reject the null hypothesis H_0. α is the significance level. For example, if we are given a p value of 0.03 or 3%, we can reject the null hypothesis at the 5% significance level, but not at the 1% significance level.

5. Prediction Using Simple Linear Regression and Prediction Intervals

We use regression equations to make predictions about a dependent variable. Let us consider the regression equation: $Y = b_0 + b_1X$. The predicted value of $\widehat{Y} = \widehat{b_0} + \widehat{b_1}X$.

The two sources of uncertainty to make a prediction are:

1. The error term
2. Uncertainty in predicting the estimated parameters b_0 and b_1

The estimated variance of the prediction error is given by:

$$s_f^2 = s^2 * \left[1 + \frac{1}{n} + \frac{(X - \overline{X})^2}{(n-1)s_x^2}\right]$$

Note: You need not memorize this formula, but understand the factors that affect s_f^2, like higher the n, lower the variance, and the better it is.

The estimated variance depends on:

- the squared standard error of estimate, s^2
- the number of observations, n
- the value of the independent variable, X
- the estimated mean $\overline{X}$
- variance, s^2, of the independent variable

Once the variance of the prediction error is known, it is easy to determine the confidence interval around the prediction. The steps are:

1. Make the prediction.
2. Compute the variance of the prediction error.
3. Determine t_c at the chosen significance level α.
4. Compute the (1-α) prediction interval using the formula below:
   $\widehat{Y} \pm t_c * s_f$

In our ROA regression model, if a company's CAPEX is 6%, its forecasted ROA is:

$$\widehat{Y} = \widehat{b_0} + \widehat{b_1}X = 4.875 + 1.25 \times 6 = 12.375\%$$

Assuming a 5% significance level (α), two sided, with n − 2 degrees of freedom (so, df = 4), the critical values for the prediction interval are ±2.776.

The standard error of the forecast is:

$$s_f = 3.49588 * \sqrt{\left[1 + \frac{1}{6} + \frac{(6 - 6.1)^2}{122.640}\right]} = 3.736912$$

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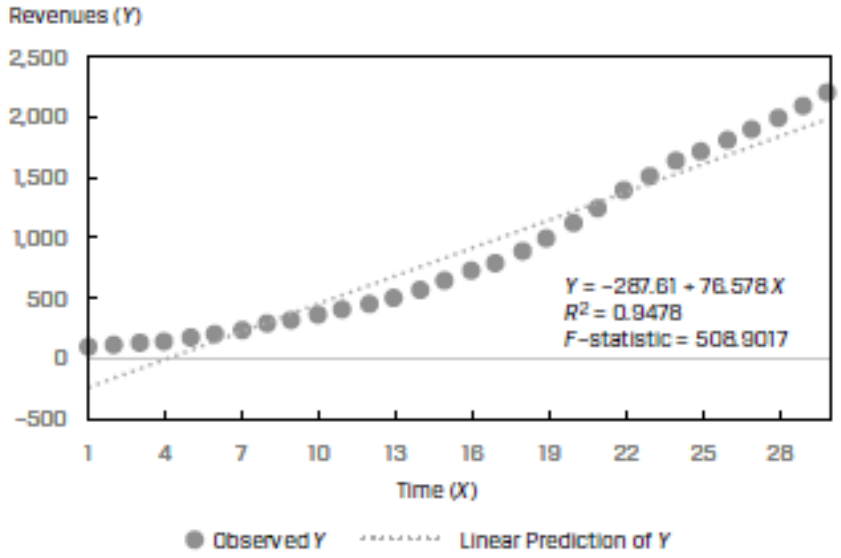
The 95% prediction interval is:

12.375 ± 2.776 (3.736912)

$2.0013 < \hat{Y} < 22.7487$

6. Functional Forms for Simple Linear Regression

Economic and financial data often exhibit non-linear relationships. For example, consider a plot of revenues of a company as the dependent variable (Y) and time as the independent variable (X). Such data will often show exponential growth. Using a simple linear model will not fit this data well as illustrated in Exhibit 33 below.



To make the simple linear regression model fit well, we will have to modify either the dependent or the independent variable. The modification process is called 'transformation' and the different types of transformations are:

- Using the log of the dependent variable
- Using the log of the independent variable
- Using the square of the independent variable
- Differencing the independent variable

In the subsequent sections, we will discuss three commonly used functional forms based on log transformations:

- Log-lin model: The dependent variable is logarithmic but the independent variable is linear.
- Lin-log model: The dependent variable is linear but the independent variable is logarithmic.
- Log-log model: Both the dependent and independent variables are in logarithmic form.

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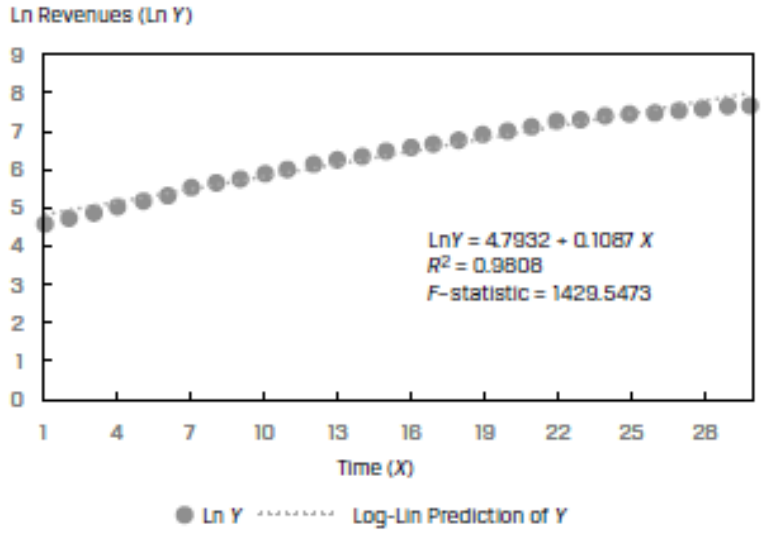
The Log-Lin Model

In the log-lin model, the dependent variable is logarithmic but the independent variable is linear. The regression equation is expressed as:

$$\ln Y_i = b_0 + b_1 X_i$$

The slope coefficient in this model is the relative change in the dependent variable for an absolute change in the independent variable.

We can transform the Y variable (revenues) in Exhibit 33 into its natural log and then fit the regression line. This is demonstrated in Exhibit 34 below:



We can see that the log-lin model is a better fitting model than the simple linear model for data with exponential growth.

Example: Making forecasts with a log-lin model

Suppose the regression model is: ln Y = −7 + 2X. If X is 2.5% what is the forecasted value of Y?

Solution:

Ln Y = -7 + 2*2.5 = -2

$Y = e^{-2} = 0.135335$

The Lin-Log Model

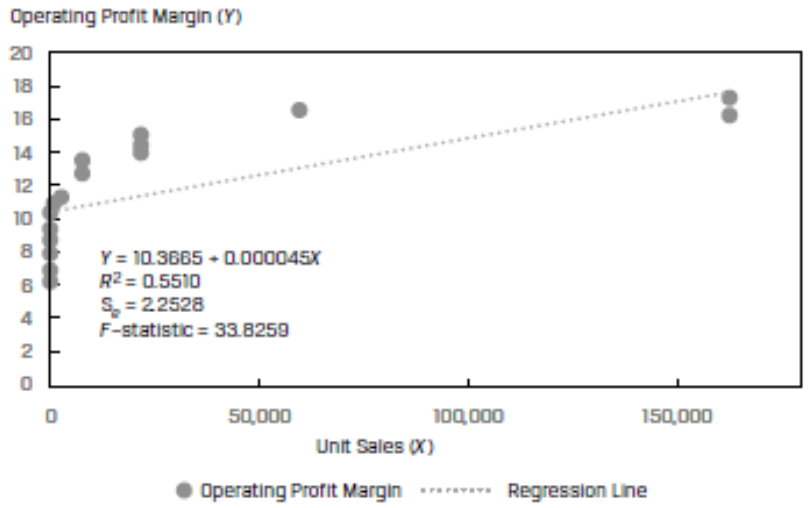
In the lin-log model, the dependent variable is linear but the independent variable is logarithmic. The regression equation is expressed as:

$$Y_i = b_0 + b_1 \ln X_i$$

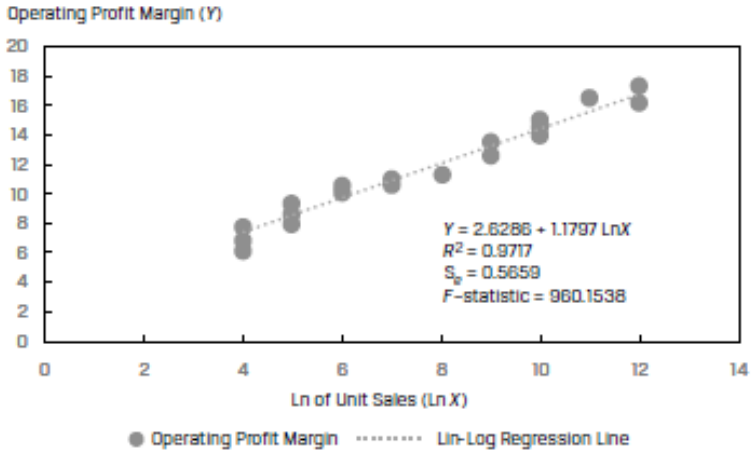
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The slope coefficient in this model is the absolute change in the dependent variable for a relative change in the independent variable.

Consider a pot of operating profit margin as the dependent variable (Y) and unit sales as the independent variable (X). The scatter plot and regression line for a sample of 30 companies is shown in Exhibit 35.



Instead of using the unit sales directly, if we transform the variable and use the natural log of unit sales as the independent variable, we get a much better fit. This is shown in Exhibit 36.



The R^2 of the model jumps to 97.17% from 55.10%. For this data, the lin-log model has a significantly higher explanatory power as compared to simple linear model.

The Log-Log Model

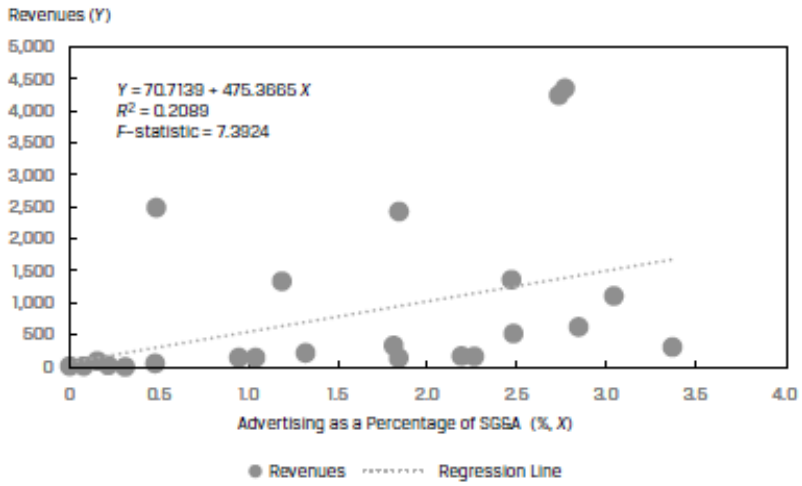
In the log-log model, both the dependent and independent variables are in logarithmic form. It is also called the 'double-log' model. The regression equation is expressed as:

$$\ln Y_i = b_0 + b_1 \ln X_i$$

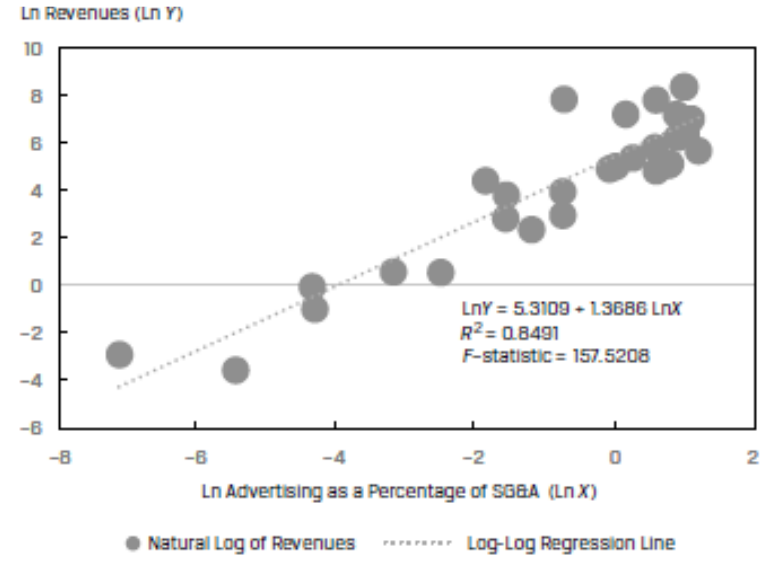
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The slope coefficient in this model is the relative change in the dependent variable for a relative change in the independent variable. The model is often used to calculate elasticities.

Consider a plot of company revenues as the dependent variable (Y) and the advertising spend as a percentage of SG&A, ADVERT, as the independent variable (X). The scatter plot and regression line for a sample company is shown in Exhibit 37 below:



The R^2 of this model is only 20.89%. If we use the natural logs of both the revenues and ADVERT variables, we get a much better fit. This is shown in Exhibit 38.



The R^2 of the model increases by more than 4 times to 84.91% and the F-stat jumps from 7.39 to 157.52. For this data, the log-log model results in a much better fit as compared to the simple linear model.

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Selecting the Correct Functional Form

To select the correct functional form, we can examine the goodness of fit measures:

- Coefficient of determination (R^2)
- F-statistic
- Standard error of estimate (SEE)

A model with a high R^2, high F-stat and low SEE is better.

In addition to these fitness measures, we can also look at the plots of residuals. A good model should show random residuals i.e. the residuals should not be correlated.

Example: Comparing Functional Forms
(*This is based on Question Set the curriculum.*)

An analyst is evaluating the relationship between the annual growth in consumer spending (CONS) in a country and the annual growth in the country's GDP (GGDP). He estimates the following two models:

	Model 1	**Model 2**
	$GGDP_i = b_0 + b_1 CONS_i + \varepsilon_i$	$GGDP_i = b_0 + b_1 \ln(CONS_i) + \varepsilon_i$
Intercept	1.040	1.006
Slope	0.669	1.994
R^2	0.788	0.867
SEE	0.404	0.320
F-stat	141.558	247.040

1. Identify the functional form used in these models.
2. Explain which model has better goodness-of-fit with the sample data.

Solution to 1:

Model 1 is a simple linear regression with no transformations. Model 2 is a lin-log model.

Solution to 2:

Model 2 has a higher R^2, higher F-stat, and a lower SEE as compared to Model 1. Therefore Model 2 has a better goodness-of-fit with the sample data.

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Summary

LO: Describe a simple linear regression model, how the least squares criterion is used to estimate regression coefficients, and the interpretation of these coefficients.

A linear regression model computes the best fit line through the scatter plot, which is the line with the smallest distance between itself and each point on the scatter plot.

The variable whose variation we want to explain is called the dependent variable.

The explanatory variable is called the independent variable.

Linear regression chooses the estimated values for intercept $\widehat{b_0}$ and slope $\widehat{b_1}$ such that the sum of the squared errors (SSE), i.e., the vertical distances between the observations and the regression line is minimized.

The intercept is the value of the dependent variable when the independent variable is zero.

The slope measures the change in the dependent variable for a one-unit change in the independent variable. If the slope is positive, the two variables move in the same direction. If the slope is negative, the two variables move in opposite directions.

LO: Explain the assumptions underlying the simple linear regression model, and describe how residuals and residual plots indicate if these assumptions may have been violated.

The simple linear regression model is based on the following four assumptions:

1. Linearity: The relationship between the dependent variable, Y, and the independent variable, X, is linear.
2. Homoskedasticity: The variance of the regression residuals is the same for all observations.
3. Independence: The observations, pairs of Ys and Xs, are independent of one another. This implies the regression residuals are uncorrelated across observations.
4. Normality: The regression residuals are normally distributed.

LO: Calculate and interpret measures of fit and formulate and evaluate tests of fit and of regression coefficients in a simple linear regression.

Measures of goodness of fit:

The coefficient of determination, denoted by R^2, measures the fraction of the total variation in the dependent variable that is explained by the independent variable.

$$R^2 = \frac{\text{Explained variation}}{\text{Total variation}} = \frac{\text{Regression sum of squares (RSS)}}{\text{Sum of squares total (SST)}}$$

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The F-statistic tests whether all the slope coefficients in a linear regression are equal to 0. In a regression with one independent variable, this is a test of the null hypothesis H_0: $b_1 = 0$ against the alternative hypothesis H_a: $b_1 \neq 0$.

The F-statistic is calculated as:

$$F = \frac{\frac{RSS}{k}}{\frac{SSE}{n-(k+1)}} = \frac{\text{Mean regression sum of squares}}{\text{Mean squared error}} = \frac{MSR}{MSE}$$

Hypothesis test of regression coefficients:

In order to test whether an estimated slope coefficient is statistically significant, we use hypothesis testing.

The steps are:

Step 1: Define the null and alternate hypothesis. Typically used definitions are: H_0: $b_1 = 0$; H_a: $b_1 \neq 0$

Step 2: Specify the level of significance. Typically, a 5% level of significance is selected. Since the alternate hypothesis contains a '≠' sign, this is a two tailed test. For a 5% level of significance, and a two tailed test the critical t-values are ±2.776.

Step 3: Compute the test statistic.

The t-statistic is calculated by subtracting the hypothesized population slope (B_1) from the estimated slope coefficient ($\widehat{b_1}$) and then dividing this difference by the standard error of the slope coefficient, $s_{\widehat{b_1}}$:

$$t = \frac{\widehat{b_1} - B_1}{s_{\widehat{b_1}}} = \frac{\text{Estimated value} - \text{Hypothesized value}}{\text{standard error}}$$

The standard error of the slope coefficient is calculated as the ratio of the model's standard error of the estimate (SEE) to the square root of the variation of the independent variable:

$$s_{\widehat{b_1}} = \frac{SEE}{\sqrt{\sum_{i=1}^{n}(X_i - \overline{X})^2}}$$

Step 4: Make a decision. Compare the absolute value of the t-statistic with the critical value and make a decision to accept or reject the null hypothesis. For a 5% significance level, our decision rule is: reject the null hypothesis if the calculated t-statistic is less than −2.776 or greater than +2.776.

LO: Describe the use of analysis of variance (ANOVA) in regression analysis, interpret ANOVA results, and calculate and interpret the standard error of estimate in a simple linear regression.

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Analysis of variance or ANOVA is a statistical procedure of dividing the total variability of a variable into components that can be attributed to different sources.

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F-statistic
Regression (explained variation)	k	RSS	$\text{MSR} = \frac{\text{RSS}}{\text{k}}$	$\text{F} = \frac{\text{MSR}}{\text{MSE}}$
Error (unexplained variation)	n - 2	SSE	$\text{MSE} = \frac{\text{SSE}}{\text{n} - \text{k} - 1}$	
Total variation	n – 1	SST		

The standard error of estimate (SEE) measures how well a given linear regression model captures the relationship between the dependent and independent variables. It is the standard deviation of the prediction errors. A low SEE implies an accurate forecast.

$$\text{Standard error of estimate (SEE)} = \sqrt{\text{MSE}}$$

LO: Calculate and interpret the predicted value for the dependent variable, and a prediction interval for it, given an estimated linear regression model and a value for the independent variable.

The estimated variance of the prediction error is given by:

$$s_f^2 = s^2 * \left[1 + \frac{1}{n} + \frac{(X - \bar{X})^2}{(n-1)s_x^2}\right]$$

Once the variance of the prediction error is known, it is easy to determine the confidence interval around the prediction. The steps are:

1. Make the prediction.
2. Compute the variance of the prediction error.
3. Determine t_c at the chosen significance level α.
4. Compute the (1-α) prediction interval using the formula below:
   $\hat{Y} \pm t_c * s_f$

LO: Describe different functional forms of simple linear regressions.

The three commonly used functional forms based on log transformations:

- Log-lin model: The dependent variable is logarithmic but the independent variable is linear.
- Lin-log model: The dependent variable is linear but the independent variable is logarithmic.

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- Log-log model: Both the dependent and independent variables are in logarithmic form.

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LM11 Introduction to Big Data Techniques

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1. Introduction

This learning module covers:

- What is 'Fintech' and how it is used in investment analysis
- A brief explanation of Big Data, artificial intelligence, and machine learning
- Applications of Big Data and Data Science to investment management

2. How Is Fintech Used in Quantitative Investment Analysis?

The term 'Fintech' comes from combining 'Finance' and 'Technology'. Fintech refers to technological innovation in the design and delivery of financial products and services.

Though the term 'Fintech' is relatively new, its earlier forms involved data processing and automation of routine tasks. Fintech later advanced into decision-making applications based on complex machine learning logic.

The major drivers of fintech have been:

- Rapid growth in data
- Technological advances

While Fintech spans the entire finance space, this learning module focuses on fintech applications that are more directly relevant to quantitative analysis in the investment industry:

- Analysis of large datasets
- Analytical tools

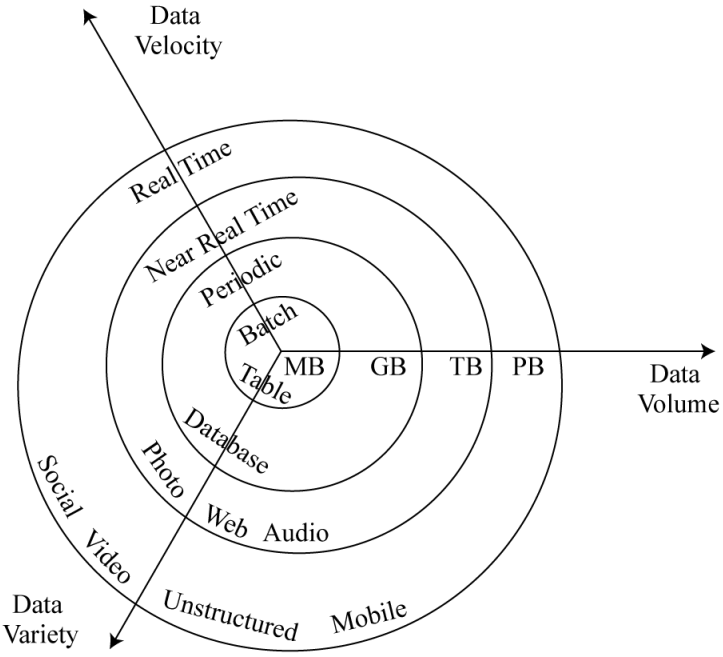
Big Data

Big Data refers to vast amount of data generated by industry, governments, individuals, and electronic devices.

Characteristics of big data typically include:

- Volume: Over the last few decades, the amount of data that we are dealing with has grown exponentially.
- Velocity: It refers to the speed at which data are communicated. In the past we often worked with batch processing; however, we are now increasingly working with real time data.
- Variety: Historically we only dealt with structured data. However, we are now also dealing with unstructured data such as text, audio, video, etc.

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Data	Volume Key	Bytes of Information
MB	Megabyte	One Million
GB	Gigabyte	One Billion
TB	Terabyte	One Trillion
PB	Petabyte	One Quadrillion

In addition to these three V's, a fourth V is becoming increasingly important, especially when using big data for drawing inferences or making predictions.

- Veracity – refers to the credibility and reliability of different data sources.

Big Data can be structured (can be organized in tables), semi-structured, or unstructured (cannot be represented in a tabular form).

Sources of Big Data

Traditional data sources include annual reports, regulatory filings, trade price and volume, etc. Alternate data include many other sources and types of data. A simple classification of alternate data sources is shown in Exhibit 2 of the curriculum.

Individuals	**Business Processes**	**Sensors**
Social media	Transaction data	Satellites
News, reviews	Corporate data	Geolocation
Web searches, personal data		Internet of Things
		Other sensors

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Big Data Challenges

While big data can be a huge asset, there are also several challenges. The quality of data may be questionable. The data may have biases, outliers, etc. The volume of data collected may not be sufficient. We might be dealing with too much data or too little data. Another concern is the appropriateness of data. In most cases working with Big Data usually involves cleansing and organizing the data before we start analyzing it.

3. Advanced Analytical Tools: Artificial Intelligence and Machine Learning

Artificial intelligence (AI) computer systems perform tasks that have traditionally required human intelligence. They exhibit cognitive and decision-making ability comparable or superior to that of human beings. An important term in this context is 'neural networks'. It refers to programming based on how the brain learns and processes information. There are examples of AI all around us. For example, chess playing computer programs, digital assistants like Apple's Siri, etc.

Machine learning (ML) refers to computer-based techniques that "extract knowledge from large amounts of data by "learning" from known examples and then generating structure or predictions" without relying on any help from a human. ML algorithms aim to "find the pattern, apply the pattern."

In ML, the dataset is divided into three distinct subsets:

1. Training dataset: It allows the algorithm to identify relationships between inputs and outputs based on historical patterns in the data.
2. Validation dataset: It is used to validate and model tune the relationships identified by training dataset.
3. Test dataset: As the name implies, this dataset is used to test the model's ability to predict well on new data.

Once an algorithm has mastered the training and validation datasets, it can be used to predict outcomes based on other datasets.

Broadly speaking there are three main approaches to machine learning:

1. Supervised learning: In supervised learning, both inputs and outputs are identified or labeled. After learning from labeled data, the trained algorithm is used to predict outcomes for new data sets.
2. Unsupervised learning: In unsupervised learning, the input and output variables are not labeled. Here we want the ML algorithm to seek relationships on its own.
3. Deep learning**:** In deep learning, (or deep learning nets), neural networks are used by the computers to perform multistage, non-linear data processing to identify patterns. Deep learning can use supervised or unsupervised machine learning approaches. With terms like AI and ML one might think that human judgment is not required, but that is

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far from the truth. For ML to work well, good human judgment is required. Human judgment is required for questions like: which data to use, how much data to use, which analytical techniques are relevant in the given context. Human judgment may also be needed to clean and filter the data before it is fed to the ML algorithm. Deep learning algorithms are used for image, pattern, and speech recognition.

Some challenges associated with machine learning are:

- Over-fitting the data: Sometimes an algorithm may try to be too precise in the way it interprets data and predicts outcomes. This leads to over-trained models and may result in data mining bias. We try to mitigate this issue by having a good validation dataset.
- Black box: ML techniques can be opaque or black box, which means we have predictions that are not very easy to understand or to explain.

Despite these challenges and weaknesses, the importance of ML in finance and investment management has been growing substantially.

4. Tackling Big Data with Data Science

Data science leverages advances in computer science, statistics, and other disciplines for the purpose of extracting information from Big Data.

Data Processing Methods

Data processing methods include:

- Capture: Refers to how data is collected from various sources and transformed into a format that can be used by the analytical process.
- Curation: Refers to the process of ensuring data quality and accuracy through data cleaning.
- Storage: Refers to how data will be recorded, archived, and accessed. It also refers to the underlying databases design. An important consideration here is whether the data is structured, unstructured, or both. We also need to be concerned whether the analytical tools need real time access to the data or not.
- Search: Refers to how we can find what we want from the vast amount of data.
- Transfer: Refers to how data will move from the underlying source to the analytical tools that are being used.

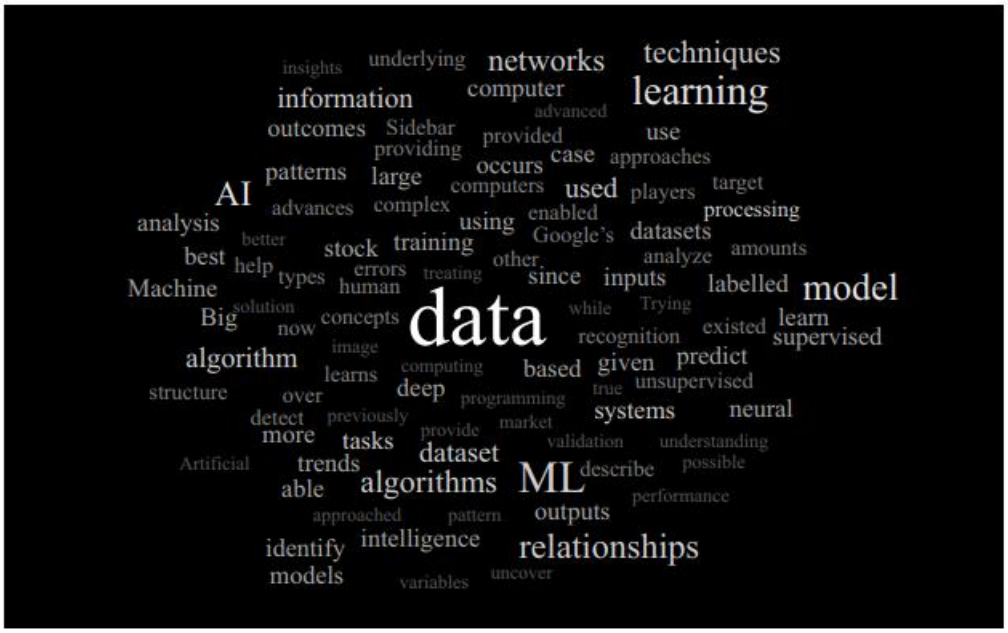
Data Visualization

Another aspect of data science is data visualization. This refers to how the data will ultimately be presented to the analyst/user. Historically, data visualization happened through graphs, charts, etc. However, in more recent times tools such as heat maps, tree diagrams, and tag clouds are also being used.

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An example of a heat map is a map of a city where routes with high traffic congestion are shown in red. A tag cloud is a technique applicable to textual data. Words that appear more often are shown in a larger font, whereas words that appear less often are shown with a smaller font. This helps us to quickly evaluate how consumers/users are talking about a given product.

Exhibit 3 from the curriculum shows an example of a 'tag cloud'.



Text Analytics and Natural Language Processing

Text analytics refers to the use of computer programs to derive meaning from large, unstructured text- or voice-based data. For example, text analytics can be used to gauge the consumer sentiment about a new product by analyzing what is being said about the product on blogs, forums, YouTube, etc. Based on this analysis, we can determine if the sentiment is very positive, positive, neutral, or negative.

Natural language processing (NLP) is an application of text analytics whereby computers analyze and interpret human language. For example, NLP analysis can be used for communications from policy makers such as the US Federal Reserve. Officials at these institutions may send subtle messages through their choice of words and inferred tone. NLP analysis can provide insights into these subtle messages. Such processing is possible because of access to Big Data and processing power.

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Summary

LO: Describe aspects of "fintech" that are directly relevant for the gathering and analyzing of financial data.

Fintech refers to the technological innovation in the design and delivery of financial products and services.

LO: Describe Big Data, artificial intelligence, and machine learning.

Big Data refers to vast amounts of data generated by industry, governments, individuals, and electronic devices.

Artificial intelligence (AI) computer systems perform tasks that have traditionally required human intelligence. They exhibit cognitive and decision-making ability comparable or superior to that of human beings.

Machine learning (ML) refers to computer-based techniques that "extract knowledge from large amounts of data by "learning" from known examples and then generating structure or predictions" without relying on any help from a human. In ML, the dataset is divided into three distinct subsets, training dataset, validation dataset, and test dataset. There are three main approaches to machine learning, i.e., supervised learning, unsupervised learning, and deep learning.

LO: Describe applications of Big Data and Data Science to investment management.

Text analytics refers to the use of computer programs to derive meaning from large, unstructured text- or voice-based data.

Natural language processing (NLP) is an application of text analytics whereby computers analyze and interpret human language.

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